

**SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4**

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)

Received**MAR 18 2016**

Well File No.

28744

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

ND Oil & Gas Division

☐ Notice of Intent	Approximate Start Date
☒ Report of Work Done	Date Work Completed September 13, 2015
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	Approximate Start Date

☐ Drilling Prognosis	☐ Spill Report
☐ Redrilling or Repair	☐ Shooting
☐ Casing or Liner	☐ Acidizing
☐ Plug Well	☐ Fracture Treatment
☐ Supplemental History	☒ Change Production Method
☐ Temporarily Abandon	☐ Reclamation
☐ Other	Well is now on pump

Well Name and Number Wade Federla 5300 41-30 9B					
Footages		Qtr-Qtr	Section	Township	Range
811 F S L 280 F W L		LOT4	30	153 N	100 W
Field Baker		Pool Bakken		County Williams	

24-HOUR PRODUCTION RATE

Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

Effective 09/13/2015 the above referenced well is on pump.

End of Tubing: 2-7/8" L-80 tubing @ 10040.83'

Pump: ESP @ 9853.11'

Company Oasis Petroleum North America LLC		Telephone Number 281-404-9436	
Address 1001 Fannin, Suite 1500			
City Houston		State TX	Zip Code 77002
Signature 		Printed Name Jennifer Swenson	
Title Regulatory Specialist		Date March 17, 2016	
Email Address jswenson@oasispetroleum.com			

FOR STATE USE ONLY

☒ Received	☐ Approved
Date 4-7-2016	
By 	
Title JARED THUNE Engineering Technician	



AUTHORIZATION TO PURCHASE AND TRANSPORT OIL FROM LEASE - Form 8
INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5698 (03-2000)



Well File No.
28744
NDIC CTB No.
To be assigned

228394

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND FOUR COPIES.

Well Name and Number WADE FEDERAL 5300 41-30 9B	Qtr-Qtr LOT4	Section 30	Township 153	Range 100	County Mckenzie
---	------------------------	----------------------	------------------------	---------------------	---------------------------

Operator Oasis Petroleum North America LLC	Telephone Number (281) 404-9573	Field BAKER
--	---	-----------------------

Address 1001 Fannin, Suite 1500	City Houston	State TX	Zip Code 77002
---	------------------------	--------------------	--------------------------

Name of First Purchaser Oasis Petroleum Marketing LLC	Telephone Number (281) 404-9627	% Purchased 100%	Date Effective June 1, 2015
---	---	----------------------------	---------------------------------------

Principal Place of Business 1001 Fannin, Suite 1500	City Houston	State TX	Zip Code 77002
---	------------------------	--------------------	--------------------------

Field Address	City	State	Zip Code
---------------	------	-------	----------

Transporter Hiland Crude, LLC	Telephone Number (580) 616-2058	% Transported 75%	Date Effective June 1, 2015
---	---	-----------------------------	---------------------------------------

Address P.O. Box 3886	City Enid	State OK	Zip Code 73702
---------------------------------	---------------------	--------------------	--------------------------

The above named producer authorizes the above named purchaser to purchase the percentage of oil stated above which is produced from the lease designated above until further notice. The oil will be transported by the above named transporter.

Other First Purchasers Purchasing From This Lease	% Purchased	Date Effective
Other First Purchasers Purchasing From This Lease	% Purchased	Date Effective
Other Transporters Transporting From This Lease	% Transported	Date Effective
Power Crude Transport	25%	June 1, 2015
Other Transporters Transporting From This Lease	% Transported	Date Effective
		June 1, 2015
Comments		

I hereby swear or affirm that the information provided is true, complete and correct as determined from all available records.		Date September 10, 2015
Signature 	Printed Name Dina Barron	Title Mktg. Contracts Administrator

Above Signature Witnessed By:		
Signature 	Printed Name Jeremy Harris	Title Marketing Scheduler

FOR STATE USE ONLY

Date Approved SEP 18 2015
By
Title Oil & Gas Production Analyst

Wade Federal 5300 41-30 9B

End of Well Report

Wade Federal 5300 41-30 9B

811' FSL - 280' FWL
SEC 30, T153N, R100W
McKenzie County, ND

Prepared by:
Heather Coutts, *well-site geologist*
Columbine Logging, Inc.
2385 S. Lipan St.
Denver CO 80223



Prepared for:



Brendan Hargrove and Kellie Rasmussen
, *operations geologists*
1001 Fannin, Suite 1500
Houston, Texas 77002

1.0 INTRODUCTION

Wade Federal 5200 41-30 9B is a East lateral Middle Bakken well located in SEC 30, T153N, R100W in McKenzie County, North Dakota. The primary pay zone was approximately 20' under the bottom of the Upper Bakken Shale. This pay zone was picked for its production potential and quality of reservoir rock. The objective was to steer the well within the defined pay zone and within the legal requirements of the state. Directional and MWD services were performed by Schlumberger/Pathfinder and RPM. Heather A. Coutts and Dylan E. Fowler were the primary well site geologists; providing geo-steering and mud logging from Columbine Logging, Inc.

Well Information

API #: 33-053-06051

Field: Baker

Spud Date: 09/05//2014

TD Date: 12/08/2014

Surface Location: 811' FSL & 280' FWL SEC 30, T153N, R100W, McKenzie County, North Dakota.

Intermediate Casing Point: 222' FSL & 1151' FWL; SEC 30; 11,525' MD, 10,703.20' TVD

Bottom Hole Location: 0' FSL & 212' FEL; SEC 29; 20,650' MD, 10,814.1' TVD

Surface Elevation: 2045'

KB: 2076'

Casing Shoe: 11,525' MD, 10,703.20' TVD

Total Depth Drilled: 20,650'

Operator: Oasis

Rig: Patterson 488

Company man: Tyler LaClaire

Well-site Geologist: Heather Coutts
Dylan E. Fowler

Mud logging: Columbine Logging

DD: RPM

Mud Service: Reliable

Drilling Mud: Invert, Brine

MWD: Schlumberger/Pathfinder

2.0 SERVICES

2.1 Well site Geology (*Columbine Logging, Inc.*)

Geological consulting and mud logging started on 09/09/2014. Services provided included; morning reports, evening reports, noon and midnight reports, sample examination, sample recording via pictures, production of vertical and horizontal mudlog, geo steering, sample collection and bagging, sample mailing and a final end of well report.

2.1.1 Geosteering

Our offset GR TVD logs were from the Wade Federal 5300 21-30H well, located on the same well pad as Wade Federal 5300 41-30 9B. Within the Middle Bakken, the primary objective was to stay near the middle gamma markers which marks the inter layering between dolomitic siltstone and limy sandstones. Gamma patterns were compared with the offset log and a TVD log was created while landing the curve to in order to land in the targeted zone. Steering in the lateral was accomplished by calculating dip from relevant gamma markers, as well as by using lithology, total gas and ROP to determine our position within the formation.

2.1.2 Gamma and Surveys

Gamma and survey MWD services were provided by Schlumberger/Pathfinder. The majority of the well was drilled within the target area of the Middle Bakken.

2.2 Mud Logging (*Columbine Logging, Inc.*)

2.2.1 Sample Examination

Samples were collected every 30 ft in the straight hole and build section, and every 30 ft while drilling the lateral. Descriptions included; mineralogy, color, firmness, argillaceous content, structure, texture, allochems, porosity, oil stain, and hydrocarbon fluorescence. Carbonate identification was determined with 10% dilute HCl, alizarin red and calcimeter. Hydrocarbon fluorescence was determined using a fluoroscope with a UV lamp.

2.2.2 Gas Detection

Gas was logged using a Bloodhound total gas/chromatograph system. The gas detection system uses an infra-red detector to measure total gas and the chromatograph separates and measures gases C1, C2, C3, iC4 and nC4. Gas was recorded in units where 1 unit equals 100 ppm. The gas detection system measured gases: C1, C2, C3, IC4, NC4, H2S, O2 and CO₂.

The Bloodhound Gas Detection and Chromatograph system use digital signal processing techniques and non-dispersive infrared and chemical sensors for gas detection. The system uses a proprietary chromatograph, which has the capability to detect from 0 to 10,000 gas units. This translates as 0 to 100% typical naturally-occurring hydrocarbon gas mixtures. Calibration is performed using National Institute of Standards and Technology (NIST) traceable calibration gases. Lab calibration points include 0%, 2.5%, and 100% pure methane. Complete immunity to saturation or damage in the presence of high concentrations of both light and heavy hydrocarbon gases precludes the necessity of constant re-calibration or zero referencing. This allows the Bloodhound to react to hydrocarbon based gases from zero to 100% in concentration without dilution.

Lag time was approximated from a calculation of annular velocity based on: pump output, open-hole diameter, cased hole diameter, collar diameter, drill pipe diameter and bottom hole assembly. Connection gases were monitored to confirm lag time calculations and thereby adjust lag time when needed.

3.0 GEOLOGY

3.1 Formation Tops Formation tops were picked using ROP, lithology, and gamma ray to identify markers in the curve and lateral (Table 3.1).

FORMATION TOPS						
Formation/Marker Beds	ACTUAL				Prognosis	
Vertical Section	Top MD (ft)	Top TVD (ft)	THICKNESS (ft)	Difference (ft)	TVD KB/DF(ft)	TVDSS (ft)
Kibbey Lime	8344	8344	148	-2	8342	-6266
Charles Salt	8492	8492	684	9	8501	-6425
Base Last Salt	9176	9176	209	-16	9160	-7084
Mission Canyon	9385	9385	549	-10	9375	-7299
Lodgepole	9934	9934	731	-11	9923	-7847
False Bakken	10785	10665	9	-15	10650	-8574
Upper Bakken Shale	10808	10674	19	-11	10663	-8587
Middle Bakken	10868	10693	~41	-15	10678	-8602
Middle Bakken (Top of Target)	10868	10709	11	-19	10690	-8614
Middle Bakken (Base of target)	10868	10720		-19	10701	-8625
Lower Bakken Shale					10714	-8638

Table 3.1 Wade Federal 5300 41-30 9B Formation Tops

3.2 Lithology

Sample analysis began at 8,000' MD in the Otter Formation.

3.3 Formation Dip

The formation had an average dip of 89.5°.

3.4 Shows

The vertical-build section was drilled with invert mud and the lateral was drilled with brine/production water. The oil-based mud contributed a background gas of 200-300 units, and saturated cuttings with oil, making all cuttings in the vertical show the same

cut and fluorescence. Gas shows were around 3500+ units during the drilling of the lateral.

3.5 Oil Shows

Invert mud was used in the vertical, masking any oil shows. In the lateral part of the well the oil shows were consistently a bright blue green florescence with a bright diffuse blue to blue green cut and a medium brown residue ring.

4.0 WELLBORE

The surface location is 811' FSL & 280' FWL SEC 30, T153N, R100W, McKenzie County, North Dakota. Ground elevation is 2,045' and KB elevation was 2,076', referenced to the Kelly bushing of Patterson 488. The curve was landed in the Middle Bakken at 11,525' MD, 10,703.20' TVD. The lateral was drilled to TD at 20,650' MD, 10,814.1' TVD, 994' 0' FSL & FEL; SEC 29. Figure 4.1 shows a cross-section of the lateral.

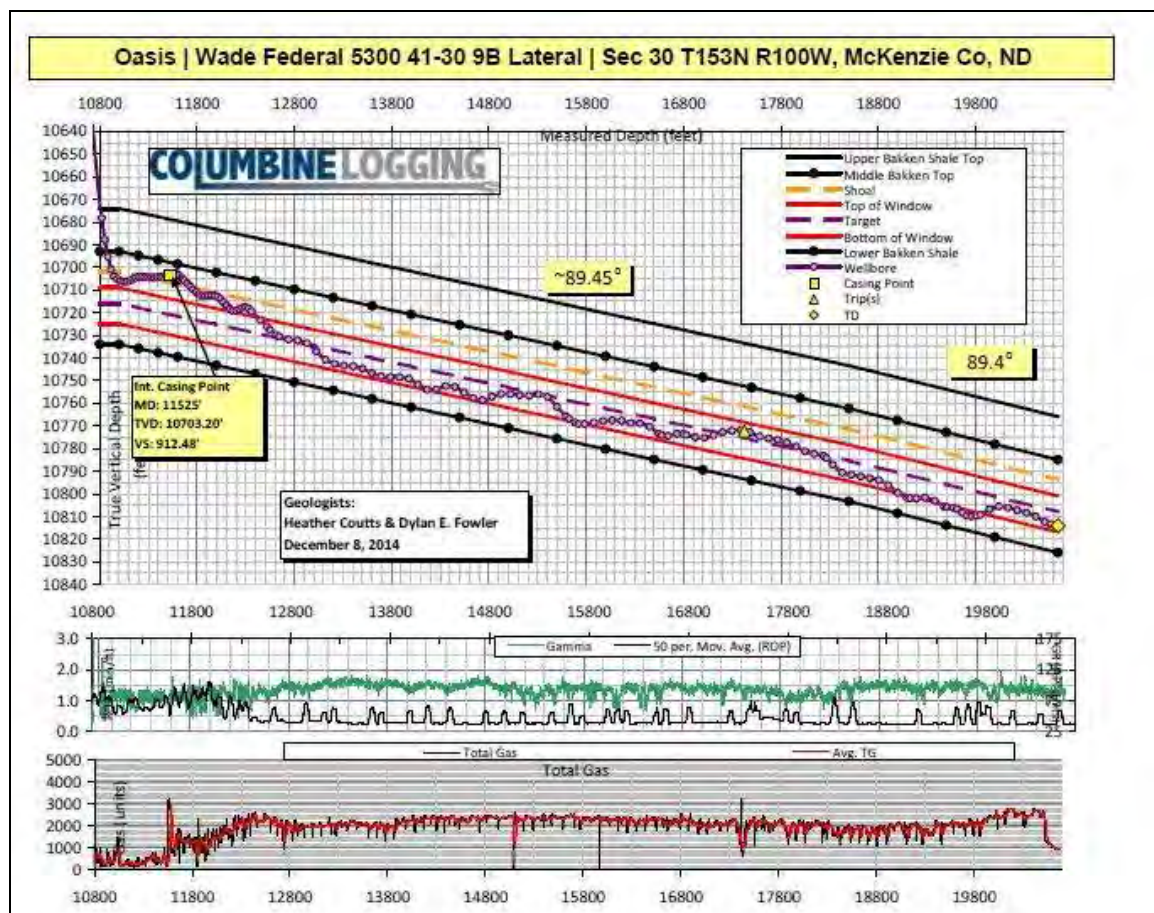


Figure 4.1 Wellbore Cross

5.0 SUMMARY AND CONCLUSION

Wade Federal 5300 41-30 9B is a South lateral Middle Bakken well located in SEC 30, T153N, R100W in McKenzie County, North Dakota. The primary pay zone was 20' under the bottom of the Upper Bakken Shale. This pay zone was picked for its production potential and quality of reservoir rock. The objective was to steer the well within the defined pay zone and within the legal requirements of the state.

The primary objective was to stay near the middle gamma markers which marks the inter layering between dolomitic siltstone and limy sandstones. Gamma patterns were compared with the offset log and a TVD log was created while landing the curve to in order to land in the targeted zone. Steering in the lateral was accomplished by calculating dip from relevant gamma markers, as well as by using lithology, total gas and ROP to determine our position within the formation.

The formation had an average dip of 89.5°. (see pg. 5, fig 4.1 for a detailed formation dip profile).

Currently the well is awaiting completion.

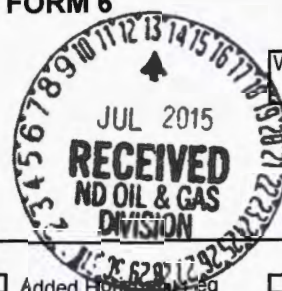
Heather Coutts, Well-Site Geologist
Columbine Logging, Inc.
9844 Titan Ct. Unit #6
Littleton, CO 80125-9354
(303) 289-7764

COLUMBINE LOGGING



WELL COMPLETION OR RECOMPLETION REPORT - FORM 6

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 2468 (04-2010)



Well File No.
28744

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.

PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

Designate Type of Completion			
☒ Oil Well	☐ EOR Well	☐ Recompletion	☐ Deepened Well
☐ Gas Well	☐ SWD Well	☐ Water Supply Well	☐ Other:
Well Name and Number Wade Federal 5300 41-30 9B		Spacing Unit Description Sec. 29/30/31/32 T153N R100W	
Operator Oasis Petroleum North America		Telephone Number (281) 404-9591	
Address 1001 Fannin, Suite 1500		Pool Bakken	
City Houston	State TX	Zip Code 77002	Permit Type ☐ Wildcat ☒ Development ☐ Extension

LOCATION OF WELL

At Surface 811 F S L	280 F W L	Qtr-Qtr LOT4	Section 30	Township 153 N	Range 100 W	County Williams
Spud Date September 5, 2014	Date TD Reached December 8, 2014	Drilling Contractor and Rig Number Patterson 488		KB Elevation (Ft) 2070	Graded Elevation (Ft) 2045	
Type of Electric and Other Logs Run (See Instructions) MWD/GR from KOP to TD; CBL from int. TD to surface						

CASING & TUBULARS RECORD (Report all strings set in well)

Well Bore	String Type	Size (Inch)	Top Set (MD Ft)	Depth Set (MD Ft)	Hole Size (Inch)	Weight (Lbs/Ft)	Anchor Set (MD Ft)	Packer Set (MD Ft)	Sacks Cement	Top of Cement
Surface Hole	Surface	13 3/8	0	2110		54.5			1178	0
Vertical Hole	Intermediate	9 5/8	0	6034		40			1168	2100
Vertical Hole	Intermediate	7	0	11552		32			250	8360
Lateral1	Liner	4 1/2	10136	20625		13.5			516	10136
Lateral1	Liner	4 1/2	0	10136		13.5			0	

PERFORATION & OPEN HOLE INTERVALS

Well Bore	Well Bore TD Drillers Depth (MD Ft)	Completion Type	Open Hole/Perforated Interval (MD Ft) Top Bottom	Kick-off Point (MD Ft)	Top of Casing Window (MD Ft)	Date Perf'd or Drilled	Date Isolated	Isolation Method	Sacks Cement
Lateral1	20650	Perforations	11552 20265	10200					

PRODUCTION

Current Producing Open Hole or Perforated Interval(s), This Completion, Top and Bottom, (MD Ft) Lateral 1- 11552' to 20625'						Name of Zone (If Different from Pool Name)			
Date Well Completed (SEE INSTRUCTIONS) June 13, 2015			Producing Method Flowing		Pumping-Size & Type of Pump		Well Status (Producing or Shut-In) Producing		
Date of Test 06/30/2015	Hours Tested 24	Choke Size 26 /64	Production for Test		Oil (Bbls) 3109	Gas (MCF) 102	Water (Bbls) 2982	Oil Gravity-API (Corr.) °	Disposition of Gas Sold
Flowing Tubing Pressure (PSI) 1500		Flowing Casing Pressure (PSI) 2800		Calculated 24-Hour Rate	Oil (Bbls) 3109	Gas (MCF) 102	Water (Bbls) 2982	Gas-Oil Ratio 33	

GEOLOGICAL MARKERS

[illegible]

PLUG BACK INFORMATION

[illegible]

CORES CUT

Top (Ft)	Bottom (Ft)	Formation	Top (Ft)	Bottom (Ft)	Formation

Drill Stem Test

Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								

Well Specific Stimulations

Date Stimulated	Stimulated Formation Bakken	Top (Ft) 11552	Bottom (Ft) 20265	Stimulation Stages 50	Volume 171968	Volume Units Barrels
Type Treatment Sand Frac	Acid %	Lbs Proppant 9289252	Maximum Treatment Pressure (PSI) 9310		Maximum Treatment Rate (BBLS/Min) 37.0	
Details 40/70 White: 1188022 20/40 White: 6518120 20/40 RC: 1395110 100 Mesh: 188000						
Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units
Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)		Maximum Treatment Rate (BBLS/Min)	
Details						
Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units
Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)		Maximum Treatment Rate (BBLS/Min)	
Details						
Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units
Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)		Maximum Treatment Rate (BBLS/Min)	
Details						
Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units
Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)		Maximum Treatment Rate (BBLS/Min)	
Details						

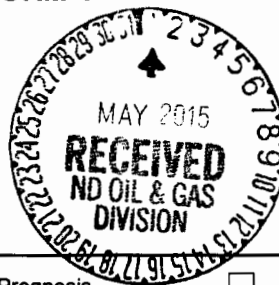
ADDITIONAL INFORMATION AND/OR LIST OF ATTACHMENTS

Frac string was pulled after frac.

I hereby swear or affirm that the information provided is true, complete and correct as determined from all available records.	Email Address jswenson@oasispetroleum.com	Date 07/10/2015
	Printed Name Jennifer Swenson	Title Regulatory Specialist

**SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4**

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)



Well File No.
28744

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☒ Notice of Intent

Approximate Start Date

May 1, 2015

☐ Report of Work Done

Date Work Completed

☐ Notice of Intent to Begin a Workover Project that may Qualify
for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.

Approximate Start Date

☐ Drilling Prognosis

☐ Spill Report

☐ Redrilling or Repair

☐ Shooting

☐ Casing or Liner

☐ Acidizing

☐ Plug Well

☐ Fracture Treatment

☐ Supplemental History

☐ Change Production Method

☐ Temporarily Abandon

☐ Reclamation

☒ Other **Change well status to CONFIDENTIAL**

Well Name and Number

Wade Federal 5300 41-30 9B

Footages

811 F S L

280 F W L

Qtr-Qtr

LOT4

Section

30

Township

153 N

Range

100 W

Field

Baker

Pool

BAKKEN

County

McKenzie

24-HOUR PRODUCTION RATE

Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)

Address

City

State

Zip Code

DETAILS OF WORK

Effective immediately, we request **CONFIDENTIAL STATUS** for the above referenced well.

This well has not been completed.

OFF CONFIDENTIAL 11/01/15.

Company Oasis Petroleum North America LLC		Telephone Number 281-404-9436	
Address 1001 Fannin, Suite 1500			
City Houston		State TX	Zip Code 77002
Signature 		Printed Name Jennifer Swenson	
Title Regulatory Specialist		Date April 30, 2015	
Email Address jswenson@oasispetroleum.com			

FOR STATE USE ONLY

☐ Received	☒ Approved
Date 5/07/15	
By 	
Title Engineering Technician	



SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)

Well File No.
28976

28744



PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☒ Notice of Intent	Approximate Start Date September 14, 2014	☐ Drilling Reanalysis	☐ Spill Report
☐ Report of Work Done	Date Work Completed	☐ Redrilling or Repair	☐ Shooting
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	Approximate Start Date	☐ Casing or Liner	☐ Acidizing
		☐ Plug Well	☐ Fracture Treatment
		☐ Supplemental History	☐ Change Production Method
		☐ Temporarily Abandon	☒ Reclamation
		☐ Other offsite pit	

Well Name and Number Wade Federal 5300 21-30 12T						
Footages 1640 F N L		270 F W L	Qtr-Qtr SWNW	Section 30	Township 153 N	Range 100 W
Field Baker		Pool Bakken		County McKenzie		

24-HOUR PRODUCTION RATE			
Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

Oasis Petroleum North America LLC respectfully requests to use an offsite pit for this well. The following wells will also use this pit:

Wade Federal 5300 31-30 2B - **28554**
Wade Federal 5300 41-30 3T2 - **28555**
Wade Federal 5300 41-30 4T - **28394**
Wade Federal 5300 41-30 5T2 - **28556**
Wade Federal 5300 41-30 6B - **28425-TH**
Wade Federal 5300 41-30 7T - **28557**
Wade Federal 5300 41-30 8T2 - **28558-TH**
Wade Federal 5300 41-30 9B - **28744**
Wade Federal 5300 21-30 13B - **28978**
Wade Federal 5300 21-30 14T2 - **28977**

Attached are the plats for the offsite pit location.

Company Oasis Petroleum North America LLC		Telephone Number 281-404-9589	
Address 1001 Fannin, Suite 1500			
City Houston		State TX	Zip Code 77002
Signature <i>Sonja Rolfes</i>		Printed Name Sonja Rolfes	
Title Regulatory Analyst		Date August 20, 2014	
Email Address			

FOR STATE USE ONLY

☐ Received	☒ Approved
Date 2-9-15	
By <i>[Signature]</i>	
Title <i>[Signature]</i>	

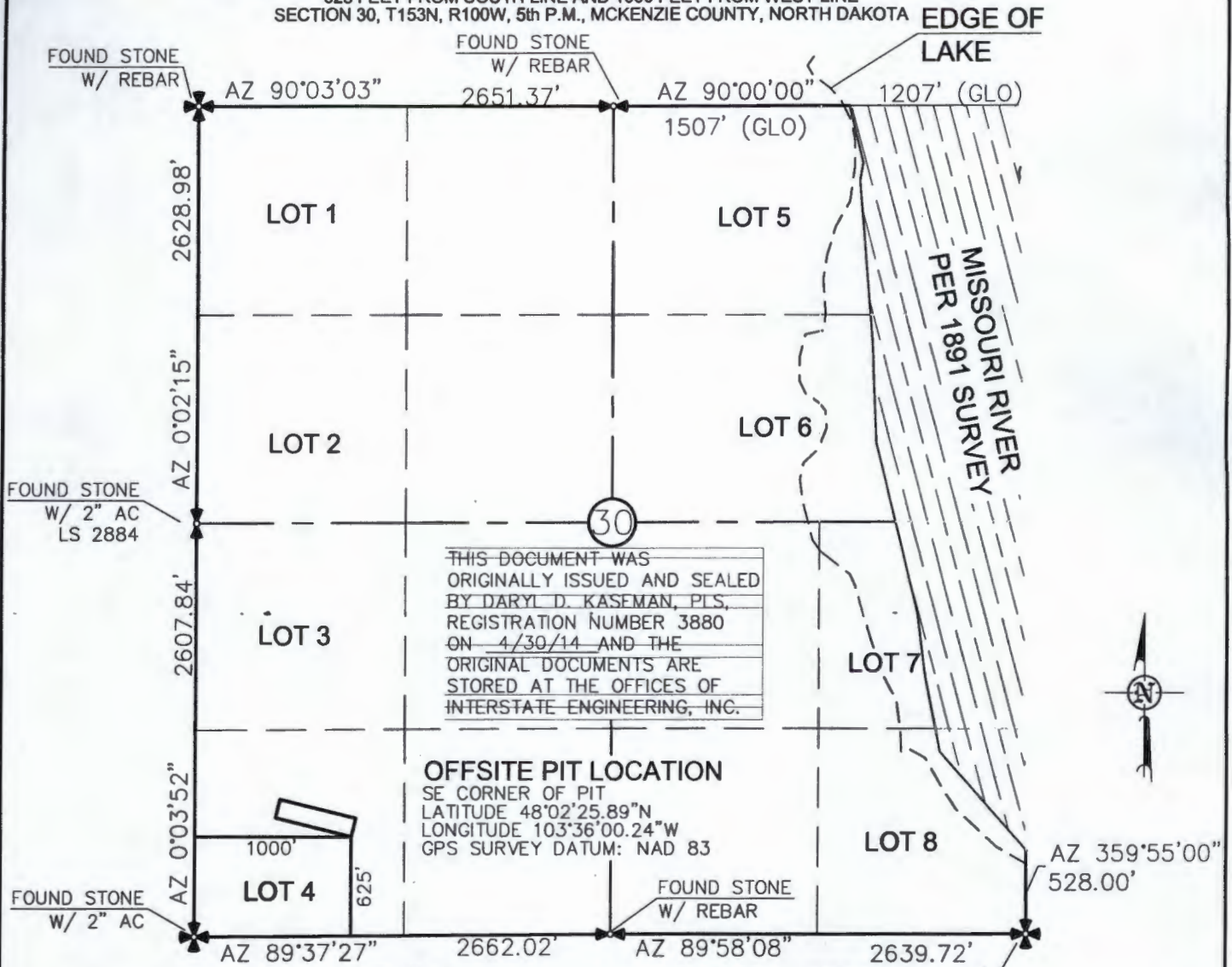
OFFSITE PIT LOCATION PLAT

OASIS PETROLEUM NORTH AMERICA, LLC

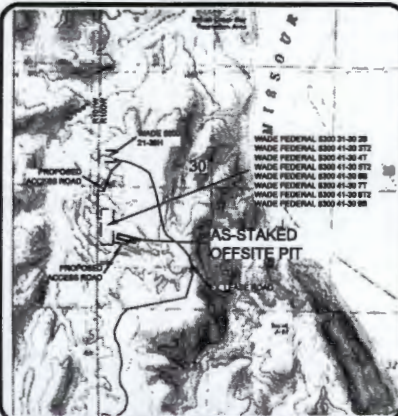
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002

AS-STAKED OFFSITE PIT FOR WADE FEDERAL 5300 31-30 2B, WADE FEDERAL 5300 41-30 3T2, WADE FEDERAL 5300 41-30 4T, WADE FEDERAL 5300 41-30 5T2, WADE FEDERAL 5300 41-30 6B, WADE FEDERAL 5300 41-30 7T, WADE FEDERAL 5300 41-30 8T2, & WADE FEDERAL 5300 41-30 9B
625 FEET FROM SOUTH LINE AND 1000 FEET FROM WEST LINE

SECTION 30, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



VICINITY MAP



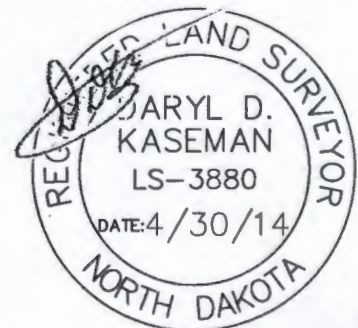
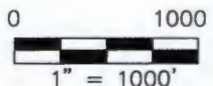
- ✕ - MONUMENT - RECOVERED
- ✕ - MONUMENT - NOT RECOVERED

STAKED ON 4/30/14
VERTICAL CONTROL DATUM WAS BASED UPON
CONTROL POINT 705 WITH AN ELEVATION OF 2158.3'

THIS SURVEY AND PLAT IS BEING PROVIDED AT THE
REQUEST OF ERIC BAYES OF OASIS PETROLEUM. I
CERTIFY THAT THIS PLAT CORRECTLY REPRESENTS
WORK PERFORMED BY ME OR UNDER MY
SUPERVISION AND IS TRUE AND CORRECT TO THE
BEST OF MY KNOWLEDGE AND BELIEF.

[Signature]

DARYL D. KASEMAN LS-3880



© 2014, INTERSTATE ENGINEERING, INC.

1/3



Interstate Engineering, Inc.
P.O. Box 648
425 East Main Street
Sidney, Montana 58270
Ph (406) 433-5617
Fax (406) 433-5618
www.interstateeng.com
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OASIS PETROLEUM NORTH AMERICA, LLC
OFFSITE PIT LOCATION PLAT
SECTION 30, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: B.H.H. Project No.: S13-09-301.09
Checked By: D.D.K. Date: APRIL 2014

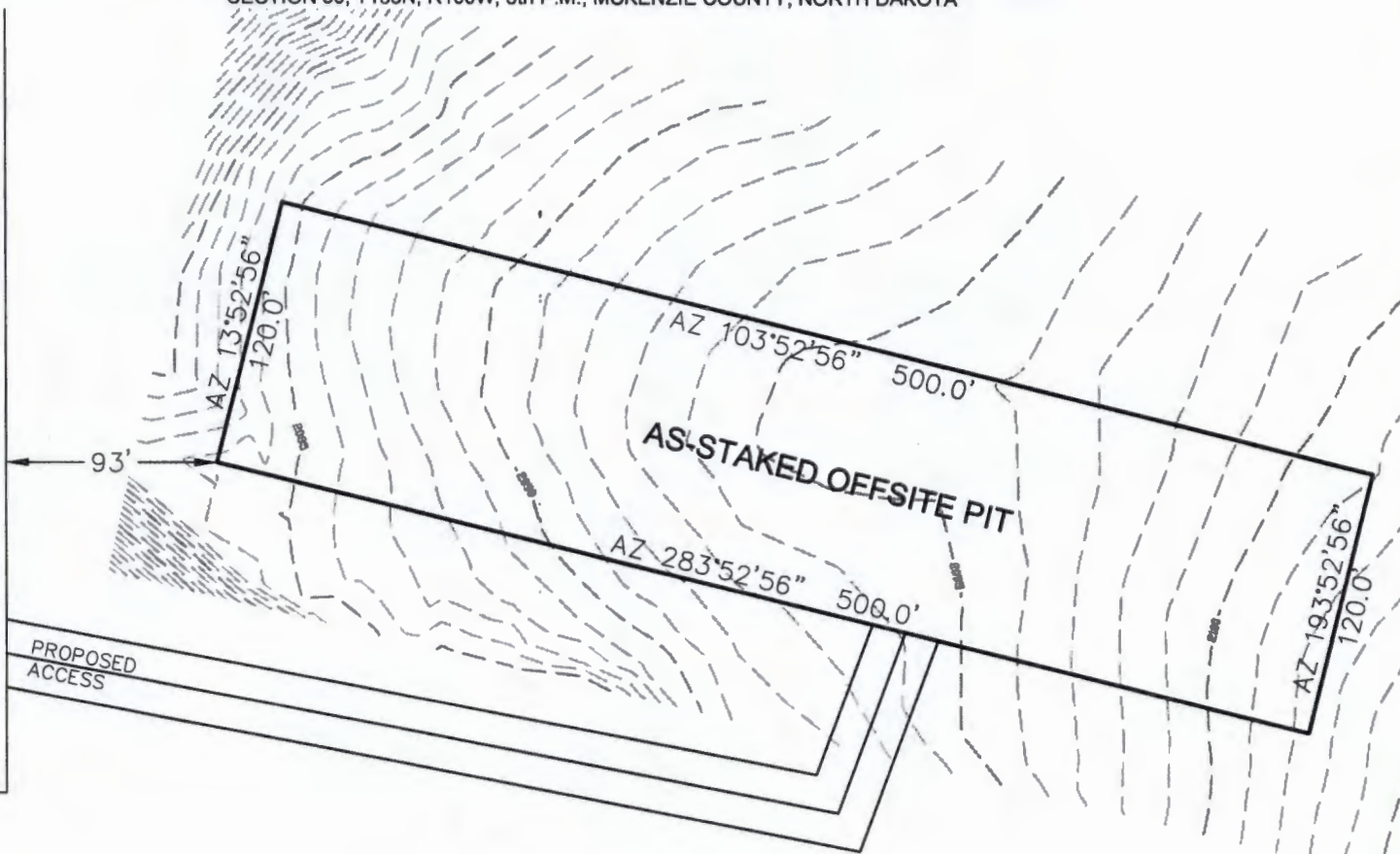
Revision No.	Date	By	Description

PAD LAYOUT

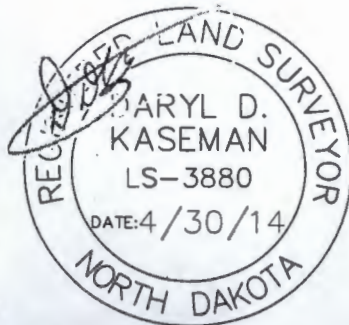
OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002

"AS-STAKED OFFSITE PIT FOR WADE FEDERAL 5300 31-30 2B, WADE FEDERAL 5300 41-30 3T2,
WADE FEDERAL 5300 41-30 4T, WADE FEDERAL 5300 41-30 5T2, WADE FEDERAL 5300 41-30 6B,
WADE FEDERAL 5300 41-30 7T, WADE FEDERAL 5300 41-30 8T2, & WADE FEDERAL 5300 41-30 9B"
625 FEET FROM SOUTH LINE AND 1000 FEET FROM WEST LINE
SECTION 30, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA

WADE FEDERAL
5300 31-30 2B
WADE FEDERAL
5300 41-30 3T2
WADE FEDERAL
5300 41-30 4T
WADE FEDERAL
5300 41-30 5T2
WADE FEDERAL
5300 41-30 6B
WADE FEDERAL
5300 41-30 7T
WADE FEDERAL
5300 41-30 8T2
WADE FEDERAL
5300 41-30 9B

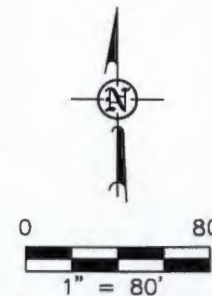


PROPOSED
ACCESS



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FAX (408) 433-5518
Web: interstate-engineering.com

Other offices in Minnesota, North Dakota and South Dakota

OASIS PETROLEUM NORTH AMERICA, LLC

PAD LAYOUT

SECTION 30, T153N, R100W

MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: BHA Project No.: 513-09-38120

Checked By: D.D.K. Date: APRIL 2014

Revision No.

Date

By

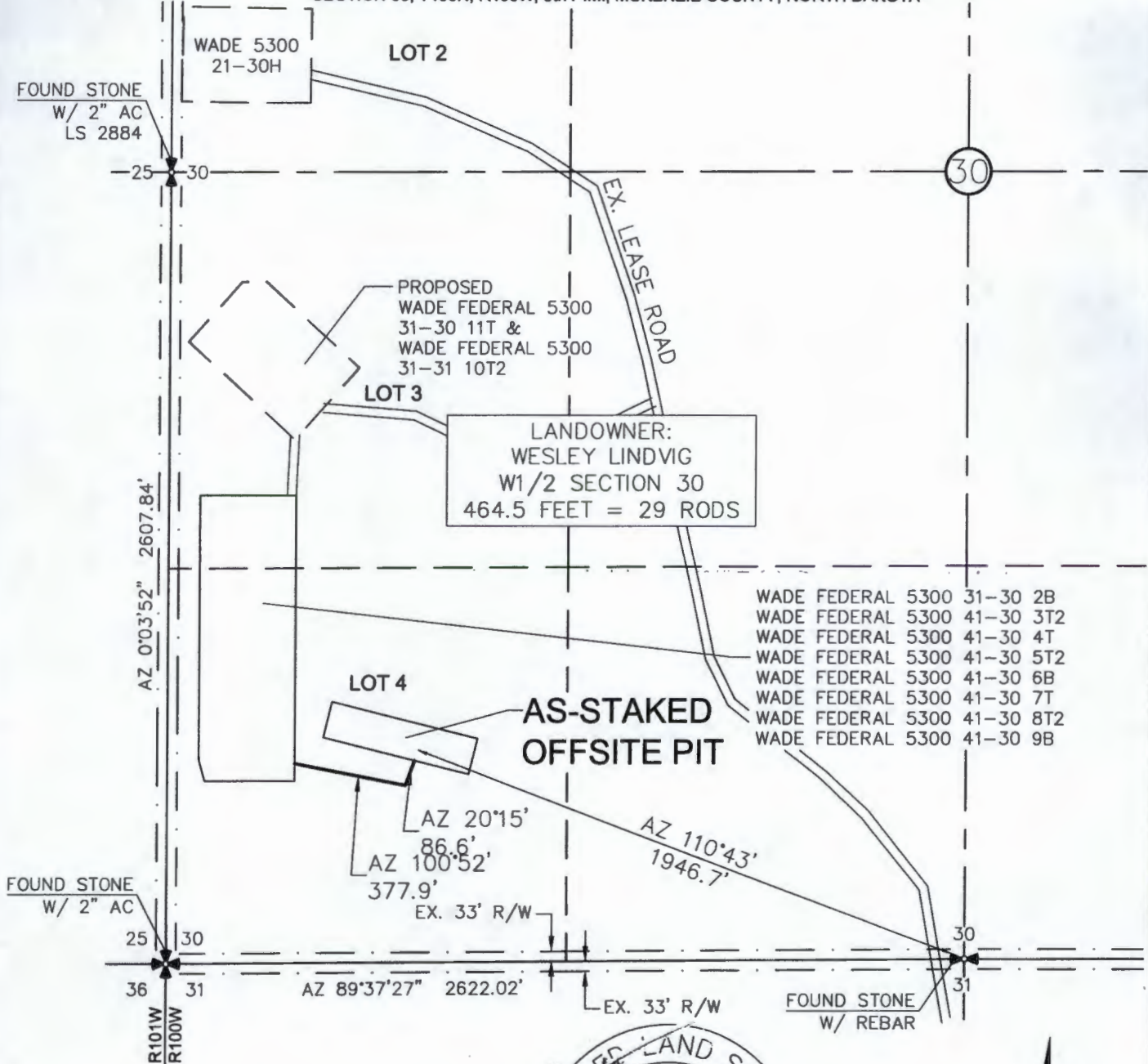
Description

ACCESS APPROACH

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002

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WADE FEDERAL 5300 41-30 9B

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OASIS PETROLEUM NORTH AMERICA, LLC
ACCESS APPROACH
SECTION 30, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: B.H.H. Project No.: S13-09-381.09
Checked By: D.D.K. Date: APRIL, 2014

Revision No.	Date	By	Description

OFF-SITE PIT AGREEMENT

In consideration of the sum of [REDACTED] paid by Oasis Petroleum North America LLC ("Oasis") the undersigned surface owners, Wesley Lindvig and Barbara Lindvig, for themselves and their heirs, successors, administrators and assigns, hereby acknowledge the receipt and sufficiency of said payment in full and complete settlement for and as a release of all claim for loss, damage or injury to the hereafter described surface property arising out of the off-site cuttings pit, in which the cuttings from the Wade Federal 5300 21-30 13B, Wade Federal 5300 21-30 14T2 wells will be buried, located on the approximately two (2.0) acre tract of land identified on the plat attached hereto as Exhibit "A" and which is situated on the following described real property located in McKenzie County, State of North Dakota, to wit:

Township 153 North, Range 100 West, 5th P.M.
Section 30: Lots 3 & 4 a/k/a W $\frac{1}{2}$ SW $\frac{1}{4}$

The undersigned knows that Oasis Petroleum North America LLC is the operator and will be drilling the Wade Federal 5300 21-30 13B, Wade Federal 5300 21-30 14T2 wells. The undersigned further states that they are fully aware that the cuttings generated from the drilling of the Wade Federal 5300 21-30 13B; Wade Federal 5300 21-30 14T2 wells will be buried in the pit on the above described location.

Dated this 19 day of May, 2014.

SURFACE OWNER(S)

Wesley Lindvig
Wesley Lindvig

Barbara J. Lindvig
Barbara Lindvig

BH W.G.L.

Location will be fenced after construction.
Pit will be reclaimed to owners satisfaction
BH W.G.L.

**SUNDRY NOTICE AND REPORTS ON WELLS - FORM**

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)



28394
28425
28554
Well File No
28555
28556
28557
28558
28744

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☐ Notice of Intent	Approximate Start Date
☐ Report of Work Done	Date Work Completed
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	Approximate Start Date

☐ Drilling Program	☐ Spill Report
☐ Redrilling or Repair	☐ Shooting
☐ Casing or Liner	☐ Acidizing
☐ Plug Well	☐ Fracture Treatment
☐ Supplemental History	☐ Change Production Method
☐ Temporarily Abandon	☐ Reclamation
☐ Other	

Well Name and Number Wade Federal 5300 41-30 4T + See Details					
Footages 1263 F S L 240 F W L		Qtr-Qtr SWSW	Section 30	Township 153 N	Range 100 W
Field	Pool Bakken	County McKenzie			

24-HOUR PRODUCTION RATE			
Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

Oasis Petroleum respectfully requests to use an offsite pit for the wells listed below. We are requesting to use an offsite pit because a pit wont fit on location with a rig anchor and the land adjacent is too rough. Attached are the plats.

Wade Federal 5300 31-30 2B - 28554
Wade Federal 5300 41-30 3T2 - 28555
Wade Federal 5300 41-30 4T - 28394
Wade Federal 5300 41-30 5T2 - 28556
Wade Federal 5300 41-30 6B - 28425
Wade Federal 5300 41-30 7T - 28557
Wade Federal 5300 41-30 8T2 - 28558
Wade Federal 5300 41-30 9B - 28744

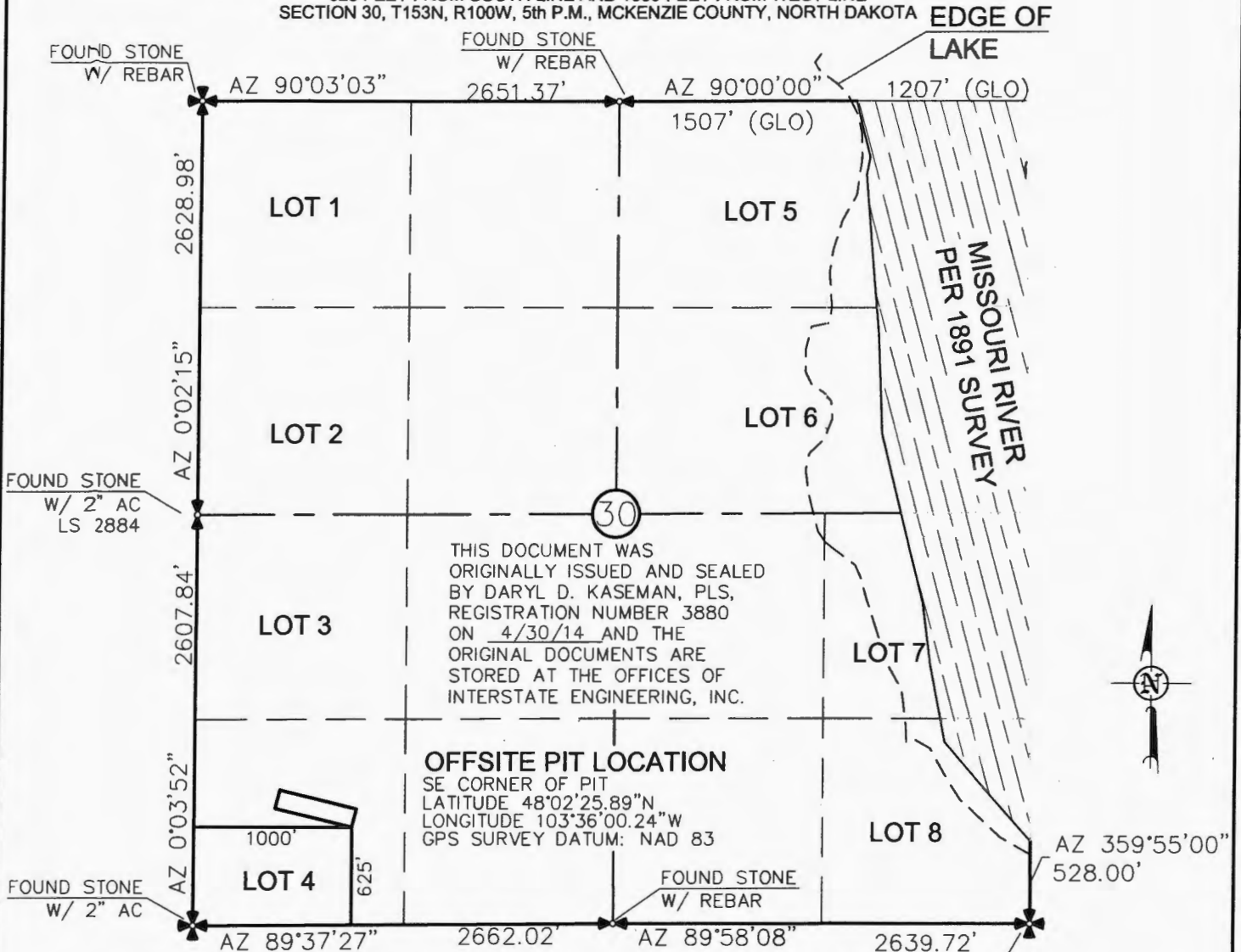
Company Oasis Petroleum North America LLC		Telephone Number 281-404-9491	
Address 1001 Fannin, Suite 1500			
City Houston	State TX	Zip Code 77002	
Signature <i>Brandi Terry</i>	Printed Name Brandi Terry		
Title Regulatory Specialist	Date May 12, 2014		
Email Address bterry@oasispetroleum.com			

FOR STATE USE ONLY	
☒ Received	☐ Approved
Date 12-23-14	
By <i>[Signature]</i>	
Title <i>[Signature]</i>	

OFFSITE PIT LOCATION PLAT

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002

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THIS DOCUMENT WAS
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BY DARYL D. KASEMAN, PLS,
REGISTRATION NUMBER 3880
ON 4/30/14 AND THE
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OFFSITE PIT LOCATION

SE CORNER OF PIT
LATITUDE 48°02'25.89"N
LONGITUDE 103°36'00.24"W
GPS SURVEY DATUM: NAD 83

VICINITY MAP



- ✖ - MONUMENT - RECOVERED
- ✖ - MONUMENT - NOT RECOVERED

STAKED ON 4/30/14
VERTICAL CONTROL DATUM WAS BASED UPON
CONTROL POINT 705 WITH AN ELEVATION OF 2158.3'

THIS SURVEY AND PLAT IS BEING PROVIDED AT THE
REQUEST OF ERIC BAYES OF OASIS PETROLEUM. I
CERTIFY THAT THIS PLAT CORRECTLY REPRESENTS
WORK PERFORMED BY ME OR UNDER MY
SUPERVISION AND IS TRUE AND CORRECT TO THE
BEST OF MY KNOWLEDGE AND BELIEF.

[Signature]
DARYL D. KASEMAN LS-3880



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MCKENZIE COUNTY, NORTH DAKOTA
Drawn By: B.J.H. Project No.: S13-09-381.09
Checked By: D.D.K. Date: APRIL 2014

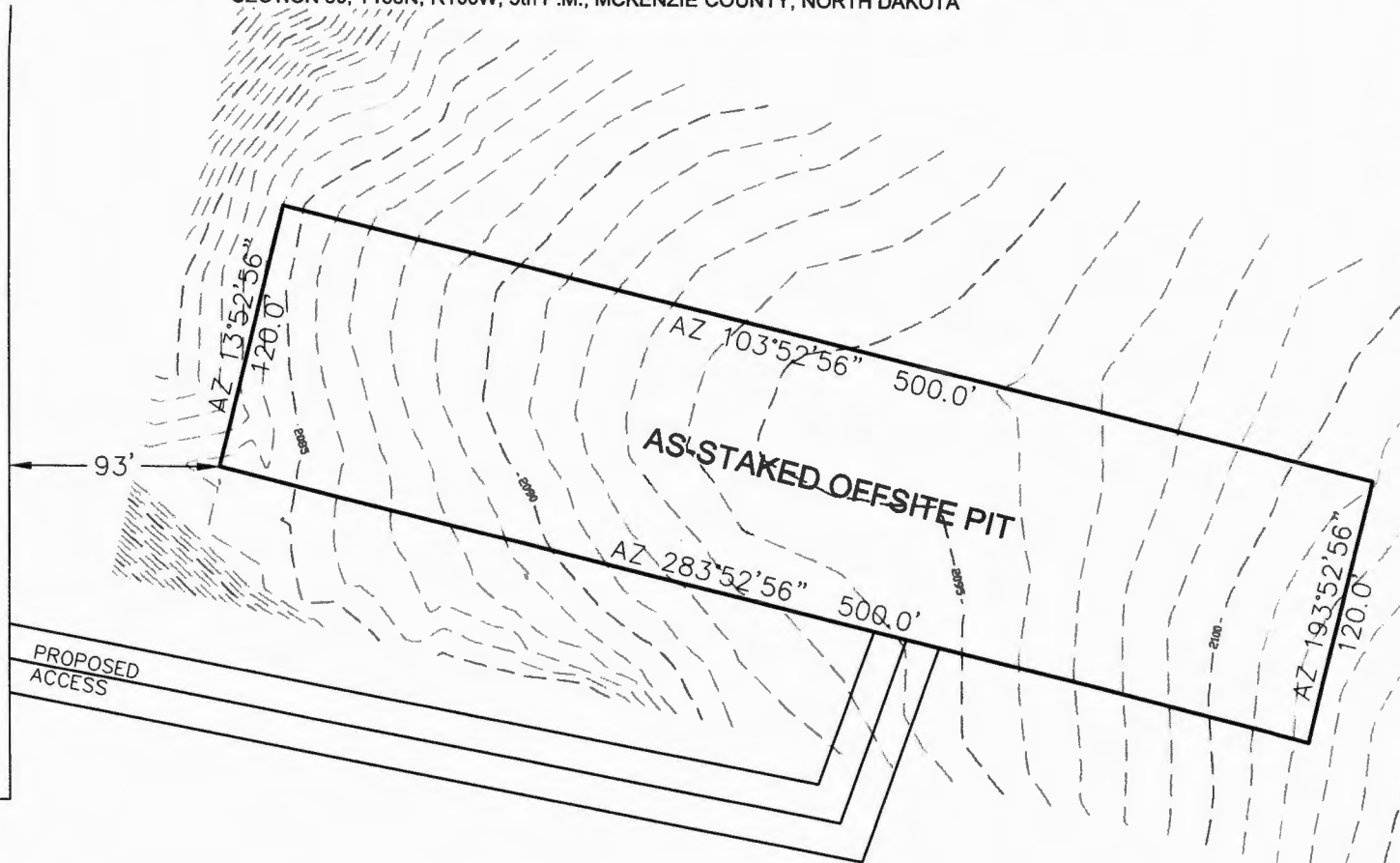
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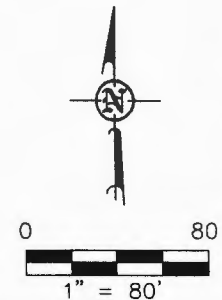
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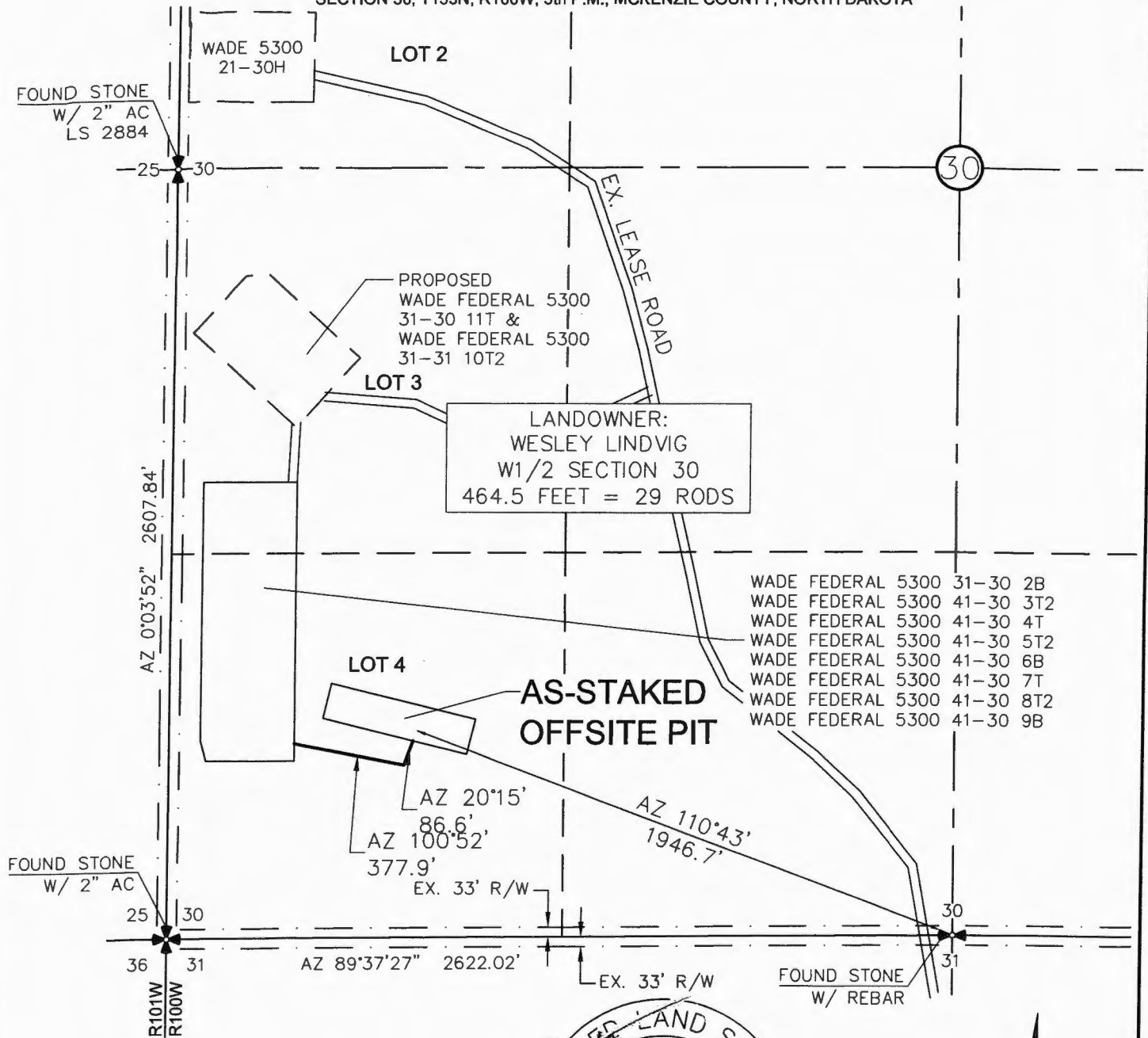
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MCKENZIE COUNTY, NORTH DAKOTA
Drawn By: B.H.H.
Checked By: D.O.K.
Project No.: S12-06-381/99
Date: APRIL, 2014

Revision No.	Date	By	Description

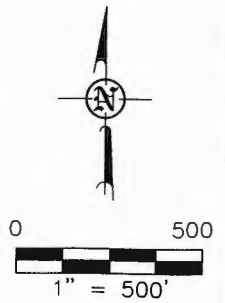
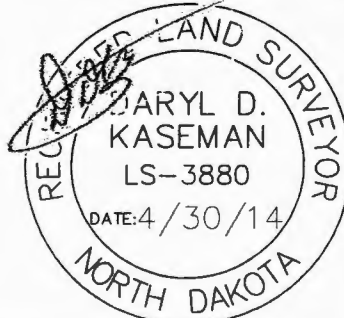
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			Drawn By: B.H.H. Checked By: D.D.K.	Project No.: S13-09-381.09 Date: APRIL 2014	<table border="1"> <tr> <th>Revision No.</th> <th>Date</th> <th>By</th> <th>Description</th> </tr> <tr><td> </td><td> </td><td> </td><td> </td></tr> <tr><td> </td><td> </td><td> </td><td> </td></tr> <tr><td> </td><td> </td><td> </td><td> </td></tr> <tr><td> </td><td> </td><td> </td><td> </td></tr> </table>	Revision No.	Date	By	Description													
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A Schlumberger Company
9251 E 104th Ave.
Commerce City, CO 80640
(303) 439-5500

Directional Survey Certification Form

Oasis Petroleum	Wade Federal 5300 41-30 #9B Original	8-Jan-2015
Company	Hole Well Name	Final report Date
ND, McKenzie County	33-053-06051	
PathFinder Job Number	County / State	API Number
N 48° 2' 27.67000"	W 103° 36' 10.82000"	811 ft FSL & 280 ft FWL
N 48.04101944	W 103.60300556	Sec 30 Twn 153 N Rng 100 W
Surface Latitude	Surface Longitude	Surface Section - Township - Range
NAD83 North Dakota State Plane, NZ, Feet	Patterson 488	KB 32ft @ 2077.00 ft / GL: 2045.00 ft MSL
Datum & Coordinate System	Rig Contractor	Height Reference
Survey Depth	108.00 Depth From	to 20577.00 Depth To
Measurement While Drilling		
Type of Survey		
Survey Depth	20577.00 Depth From	to 20650.00 Depth To
Straight line projection to Bit/TD		
Type of Survey		
Site Supervisors	B. Feddersen - FE1	
Directional Driller 1	MWD Surveyor 1	
	J. Holt - FE	
Directional Driller 2	MWD Surveyor 2	

The data submitted in this report conforms to the standards and procedures as set forth by Schlumberger. This report represents a true and correct directional wellbore survey based on original survey data obtained at the well site.

Matt VanderSchaaf

Matt VanderSchaaf
PathFinder Well Planner III

1/8/2015

Date



Wade Federal 5300 41-30 #9B MWD 0' to 20650' Definitive Survey Geodetic Report (Def Survey)



Report Date: January 08, 2015 - 01:18 PM
Client: Oasis Petroleum
Field: ND, McKenzie County (NAD 83 NZ) Oasis 2014
Structure / Slot: Oasis 30-153N-100W (Wade Federal 5300 41-30 6-9 Pad) - Patterson
Well: Wade Federal 5300 41-30 #9B
Borehole: Original Hole
UWI / API#: Unknown / Unknown
Survey Name: Wade Federal 5300 41-30 #9B MWD 0' to 20650' Definitive
Survey Date: December 10, 2014
Tort / AHD / DDI / ERD Ratio: 291.739 ° / 10262.666 ft / 6.757 / 0.949
Coordinate Reference System: NAD83 North Dakota State Plane, Northern Zone, Feet
Location Lat / Long: N 48° 2' 27.67000", W 103° 36' 10.82000"
Location Grid N/E Y/X: N 395022.017 ft, E 1209639.562 ft
CRS Grid Convergence Angle: -2.3091 °
Grid Scale Factor: 0.99993613
Version / Patch: 2.8.572.0

Survey / DLS Computation: Minimum Curvature / Lubinski
Vertical Section Azimuth: 94.623 ° (True North)
Vertical Section Origin: 0.000 ft, 0.000 ft
TVD Reference Datum: KB 32ft
TVD Reference Elevation: 2077.000 ft above MSL
Seabed / Ground Elevation: 2045.000 ft above MSL
Magnetic Declination: 8.394 °
Total Gravity Field Strength: 1000.0448mgn (9.80665 Based)
Gravity Model: GARM
Total Magnetic Field Strength: 56235.543 nT
Magnetic Dip Angle: 72.903 °
Declination Date: December 10, 2014
Magnetic Declination Model: BGGM 2014
North Reference: True North
Grid Convergence Used: 0.0000 °
Total Corr Mag North->True North: 8.3939 °
Local Coord Referenced To: Well Head

Comments	MD (ft)	Incl (°)	Azim True (°)	TVD (ft)	VSEC (ft)	NS (ft)	EW (ft)	DLS (°/100ft)	Northing (ft)	Easting (ft)	Latitude (N/S ° ' ")	Longitude (E/W ° ' ")
Surface	0.00	0.00	180.00	0.00	0.00	0.00	0.00	N/A	395022.02	1209639.56	N 48 2 27.67	W 103 36 10.82
Begin MWD	108.00	0.62	354.65	108.00	-0.10	0.58	-0.05	0.57	395022.60	1209639.53	N 48 2 27.68	W 103 36 10.82
Survey	203.00	0.35	144.37	203.00	0.00	0.86	0.07	0.99	395022.87	1209639.66	N 48 2 27.68	W 103 36 10.82
	294.00	0.62	206.80	293.99	-0.01	0.19	0.01	0.61	395022.21	1209639.58	N 48 2 27.67	W 103 36 10.82
	386.00	0.79	162.10	385.99	0.05	-0.86	-0.02	0.61	395021.16	1209639.51	N 48 2 27.66	W 103 36 10.82
	476.00	0.97	160.44	475.98	0.60	-2.16	0.42	0.20	395019.84	1209639.90	N 48 2 27.65	W 103 36 10.81
	566.00	0.88	183.12	565.97	0.93	-3.57	0.64	0.42	395018.42	1209640.06	N 48 2 27.63	W 103 36 10.81
	657.00	0.97	175.69	656.95	1.06	-5.04	0.66	0.16	395016.96	1209640.02	N 48 2 27.62	W 103 36 10.81
	747.00	1.06	328.36	746.95	0.69	-5.09	0.28	2.19	395016.92	1209639.64	N 48 2 27.62	W 103 36 10.82
	838.00	1.41	331.83	837.93	-0.41	-3.38	-0.69	0.39	395018.66	1209638.74	N 48 2 27.64	W 103 36 10.83
	928.00	0.35	228.74	927.92	-1.21	-2.59	-1.42	1.70	395019.49	1209638.04	N 48 2 27.64	W 103 36 10.84
	1019.00	0.97	232.92	1018.91	-1.97	-3.24	-2.24	0.68	395018.87	1209637.19	N 48 2 27.64	W 103 36 10.85
	1109.00	0.88	215.20	1108.90	-2.89	-4.26	-3.25	0.33	395017.89	1209636.15	N 48 2 27.63	W 103 36 10.87
	1200.00	1.06	225.82	1199.89	-3.80	-5.42	-4.25	0.28	395016.77	1209635.09	N 48 2 27.62	W 103 36 10.88
	1290.00	1.49	1.32	1289.88	-4.42	-4.83	-4.83	2.63	395017.39	1209634.55	N 48 2 27.62	W 103 36 10.89
	1395.00	0.79	348.30	1394.86	-4.70	-2.76	-4.94	0.71	395019.46	1209634.52	N 48 2 27.64	W 103 36 10.89
	1490.00	0.62	6.01	1489.85	-4.87	-1.60	-5.02	0.29	395020.62	1209634.48	N 48 2 27.65	W 103 36 10.89
	1585.00	0.44	335.88	1584.84	-5.04	-0.76	-5.11	0.34	395021.46	1209634.42	N 48 2 27.66	W 103 36 10.90
	1680.00	0.62	26.88	1679.84	-5.02	0.03	-5.03	0.51	395022.25	1209634.54	N 48 2 27.67	W 103 36 10.89
	1775.00	0.70	18.71	1774.83	-4.68	1.04	-4.61	0.13	395023.24	1209635.00	N 48 2 27.68	W 103 36 10.89
	1870.00	0.26	353.94	1869.83	-4.58	1.80	-4.45	0.50	395024.00	1209635.19	N 48 2 27.69	W 103 36 10.89
	1966.00	0.62	76.38	1965.83	-4.13	2.14	-3.97	0.67	395024.32	1209635.68	N 48 2 27.69	W 103 36 10.88
	2061.00	0.53	8.39	2060.82	-3.61	2.70	-3.40	0.68	395024.85	1209636.27	N 48 2 27.70	W 103 36 10.87
9 5/8" Casing Point	2110.00	0.42	11.31	2109.82	-3.57	3.10	-3.34	0.23	395025.25	1209636.35	N 48 2 27.70	W 103 36 10.87
	2183.00	0.26	20.21	2182.82	-3.50	3.52	-3.23	0.23	395025.66	1209636.48	N 48 2 27.70	W 103 36 10.87
	2278.00	0.35	3.87	2277.82	-3.44	4.01	-3.13	0.13	395026.15	1209636.59	N 48 2 27.71	W 103 36 10.87
	2372.00	0.44	345.60	2371.82	-3.57	4.64	-3.20	0.16	395026.79	1209636.55	N 48 2 27.72	W 103 36 10.87
	2467.00	0.53	100.79	2466.82	-3.25	4.92	-2.86	0.86	395027.04	1209636.90	N 48 2 27.72	W 103 36 10.86
	2561.00	0.79	102.40	2560.81	-2.17	4.69	-1.80	0.28	395026.78	1209637.95	N 48 2 27.72	W 103 36 10.85
	2656.00	0.88	110.06	2655.80	-0.82	4.30	-0.48	0.15	395026.34	1209639.26	N 48 2 27.71	W 103 36 10.83
	2751.00	0.79	104.25	2750.79	0.53	3.89	0.84	0.13	395025.87	1209640.56	N 48 2 27.71	W 103 36 10.81
	2847.00	0.53	107.76	2846.78	1.61	3.59	1.91	0.27	395025.53	1209641.61	N 48 2 27.71	W 103 36 10.79
	2942.00	0.44	93.92	2941.78	2.40	3.44	2.69	0.15	395025.34	1209642.39	N 48 2 27.70	W 103 36 10.78
	3037.00	0.97	168.14	3036.77	3.00	2.62	3.22	1.00	395024.51	1209642.88	N 48 2 27.70	W 103 36 10.77
	3132.00	1.06	170.44	3131.76	3.44	0.97	3.53	0.10	395022.84	1209643.13	N 48 2 27.68	W 103 36 10.77
	3227.00	0.62	213.45	3226.75	3.41	-0.33	3.39	0.78	395021.56	1209642.94	N 48 2 27.67	W 103 36 10.77
	3322.00	0.62	211.52	3321.74	2.93	-1.19	2.84	0.02	395020.71	1209642.35	N 48 2 27.66	W 103 36 10.78
	3417.00	0.62	225.75	3416.74	2.36	-1.99	2.20	0.16	395019.94	1209641.68	N 48 2 27.65	W 103 36 10.79
	3512.00	0.70	217.39	3511.73	1.71	-2.81	1.48	0.13	395019.15	1209640.93	N 48 2 27.64	W 103 36 10.80
	3607.00	0.62	227.02	3606.73	1.04	-3.62	0.76	0.14	395018.37	1209640.17	N 48 2 27.63	W 103 36 10.81
	3702.00	0.70	216.87	3701.72	0.39	-4.44	0.03	0.15	395017.58	1209639.41	N 48 2 27.63	W 103 36 10.82
	3797.00	0.62	234.53	3796.71	-0.31	-5.20	-0.74	0.23	395016.85	1209638.62	N 48 2 27.62	W 103 36 10.83
	3892.00	0.18	274.45	3891.71	-0.86	-5.48	-1.30	0.52	395016.59	1209638.04	N 48 2 27.62	W 103 36 10.84
	3987.00	0.35	299.02	3986.71	-1.27	-5.33	-1.71	0.21	395016.76	1209637.64	N 48 2 27.62	W 103 36 10.85
	4082.00	0.09	94.55	4081.71	-1.46	-5.20	-1.89	0.46	395016.90	1209637.47	N 48 2 27.62	W 103 36 10.85
	4177.00	0.44	85.19	4176.71	-1.03	-5.17	-1.45	0.37	395016.91	1209637.91	N 48 2 27.62	W 103 36 10.84
	4272.00	0.53	89.11	4271.70	-0.23	-5.14	-0.64	0.10	395016.91	1209638.71	N 48 2 27.62	W 103 36 10.83
	4367.00	0.44	107.33	4366.70	0.56	-5.24	0.14	0.19	395016.78	1209639.49	N 48 2 27.62	W 103 36 10.82
	4462.00	0.53	94.71	4461.70	1.36	-5.38	0.93	0.15	395016.60	1209640.27	N 48 2 27.62	W 103 36 10.81
	4558.00	0.70	111.42	4557.69	2.37	-5.63	1.92	0.26	395016.31	1209641.25	N 48 2 27.61	W 103 36 10.79
	4652.00	0.79	118.50	4651.68	3.51	-6.15	3.02	0.14	395015.75	1209642.33	N 48 2 27.61	W 103 36 10.78
	4747.00	0.35	141.90	4746.68	4.30	-6.69	3.78	0.51	395015.18	1209643.07	N 48 2 27.60	W 103 36 10.76
	4842.00	0.62	177.39	4841.68	4.56	-7.43	3.98	0.41	395014.43	1209643.24	N 48 2 27.60	W 103 36 10.76
	4937.00	0.70	168.89	4936.67	4.79	-8.52	4.11	0.13	395013.34	1209643.33	N 48 2 27.59	W 103 36 10.76
	5032.00	0.88	162.36	5031.66	5.22	-9.78	4.45	0.21	395012.07	1209643.61	N 48 2 27.57	W 103 36 10.75
	5127.00	1.23	151.98	5126.65	6.05	-11.38	5.15	0.42	395010.44	1209644.25	N 48 2 27.56	W 103 36 10.74
	5222.00	1.23	170.34	5221.62	6.85	-13.28	5.80	0.41	395008.51	1209644.82	N 48 2 27.54	W 103 36 10.73
	5317.00	1.06	263.97	5316.61	6.24	-14.38	5.09	1.76	395007.45	1209644.07	N 48 2 27.53	W 103 36 10.75
	5413.00	0.62	300.59	5412.60	4.90	-14.21	3.76	0.70	395007.67	1209642.75	N 48 2 27.53	W 103 36 10.76
	5507.00	0.53	305.10	5506.60	4.07	-13.70	2.97	0.11	395008.21	1209641.98	N 48 2 27.53	W 103 36 10.78
	5602.00	0.44	314.85	5601.59	3.41	-13.19	2.35	0.13	395008.74	1209641.38	N 48 2 27.54	W 103 36 10.79
	5697.00	0.35	327.41	5696.59	2.95	-12.69	1.94	0.13	395009.26	1209640.99	N 48 2 27.54	W 103 36 10.79
	5792.00	0.18	347.03	5791.59	2.73	-12.30	1.75	0.20	395009.66	1209640.81	N 48 2 27.55	W 103 36 10.79
	5887.00	0.00	243.83	5886.59	2.69	-12.15	1.71	0.19	395009.81	1209640.79	N 48 2 27.55	W 103 36 10.79
	5982.00	0.09	56.00	5981.59	2.75	-12.11	1.78	0.09	395009.85	1209640.85	N 48 2 27.55	W 103 36 10.79
	6073.00	0.09	345.03	6072.59	2.78	-12.00	1.82	0.11	395009.95	1209640.89	N 48 2 27.55	W 103 36 10.79
	6168.00	0.79	180.22	6167.59	2.80	-12.58	1.80	0.92	395009.37	1209640.85	N 48 2 27.55	W 103 36 10.79
	6264.00	0.79	178.03	6263.58	2.93	-13.91	1.82	0.03	395008.05	1209640.82	N 48 2 27.53	W 103 36 10.79
	6359.00	0.62	137.18	6358.57	3.38	-14.94	2.19	0.54	395007.00	1209641.15	N 48 2 27.52	W 103 36 10.79
	6454.00	0.70	118.36	6453.56	4.29	-15.59	3.05	0.24	395006.32	1209641.98	N 48 2 27.52	W 103 36 10.78
	6549.00	0.70	136.50	6548.56	5.26	-16.29	3.96	0.23	395005.58	1209642.86	N 48 2 27.51	W 103 36 10.76
	6644.00	0.62	145.71	6643.55	6.01	-17.13	4.65	0.14	395004.71	1209643.51	N 48 2 27.50	W 103 36 10.75
	6739.00	0.44	156.29	6738.55	6.51	-17.89	5.08	0.21	395003.94	1209643.92	N 48 2 27.49	W 103 36 10.75
	6834.00	0.62	150.39	6833.54	6.97	-18.67	5.48	0.20	395003.14	1209644.29	N 48 2 27.49	

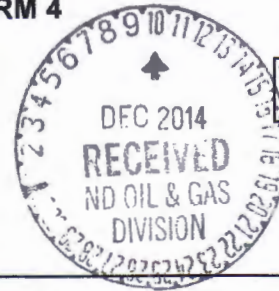
Comments	MD (ft)	Incl (°)	Azim True (°)	TVD (ft)	VSEC (ft)	NS (ft)	EW (ft)	DLS (°/100ft)	Northing (ft)	Easting (ft)	Latitude (N/S ° ' ")	Longitude (E/W ° ' ")
	7119.00	0.62	183.07	7118.52	7.44	-22.10	5.68	0.18	394999.70	1209644.35	N 48° 2' 27.45" W	103° 36' 10.74"
	7214.00	0.70	187.52	7213.52	7.43	-23.19	5.58	0.10	394998.62	1209644.20	N 48° 2' 27.44" W	103° 36' 10.74"
	7310.00	0.70	179.94	7309.51	7.45	-24.36	5.50	0.10	394997.46	1209644.08	N 48° 2' 27.43" W	103° 36' 10.74"
	7405.00	0.88	212.55	7404.50	7.15	-25.56	5.11	0.50	394996.28	1209643.64	N 48° 2' 27.42" W	103° 36' 10.74"
	7500.00	0.79	212.90	7499.49	6.50	-26.72	4.36	0.09	394995.15	1209642.84	N 48° 2' 27.41" W	103° 36' 10.76"
	7595.00	0.53	204.77	7594.48	6.04	-27.67	3.82	0.29	394994.22	1209642.27	N 48° 2' 27.40" W	103° 36' 10.76"
	7690.00	0.44	205.70	7689.48	5.76	-28.40	3.48	0.10	394993.51	1209641.90	N 48° 2' 27.39" W	103° 36' 10.77"
	7785.00	0.44	213.92	7784.48	5.45	-29.03	3.12	0.07	394992.89	1209641.51	N 48° 2' 27.38" W	103° 36' 10.77"
	7880.00	0.44	237.18	7879.47	4.98	-29.53	2.61	0.19	394992.41	1209640.98	N 48° 2' 27.38" W	103° 36' 10.78"
	7975.00	0.44	224.94	7974.47	4.45	-29.98	2.04	0.10	394991.98	1209640.40	N 48° 2' 27.37" W	103° 36' 10.79"
	8070.00	0.44	260.79	8069.47	3.86	-30.30	1.43	0.29	394991.69	1209639.77	N 48° 2' 27.37" W	103° 36' 10.80"
	8164.00	0.44	283.27	8163.47	3.16	-30.28	0.72	0.18	394991.74	1209639.06	N 48° 2' 27.37" W	103° 36' 10.81"
	8259.00	0.44	226.75	8258.46	2.55	-30.44	0.10	0.44	394991.60	1209638.43	N 48° 2' 27.37" W	103° 36' 10.82"
	8354.00	0.35	182.13	8353.46	2.32	-30.98	-0.18	0.33	394991.07	1209638.14	N 48° 2' 27.36" W	103° 36' 10.82"
	8449.00	0.26	182.50	8448.46	2.34	-31.49	-0.20	0.09	394990.57	1209638.10	N 48° 2' 27.36" W	103° 36' 10.82"
	8544.00	0.53	344.92	8543.46	2.20	-31.28	-0.32	0.82	394990.78	1209637.98	N 48° 2' 27.36" W	103° 36' 10.82"
	8640.00	0.53	1.20	8639.46	2.02	-30.41	-0.43	0.16	394991.66	1209637.91	N 48° 2' 27.37" W	103° 36' 10.83"
	8735.00	0.70	353.96	8734.45	1.89	-29.39	-0.48	0.20	394992.67	1209637.90	N 48° 2' 27.38" W	103° 36' 10.83"
	8830.00	0.79	350.03	8829.44	1.62	-28.17	-0.65	0.11	394993.90	1209637.77	N 48° 2' 27.39" W	103° 36' 10.83"
	8925.00	0.62	341.50	8924.43	1.25	-27.03	-0.93	0.21	394995.04	1209637.54	N 48° 2' 27.40" W	103° 36' 10.83"
	9020.00	0.35	8.19	9019.43	1.07	-26.26	-1.05	0.36	394995.82	1209637.45	N 48° 2' 27.41" W	103° 36' 10.84"
	9115.00	0.44	174.73	9114.43	1.15	-26.34	-0.98	0.63	394995.74	1209637.52	N 48° 2' 27.41" W	103° 36' 10.83"
	9210.00	0.44	194.72	9209.43	1.15	-27.05	-1.04	0.16	394995.03	1209637.44	N 48° 2' 27.40" W	103° 36' 10.84"
	9305.00	0.53	207.80	9304.42	0.91	-27.79	-1.34	0.15	394994.30	1209637.11	N 48° 2' 27.40" W	103° 36' 10.84"
	9400.00	0.62	210.93	9399.42	0.51	-28.62	-1.80	0.10	394993.49	1209636.61	N 48° 2' 27.39" W	103° 36' 10.85"
	9495.00	0.53	220.26	9494.42	0.02	-29.40	-2.35	0.14	394992.74	1209636.03	N 48° 2' 27.38" W	103° 36' 10.85"
	9590.00	0.53	223.92	9589.41	-0.51	-30.05	-2.94	0.04	394992.11	1209635.41	N 48° 2' 27.37" W	103° 36' 10.86"
	9685.00	0.70	209.58	9684.41	-1.03	-30.87	-3.53	0.24	394991.31	1209634.79	N 48° 2' 27.37" W	103° 36' 10.87"
	9779.00	0.70	224.05	9778.40	-1.64	-31.78	-4.22	0.19	394990.43	1209634.07	N 48° 2' 27.36" W	103° 36' 10.88"
	9874.00	0.62	210.51	9873.39	-2.23	-32.64	-4.88	0.18	394989.60	1209633.37	N 48° 2' 27.35" W	103° 36' 10.89"
	9969.00	0.62	248.94	9968.39	-2.92	-33.27	-5.62	0.43	394989.00	1209632.61	N 48° 2' 27.34" W	103° 36' 10.90"
	10063.00	0.70	276.29	10062.38	-3.95	-33.39	-6.67	0.34	394988.92	1209631.56	N 48° 2' 27.34" W	103° 36' 10.92"
	10156.00	0.62	273.31	10155.38	-5.02	-33.30	-7.73	0.09	394989.06	1209630.50	N 48° 2' 27.34" W	103° 36' 10.93"
	10188.00	0.26	210.30	10187.37	-5.23	-33.35	-7.94	1.73	394989.01	1209630.28	N 48° 2' 27.34" W	103° 36' 10.94"
	10219.00	2.37	122.47	10218.37	-4.69	-33.76	-7.44	7.66	394988.59	1209630.77	N 48° 2' 27.34" W	103° 36' 10.93"
	10251.00	5.45	116.65	10250.29	-2.70	-34.79	-5.52	9.69	394987.48	1209632.65	N 48° 2' 27.33" W	103° 36' 10.90"
	10283.00	9.58	114.52	10282.01	1.22	-36.58	-1.74	12.93	394985.54	1209636.35	N 48° 2' 27.31" W	103° 36' 10.85"
	10314.00	13.19	115.91	10312.39	6.94	-39.20	3.79	11.68	394982.70	1209641.77	N 48° 2' 27.28" W	103° 36' 10.76"
	10346.00	17.15	118.21	10343.27	14.67	-43.03	11.24	12.51	394978.58	1209649.06	N 48° 2' 27.25" W	103° 36' 10.65"
	10378.00	21.10	119.86	10373.50	24.21	-48.13	20.39	12.46	394973.11	1209658.00	N 48° 2' 27.20" W	103° 36' 10.52"
	10409.00	24.45	121.94	10402.08	34.96	-54.30	30.68	11.11	394966.53	1209668.03	N 48° 2' 27.13" W	103° 36' 10.37"
	10441.00	27.35	122.02	10430.86	47.37	-61.70	42.54	9.06	394958.65	1209679.57	N 48° 2' 27.06" W	103° 36' 10.19"
	10473.00	30.51	122.96	10458.86	61.05	-70.02	55.59	9.98	394949.82	1209692.28	N 48° 2' 26.98" W	103° 36' 10.00"
	10504.00	33.50	123.49	10485.15	75.47	-79.03	69.33	9.69	394940.27	1209705.65	N 48° 2' 26.89" W	103° 36' 9.80"
	10536.00	36.67	123.84	10511.33	91.55	-89.22	84.63	9.93	394929.46	1209720.53	N 48° 2' 26.79" W	103° 36' 9.57"
	10568.00	40.89	123.43	10536.27	109.07	-100.32	101.32	13.21	394917.70	1209736.75	N 48° 2' 26.68" W	103° 36' 9.33"
	10600.00	45.02	122.68	10559.69	128.25	-112.20	119.59	13.00	394905.09	1209754.53	N 48° 2' 26.56" W	103° 36' 9.06"
	10631.00	48.63	122.53	10580.89	148.21	-124.38	138.63	11.65	394892.16	1209773.06	N 48° 2' 26.44" W	103° 36' 8.78"
	10663.00	51.88	122.14	10601.35	169.99	-137.54	159.42	10.20	394878.17	1209793.30	N 48° 2' 26.31" W	103° 36' 8.47"
	10694.00	54.34	123.12	10619.96	191.87	-150.91	180.30	8.33	394863.97	1209813.62	N 48° 2' 26.18" W	103° 36' 8.17"
	10726.00	58.30	123.40	10637.70	215.24	-165.51	202.56	12.40	394848.49	1209835.28	N 48° 2' 26.04" W	103° 36' 7.84"
	10758.00	62.61	124.22	10653.48	239.54	-181.00	225.68	13.65	394832.08	1209857.76	N 48° 2' 25.88" W	103° 36' 7.50"
	10790.00	66.83	124.79	10667.14	264.62	-197.39	249.52	13.29	394814.74	1209880.91	N 48° 2' 25.72" W	103° 36' 7.15"
	10821.00	70.79	124.68	10678.34	289.62	-213.86	273.27	12.78	394797.34	1209903.98	N 48° 2' 25.56" W	103° 36' 6.80"
	10853.00	74.83	124.99	10687.80	316.03	-231.32	298.36	12.66	394778.88	1209928.34	N 48° 2' 25.39" W	103° 36' 6.43"
	10885.00	77.91	124.07	10695.34	342.98	-248.94	323.97	10.02	394760.24	1209953.22	N 48° 2' 25.21" W	103° 36' 6.05"
	10917.00	81.78	123.57	10700.98	370.48	-266.47	350.14	12.19	394741.67	1209978.66	N 48° 2' 25.04" W	103° 36' 5.67"
	10948.00	86.70	122.12	10704.09	397.64	-283.19	376.04	16.54	394723.92	1210003.87	N 48° 2' 24.88" W	103° 36' 5.29"
	10980.00	88.99	120.40	10705.29	426.22	-299.78	403.38	8.95	394706.25	1210030.51	N 48° 2' 24.71" W	103° 36' 4.88"
7" Casing Point	11000.00	88.87	119.93	10705.67	444.27	-309.83	420.66	2.42	394695.51	1210047.38	N 48° 2' 24.61" W	103° 36' 4.63"
	11011.00	88.81	119.67	10705.89	454.22	-315.30	430.21	2.42	394689.66	1210056.69	N 48° 2' 24.56" W	103° 36' 4.49"
	11043.00	89.78	119.10	10706.28	483.27	-331.00	458.09	3.52	394672.85	1210083.91	N 48° 2' 24.40" W	103° 36' 4.08"
	11075.00	90.31	119.37	10706.26	512.37	-346.63	486.01	1.86	394656.11	1210111.18	N 48° 2' 24.25" W	103° 36' 3.67"
	11107.00	90.66	119.59	10705.99	541.40	-362.37	513.87	1.29	394639.26	1210138.38	N 48° 2' 24.09" W	103° 36' 3.26"
	11138.00	91.71	119.76	10705.34	569.48	-377.72	540.80	3.43	394622.84	1210164.67	N 48° 2' 23.94" W	103° 36' 2.86"
	11169.00	90.84	121.79	10704.65	597.30	-393.57	567.42	7.12	394605.93	1210190.63	N 48° 2' 23.79" W	103° 36' 2.47"
	11200.00	90.75	122.41	10704.22	624.80	-410.05	593.68	2.02	394588.41	1210216.20	N 48° 2' 23.62" W	103° 36' 2.08"
	11231.00	89.60	122.72	10704.13	652.18	-426.73	619.81	3.84	394570.69	1210241.63	N 48° 2' 23.46" W	103° 36' 1.70"
	11263.00	89.60	122.52	10704.35	680.44	-443.98	646.76	0.62	394552.37	1210267.87	N 48° 2' 23.29" W	103° 36' 1.30"
	11294.00	90.04	122.55	10704.45	707.83	-460.65	672.89	1.42	394534.66	1210293.31	N 48° 2' 23.12" W	103° 36' 0.92"
	11326.00	89.87	121.84	10704.48	736.20	-477.70	699.97	2.28	394516.53	1210319.68	N 48° 2' 22.96" W	103° 36' 0.52"
	11357.00	89.96	121.77	10704.52	763.77	-494.04	726.32	0.37	394499.15	1210345.34	N 48° 2' 22.79" W	103° 36' 0.13"
	11388.00	90.22	121.40	10704.47	791.40	-510.28	752.72	1.46	394481.86	1210371.07	N 48° 2' 22.63" W	103° 36' 59.74"
	11420.00	90.48	122.14	10704.28	819.88	-527.12	779.93	2.45	394463.93	1210397.57	N 48° 2' 22.47" W	103° 36' 59.34"
	11451.00	90.48	122.22	10704.02	847.36	-543.63	806.17	0.26	394446.38	1210423.12	N 48° 2' 22.31" W	103° 36' 58.96"
	11482.00	90.75	122.75	10703.68	874.76	-560.28	83					

Comments	MD (ft)	Incl (°)	Azim True (°)	TVD (ft)	VSEC (ft)	NS (ft)	EW (ft)	DLS (°/100ft)	Northing (ft)	Easting (ft)	Latitude (N/S ° ' ")	Longitude (E/W ° ' ")
	12646.00	88.99	90.38	10730.54	1999.83	-808.93	1940.95	1.51	394135.60	1211546.22	N 48	2 19.69 W 103 35 42.26
	12741.00	89.43	90.18	10731.85	2094.55	-809.40	2035.94	0.51	394131.31	1211641.11	N 48	2 19.68 W 103 35 40.86
	12836.00	90.13	90.17	10732.22	2189.26	-809.69	2130.93	0.74	394127.19	1211736.01	N 48	2 19.68 W 103 35 39.46
	12932.00	88.29	90.37	10733.54	2284.97	-810.14	2226.92	1.93	394122.87	1211831.89	N 48	2 19.67 W 103 35 38.05
	13027.00	87.32	89.91	10737.18	2379.61	-810.37	2321.85	1.13	394118.82	1211926.73	N 48	2 19.67 W 103 35 36.65
	13122.00	88.37	89.01	10740.75	2474.15	-809.48	2416.78	1.46	394115.89	1212021.61	N 48	2 19.68 W 103 35 35.26
	13217.00	89.34	89.35	10742.65	2568.70	-808.12	2511.75	1.08	394113.42	1212116.55	N 48	2 19.69 W 103 35 33.86
	13312.00	89.78	89.40	10743.38	2663.30	-807.08	2606.74	0.47	394110.63	1212211.50	N 48	2 19.70 W 103 35 32.46
	13407.00	89.87	88.74	10743.67	2757.86	-805.54	2701.72	0.70	394108.34	1212306.47	N 48	2 19.72 W 103 35 31.06
	13503.00	88.90	87.72	10744.70	2853.25	-802.57	2797.67	1.47	394107.44	1212402.45	N 48	2 19.75 W 103 35 29.65
	13598.00	88.64	89.28	10746.74	2947.68	-800.09	2892.61	1.66	394106.10	1212497.41	N 48	2 19.77 W 103 35 28.26
	13693.00	89.60	90.67	10748.20	3042.36	-800.05	2987.60	1.78	394102.31	1212592.31	N 48	2 19.77 W 103 35 26.86
	13789.00	89.96	90.44	10748.57	3138.11	-800.98	3083.59	0.44	394097.51	1212688.18	N 48	2 19.76 W 103 35 25.45
	13884.00	90.04	89.82	10748.57	3232.82	-801.19	3178.59	0.66	394093.47	1212783.09	N 48	2 19.76 W 103 35 24.05
	13978.00	89.16	90.02	10749.22	3326.50	-801.06	3272.59	0.96	394089.82	1212877.01	N 48	2 19.76 W 103 35 22.66
	14072.00	87.67	90.39	10751.82	3420.18	-801.40	3366.55	1.63	394085.70	1212970.88	N 48	2 19.76 W 103 35 21.28
	14167.00	89.69	89.96	10754.01	3514.87	-801.69	3461.52	2.17	394081.58	1213065.75	N 48	2 19.76 W 103 35 19.88
	14262.00	90.57	89.99	10753.80	3609.55	-801.64	3556.52	0.93	394077.79	1213160.67	N 48	2 19.76 W 103 35 18.49
	14357.00	91.10	89.45	10752.41	3704.20	-801.18	3651.50	0.80	394074.43	1213255.59	N 48	2 19.76 W 103 35 17.09
	14452.00	88.46	90.04	10752.78	3798.84	-800.76	3746.49	2.85	394071.03	1213350.52	N 48	2 19.76 W 103 35 15.69
	14547.00	88.55	90.01	10755.26	3893.50	-800.80	3841.46	0.10	394067.16	1213445.40	N 48	2 19.76 W 103 35 14.29
	14642.00	88.46	89.97	10757.73	3988.16	-800.78	3936.43	0.10	394063.35	1213540.28	N 48	2 19.76 W 103 35 12.90
	14737.00	90.40	90.01	10758.68	4082.84	-800.77	4031.42	2.04	394059.54	1213635.19	N 48	2 19.76 W 103 35 11.50
	14833.00	91.45	90.02	10757.13	4178.52	-800.79	4127.41	1.09	394055.65	1213731.09	N 48	2 19.76 W 103 35 10.09
	14928.00	89.96	89.68	10755.96	4273.18	-800.54	4222.40	1.61	394052.07	1213826.01	N 48	2 19.77 W 103 35 8.69
	15023.00	90.04	89.66	10755.96	4367.82	-799.99	4317.39	0.09	394048.79	1213920.95	N 48	2 19.77 W 103 35 7.29
	15118.00	89.34	90.67	10756.47	4462.53	-800.27	4412.39	1.29	394044.69	1214015.85	N 48	2 19.77 W 103 35 5.89
	15214.00	90.31	91.23	10756.77	4558.33	-801.86	4508.38	1.17	394039.23	1214111.68	N 48	2 19.75 W 103 35 4.48
	15309.00	90.57	90.11	10756.04	4653.10	-802.97	4603.36	1.21	394034.29	1214206.55	N 48	2 19.74 W 103 35 3.08
	15404.00	87.93	89.98	10757.28	4747.78	-803.05	4698.35	2.78	394030.39	1214301.44	N 48	2 19.74 W 103 35 1.68
	15499.00	86.88	90.97	10761.58	4842.44	-803.83	4793.24	1.52	394025.78	1214396.23	N 48	2 19.73 W 103 35 0.29
	15594.00	87.93	90.82	10765.88	4937.14	-805.31	4888.13	1.12	394020.48	1214490.97	N 48	2 19.72 W 103 34 58.89
	15626.00	88.37	89.75	10766.92	4969.03	-805.47	4920.12	3.61	394019.03	1214522.92	N 48	2 19.71 W 103 34 58.42
	15689.00	88.55	89.83	10768.61	5031.78	-805.24	4983.09	0.31	394016.73	1214585.85	N 48	2 19.72 W 103 34 57.49
	15721.00	89.69	89.96	10769.10	5063.67	-805.18	5015.09	3.59	394015.50	1214617.82	N 48	2 19.72 W 103 34 57.02
	15784.00	90.04	89.88	10769.25	5126.46	-805.10	5078.09	0.57	394013.05	1214680.77	N 48	2 19.72 W 103 34 56.10
	15879.00	90.75	89.37	10768.59	5221.09	-804.47	5173.08	0.92	394009.84	1214775.71	N 48	2 19.72 W 103 34 54.70
	15974.00	90.04	89.50	10767.94	5315.70	-803.54	5268.08	0.76	394006.95	1214870.65	N 48	2 19.73 W 103 34 53.30
	16070.00	90.48	90.10	10767.50	5411.36	-803.20	5364.07	0.78	394003.42	1214966.58	N 48	2 19.74 W 103 34 51.89
	16165.00	89.25	91.44	10767.73	5506.14	-804.48	5459.06	1.91	393998.31	1215061.43	N 48	2 19.72 W 103 34 50.49
	16261.00	89.52	91.39	10768.76	5601.98	-806.85	5555.03	0.29	393992.08	1215157.22	N 48	2 19.70 W 103 34 49.08
	16355.00	90.40	90.99	10768.82	5695.81	-808.80	5649.00	1.03	393986.34	1215251.04	N 48	2 19.68 W 103 34 47.70
	16451.00	87.76	91.56	10770.37	5791.63	-810.94	5744.96	2.81	393980.34	1215346.82	N 48	2 19.66 W 103 34 46.28
	16546.00	88.46	90.35	10773.50	5886.38	-812.52	5839.89	1.47	393974.94	1215441.61	N 48	2 19.64 W 103 34 44.89
	16578.00	89.43	89.52	10774.09	5918.27	-812.48	5871.89	3.99	393973.69	1215473.57	N 48	2 19.64 W 103 34 44.42
	16642.00	90.13	89.11	10774.33	5981.99	-811.72	5935.88	1.27	393971.87	1215537.54	N 48	2 19.65 W 103 34 43.47
	16735.00	90.92	88.57	10773.48	6074.51	-809.83	6028.86	1.03	393970.01	1215630.51	N 48	2 19.67 W 103 34 42.11
	16798.00	89.08	89.56	10773.48	6137.21	-808.81	6091.84	3.32	393968.50	1215693.49	N 48	2 19.68 W 103 34 41.18
	16829.00	89.08	89.40	10773.98	6168.08	-808.53	6122.84	0.52	393967.53	1215724.47	N 48	2 19.68 W 103 34 40.72
	16923.00	89.69	89.80	10774.99	6261.72	-807.87	6216.83	0.78	393964.40	1215818.40	N 48	2 19.69 W 103 34 39.34
	17017.00	89.96	89.87	10775.28	6355.39	-807.60	6310.83	0.30	393960.88	1215912.33	N 48	2 19.69 W 103 34 37.96
	17111.00	91.36	89.68	10774.19	6449.04	-807.23	6404.82	1.50	393957.46	1216006.25	N 48	2 19.69 W 103 34 36.57
	17205.00	90.22	89.83	10772.90	6542.69	-806.83	6498.81	1.22	393954.08	1216100.18	N 48	2 19.70 W 103 34 35.19
	17299.00	90.66	89.76	10772.17	6636.36	-806.49	6592.81	0.47	393950.63	1216194.10	N 48	2 19.70 W 103 34 33.81
	17393.00	89.78	89.97	10771.81	6730.03	-806.27	6686.80	0.96	393947.06	1216288.03	N 48	2 19.70 W 103 34 32.42
	17487.00	88.90	90.89	10772.90	6823.77	-806.98	6780.79	1.35	393942.57	1216381.91	N 48	2 19.69 W 103 34 31.04
	17581.00	88.99	90.83	10774.63	6917.55	-808.39	6874.77	0.12	393937.38	1216475.74	N 48	2 19.68 W 103 34 29.66
	17676.00	89.87	90.86	10775.57	7012.34	-809.79	6969.75	0.93	393932.15	1216570.59	N 48	2 19.67 W 103 34 28.26
	17770.00	89.43	91.43	10776.15	7106.17	-811.67	7063.73	0.77	393926.49	1216664.41	N 48	2 19.65 W 103 34 26.88
	17864.00	89.16	91.94	10777.30	7200.04	-814.43	7157.68	0.61	393919.94	1216758.16	N 48	2 19.62 W 103 34 25.50
	17958.00	88.64	92.11	10779.11	7293.92	-817.75	7251.61	0.58	393912.84	1216851.87	N 48	2 19.59 W 103 34 24.11
	18051.00	88.72	91.97	10781.25	7386.80	-821.06	7344.52	0.17	393905.79	1216944.57	N 48	2 19.55 W 103 34 22.75
	18145.00	89.96	92.66	10782.33	7480.72	-824.86	7438.44	1.51	393898.21	1217038.25	N 48	2 19.52 W 103 34 21.36
	18240.00	88.90	91.83	10783.28	7575.63	-828.58	7533.36	1.42	393890.67	1217132.94	N 48	2 19.48 W 103 34 19.97
	18272.00	87.76	90.32	10784.21	7607.55	-829.18	7565.34	5.91	393888.78	1217164.87	N 48	2 19.47 W 103 34 19.50
	18336.00	86.88	91.00	10787.20	7671.33	-829.91	7629.26	1.74	393885.47	1217228.71	N 48	2 19.46 W 103 34 18.56
	18431.00	89.08	89.63	10790.55	7765.99	-830.44	7724.19	2.73	393881.12	1217323.53	N 48	2 19.46 W 103 34 17.16
	18526.00	89.69	89.49	10791.57	7860.62	-829.71	7819.18	0.66	393878.03	1217418.47	N 48	2 19.47 W 103 34 15.76
	18622.00	89.43	89.87	10792.31	7956.26	-829.17	7915.18	0.48	393874.69	1217514.40	N 48	2 19.47 W 103 34 14.35
	18717.00	89.60	89.85	10793.11	8050.93	-828.94	8010.17	0.18	393871.10	1217609.33	N 48	2 19.47 W 103 34 12.95
	18812.00	89.25	89.92	10794.07	8145.60	-828.75	8105.17	0.38	393867.46	1217704.25	N 48	2 19.47 W 103 34 11.55
	18907.00	88.11	89.68	10796.25	8240.23	-828.42	8200.14	1.23	393863.97	1217799.15	N 48	2 19.48 W 103 34 10.16
	19003.00	87.93	89.52	10799.57	8335.81	-827.75	8296.08	0.25	393860.77	1217895.03	N 48	2 19.48 W 103 34 8.74
	19098.00	89.43	89.19	10801.76	8430.38	-826.68	8391.05	1.62	393858.01	1217989.96	N 48	2 19.49 W 103 34 7.35
	19130.00	89.87	89.12	10801.96	8462.23	-826.20	8423.04	1.39	393857.20	1218021.94	N 48	2 19.50 W 103 34 6.88
	19193.00	90.04	89.13	10802.00	8524.94	-825						

Comments	MD (ft)	Incl (°)	Azim True (°)	TVD (ft)	VSEC (ft)	NS (ft)	EW (ft)	DLS (°/100ft)	Northing (ft)	Easting (ft)	Latitude (N/S ° ' ")	Longitude (E/W ° ' ")
		1	0.000	32.000	1/98.425	30.000	30.000	SLB_MWD-STD-Depth Only		Original Hole / Wade Federal 5300 41-30 #9B MWD 0' to 20650'		
		1	32.000	32.000	Act Stns	30.000	30.000	SLB_MWD-STD-Depth Only		Original Hole / Wade Federal 5300 41-30 #9B MWD 0' to 20650'		
		1	32.000	20577.000	Act Stns	30.000	30.000	SLB_MWD-STD		Original Hole / Wade Federal 5300 41-30 #9B MWD 0' to 20650'		
		1	20577.000	20650.000	Act Stns	30.000	30.000	SLB_BLIND+TREND		Original Hole / Wade Federal 5300 41-30 #9B MWD 0' to 20650'		

**SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4**

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)



Well File No.

28744

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☒ Notice of Intent	Approximate Start Date December 9, 2014
☐ Report of Work Done	Date Work Completed
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	Approximate Start Date

☐ Drilling Prognosis	☐ Spill Report
☐ Redrilling or Repair	☐ Shooing
☐ Casing or Liner	☐ Acidizing
☐ Plug Well	☐ Fracture Treatment
☐ Supplemental History	☐ Change Production Method
☐ Temporarily Abandon	☐ Reclamation
☐ Other	Waiver from tubing/packer requirement

Well Name and Number Wade Federal 5300 41-30 9B					
Footages 811 F S L 280 F W L		Qtr-Qtr SWSW	Section 30	Township 153 N	Range 100 W
Field Baker	Pool Bakken	County McKenzie			

24-HOUR PRODUCTION RATE			
Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

Oasis Petroleum North America LLC requests a variance to NDAC 43-02-03-21 for the tubing/packer requirement: Casing, tubing, and cementing requirements during the completion period immediately following the upcoming fracture stimulation.

The following assurances apply:

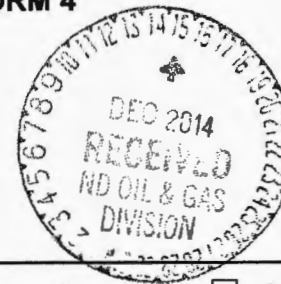
1. the well is equipped with new 29# and 32# casing at surface with an API burst rating of 11,220 psi;
2. The Frac design will use a safety factor of 0.85 API burst rating to determine the maximum pressure;
3. Damage to the casing during the frac would be detected immediately by monitoring equipment;
4. The casing is exposed to significantly lower rates and pressures during flowback than during the frac job;
5. The frac fluid and formation fluids have very low corrosion and erosion rates;
6. Production equipment will be installed as soon as possible after the well ceases flowing;
7. A 300# gauge will be installed on the surface casing during the flowback period

Company Oasis Petroleum North America LLC		Telephone Number 281-404-9436	
Address 1001 Fannin, Suite 1500			
City Houston		State TX	Zip Code 77002
Signature 	Printed Name Jennifer Swenson		
Title Regulatory Assistant	Date December 9, 2014		
Email Address jswenson@oasispetroleum.com			

FOR STATE USE ONLY	
☐ Received	☒ Approved
Date December 11, 2014	
By 	
Title PETROLEUM ENGINEER	

**SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4**

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)



Well File No.

28744

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☒ Notice of Intent

Approximate Start Date
December 11, 2014

☐ Report of Work Done

Date Work Completed

☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.

Approximate Start Date

☐ Drilling Prognosis☐ Spill Report☐ Redrilling or Repair☐ Shooting☐ Casing or Liner☐ Acidizing☐ Plug Well☐ Fracture Treatment☐ Supplemental History☐ Change Production Method☐ Temporarily Abandon☐ Reclamation☒ Other **Proposed Frac String & Gauge**

Well Name and Number

Wade Federal 5300 41-30 9B

Footages

811 F S L**280 F W L**

Qtr-Qtr

Lot 4

Section

30

Township

153 N

Range

100 W

Field

Baker

Pool

Bakken

County

McKenzie**24-HOUR PRODUCTION RATE**

Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)

Address

City

State

Zip Code

DETAILS OF WORK

As requested, Oasis Petroleum respectfully requests to permission to run frac string prior to performing a hydraulic fracture simulation and install a surface casing gauge to monitor pressures.

Approved pursuant to 43-02-03-27.1 hydraulic fracturing rule

Company Oasis Petroleum North America LLC		Telephone Number 281-404-9652	
Address 1001 Fannin, Suite 1500			
City Houston		State TX	Zip Code 77002
Signature 		Printed Name Victoria Siemieniewski	
Title Regulatory Specialist		Date December 12, 2014	
Email Address vsiemieniewski@oasispetroleum.com			

FOR STATE USE ONLY

☐ Received	☒ Approved
Date 12/16/14	
By 	
Title Regulatory Specialist	

**SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4**

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)



Well File No.
28744

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☒ Notice of Intent	Approximate Start Date September 1, 2014	☐ Drilling Prognosis	☐ Spill Report
☐ Report of Work Done	Date Work Completed	☐ Redrilling or Repair	☐ Shooting
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	Approximate Start Date	☒ Casing or Liner	☐ Acidizing
		☐ Plug Well	☐ Fracture Treatment
		☐ Supplemental History	☐ Change Production Method
		☐ Temporarily Abandon	☐ Reclamation
		☐ Other	

Well Name and Number Wade Federal 5300 41-30 9B							
Footages		Qtr-Qtr	Section	Township	Range		
811 F S L 280 F W L		lot4	30	153 N	100 W		
Field		Pool		County			
		Bakken		McKenzie			

24-HOUR PRODUCTION RATE			
Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

Oasis Petroleum respectfully requests to revise the casing plan for the subject well as follows:

13 3/8" 54.5# surface casing will be ran to 2,050'
Contingency 9 5/8" 40# will be ran to 6,400' in order to isolate the Dakota
7" 32# intermediate casing will be ran to 11,529'
4.5" 13.5# liner will be ran to 20,729'

Attached is a revised drill plan, directional plan/plot and well summary.

Company Oasis Petroleum North America LLC		Telephone Number 281-404-9491	
Address 1001 Fannin, Suite 1500			
City Houston	State TX	Zip Code 77002	
Signature 	Printed Name Brandi Terry		
Title Regulatory Specialist	Date July 30, 2014		
Email Address bterry@oasispetroleum.com			

FOR STATE USE ONLY	
☐ Received	☒ Approved
Date 8-18-14	
By 	
Title Petroleum Resource Specialist	

DRILLING PLAN									
OPERATOR		Oasis Petroleum				COUNTY/STATE		McKenzie Co., ND	
WELL NAME		Wade Federal 5300 41-30 9B				RIG		B22	
WELL TYPE		Horizontal Middle Bakken							
LOCATION		SW SW 30-153N-100W				Surface Location (survey plat):		811' FSL 280' FWL	
EST. T.D.		20,729'				GROUND ELEV:		2,045'	
						KB ELEV:		2,070'	
		TOTAL LATERAL: 9,200'						Sub Height: 25'	
MARKER		TVD	Subsea TVD	LOGS:	Type	Interval			
Pierre	NDIC MAP	1,920	150	OH Logs: Request Log waiver based on the Wade Federal 5300 21-30H 2,150' N of surface location					
Greenhorn		4,566	-2,496	CBL/GR: Above top of cement/GR to base of casing					
Mowry		4,969	-2,899	MWD GR: KOP to lateral TD					
Dakota		5,391	-3,321	DEVIATION: Surf: 3 deg. max., 1 deg / 100'; srvy every 500'					
Rierdon		6,407	-4,337	Prod: 5 deg. max., 1 deg / 100'; srvy every 100'					
Durham Salt		6,896	-4,826						
Durham Salt Base		6,942	-4,872						
Pine Salt		7,205	-5,135						
Pine Salt Base		7,229	-5,159						
Opeche Salt		7,291	-5,221						
Opeche Salt Base		7,371	-5,301						
Amsden		7,615	-5,545						
Tyler		7,771	-5,701						
Otter/Base Minnelusa		7,994	-5,924	TESTS: None planned					
Kibbey Lime		8,336	-6,266	CORES: None planned					
Charles Salt		8,484	-6,414						
Base Last Salt		9,163	-7,093						
Mission Canyon		9,377	-7,307						
Lodgepole		9,926	-7,856						
False Bakken		10,656	-8,586	MUDLOGGING: Two-Man: Begin 200' above Kibbey					
Upper Bakken Shale		10,666	-8,596	30' samples in curve and lateral					
Middle Bakken (Top of Target)		10,695	-8,625						
Middle Bakken (Base of target)		10,706	-8,636						
Lower Bakken Shale		10,719	-8,649						
Threeforks		10,744	-8,674						
				BOP: 11" 5000 psi blind, pipe & annular					
Est. Dip Rate: 0.33									
Max. Anticipated BHP: 4845				Surface Formation: Glacial till					
MUD:	Interval	Type	WT	Vis	WL	Remarks			
Surface:	0' - 2,050'	FW/Gel - Lime Sweeps	8.4-9.0	28-32	NC	Circ Mud Tanks			
Intermediate:	2,050' - 11,529'	Invert	9.5-10.4	40-50	30+HtHp	Circ Mud Tanks			
Lateral:	11,529' - 20,729'	Salt Water	9.8-10.2	28-32	NC	Circ Mud Tanks			
CASING:	Size	Wt ppf	Hole	Depth	Cement	WOC	Remarks		
Surface:	13-3/8"	54.5#	17-1/2"	2,050'	To Surface	12	100' into Pierre		
Intermediate (Dakota):	9-5/8"	40#	12-1/4"	6,400'	To Surface	24	Set Casing across Dakota		
Intermediate:	7"	32#	8-3/4"	11,529'	4891	24	500' above Dakota		
Production Liner:	4.5"	13.5#	6"	20,729'	TOL @ 10,178'				
PROBABLE PLUGS, IF REQ'D:									
OTHER:	MD	TVD	FWL FSL	FWL FWL	S-C-R	AZI			
Surface:	2,050'	2,050'	811' FSL	280' FWL	S 30 T153N R100W	Survey Company:			
KOP:	10,229'	10,229'	766' FSL	280' FWL	S 30 T153N R100W	Build Rate:	12 deg / 100'		
EOC:	10,979'	10,706'	513' FSL	885' FWL	S 30 T153N R100W				
Casing Point:	11,529'	10,706'	222' FSL	1151' FWL	S 30 T153N R100W				
Middle Bakken Lateral TD:	20,729'	10,706'	0' FSL	210' FEL	S 29 T153N R100W				
Comments:									
Request Log waiver based on the Wade Federal 5300 21-30H 2,150' N of surface location									
Oasis Petroleum does not use Diesel Fuel, as defined by the US EPA in the list below, in our hydraulic fracture operations.									
68334-30-5 (Primary Name: Fuels, diesel), 68476-34-6 (Primary Name: Fuels, diesel, No. 2); 68476-30-2 (Primary Name: Fuel oil No. 2); 68476-31-3 (Primary Name: Fuel oil, No. 4)									
8008-20-6 (Primary Name: Kerosene)									
 35 Packers; 35 Sleeves No Frac String Planned									
Geology: N. Gabelman		1/20/2014		Engineering: DAD 5/19/14					

**Oasis Petroleum
Well Summary
Wade Federal 5300 41-30 9B
Section 30 T153N R100W
McKenzie County, ND**

SURFACE CASING AND CEMENT DESIGN

							Make-up Torque (ft-lbs)		
Size	Interval	Weight	Grade	Coupling	I.D.	Drift	Minimum	Optimum	Max
13-3/8"	0' to 2,050'	54.5	J-55	STC	12.615"	12.459"	4,100	5,470	6,840

Interval	Description	Collapse	Burst	Tension
		(psi) a	(psi) b	(1000 lbs) c
0' to 2,050'	13-3/8", 54.5#, J-55, STC, 8rd	1130 / 1.18	2730 / 2.84	514 / 2.62

API Rating & Safety Factor

- a) Based on full casing evacuation with 9 ppg fluid on backside (2,050' setting depth).
- b) Burst pressure based on 9 ppg fluid with no fluid on backside (2,050' setting depth).
- c) Based on string weight in 9 ppg fluid at 2,050' TVD plus 100k# overpull. (Buoyed weight equals 96k lbs.)

Cement volumes are based on 13-3/8" casing set in 17-1/2" hole with 50% excess to circulate cement back to surface. Mix and pump the following slurry.

Pre-flush (Spacer): **20 bbls** fresh water

Lead Slurry: **596 sks** (308 bbls) 2.9 yield conventional system with 94 lb/sk cement, .25 lb/sk D130 Lost Circulation Control Agent, 2% CaCl₂, 4% D079 Extender and 2% D053 Expanding Agent.

Tail Slurry: **349 sks** (72 bbls) 1.16 yield conventional system with 94 lb/sk cement, .25% CaCl₂ and 0.25 lb/sk Lost Circulation Control Agent

**Oasis Petroleum
Well Summary
Wade Federal 5300 41-30 9B
Section 30 T153N R100W
McKenzie County, ND**

CONTINGENCY SURFACE CASING AND CEMENT DESIGN

Size	Interval	Weight	Grade	Coupling	I.D.	Drift	Make-up Torque (ft-lbs)		
							Minimum	Optimum	Max
9-5/8"	0' to 6,400'	40	L-80	LTC	8.835"	8.75"	5,450	7,270	9,090

Interval	Description	Collapse	Burst	Tension
		(psi) a	(psi) b	(1000 lbs) c
0' to 6,400'	9-5/8", 40#, L-80, LTC, 8rd	3090 / 3.71	5750 / 1.24	837 / 3.86

API Rating & Safety Factor

- a) Collapse pressure based on 11.5 ppg fluid on the backside and 9 ppg fluid inside of casing.
- b) Burst pressure calculated from a gas kick coming from the production zone (Bakken Pool) at 9,000 psi and a subsequent breakdown at the 9-5/8" shoe, based on a 13.5#/ft fracture gradient. Backup of 9 ppg fluid.
- c) Yield based on string weight in 10 ppg fluid, (217k lbs buoyed weight) plus 100k lbs overpull.

Cement volumes are based on 9-5/8" casing set in 12-1/4" hole with 10% excess in OH and 0% excess inside surface casing. TOC at surface.

Pre-flush (Spacer): 20 bbls Chem wash

Lead Slurry: 592 sks (210 bbls) Conventional system with 75 lb/sk cement, 0.5 lb/sk lost circulation, 10% expanding agent, 2% extender, 2% CaCl₂, 0.2% anti-foam and 0.4% fluid loss agent.

Tail Slurry: 521 sks (108 bbls) Conventional system with 94 lb/sk cement, 0.3% anti-settling agent, 0.3% fluid loss agent, 0.3 lb/sk lost circulation control agent, 0.2% anti-foam and 0.1% retarder.

**Oasis Petroleum
Well Summary
Wade Federal 5300 41-30 9B
Section 30 T153N R100W
McKenzie County, ND**

INTERMEDIATE CASING AND CEMENT DESIGN

Size	Interval	Weight	Grade	Coupling	I.D.	Drift	Make-up Torque (ft-lbs)		
							Minimum	Optimum	Max
7"	0' – 11,529'	32	HCP-110	LTC	6.094"	6.000"	6730	8970	9870

***Special drift

Interval	Length	Description	Collapse	Burst	Tension
			(psi) a	(psi) b	(1000 lbs) c
0' – 11,529'	11,529'	7", 32#, HCP-110, LTC, 8rd	11820 / 2.12*	12460 / 1.29	897 / 2.30
6,896' – 10,229'	3,333'	7", 32#, HCP-110, LTC, 8rd	11820 / 1.45*	12460 / 1.29	

API Rating & Safety Factor

- a) *Assume full casing evacuation with 10 ppg fluid on backside. **Assume full casing evacuation with 1.2 psi/ft equivalent fluid gradient across salt intervals.
- b) Burst pressure based on 9000 psig max press for stimulation plus 10.2 ppg fluid in casing and 9 ppg fluid on backside-to 10,706' TVD.
- c) Based on string weight in 10 ppg fluid, (290k lbs buoyed weight) plus 100k

Cement volumes are estimates based on 7" casing set in an 8-3/4" hole with 30% excess.

Pre-flush (Spacer): **50 bbls Saltwater**
40 bbls Weighted MudPush Express

Lead Slurry: **189 sks** (87 bbls) 2.59 yield conventional system with 47 lb/sk cement, 37 lb/sk D035 extender, 3.0% KCl, 3.0% D154 extender, 0.3% D208 viscosifier, 0.07% retarder, 0.2% anti-foam, 0.5 lb/sk, D130 LCM.

Tail Slurry: **592 sks** (184 bbls) 1.55 yield conventional system with 94 lb/sk cement, 3.0% KCl, 35.0% Silica, 0.5% retarder, 0.2% fluid loss, 0.2% anti-foam and 0.5 lb/sk LCM.

**Oasis Petroleum
Well Summary
Wade Federal 5300 41-30 9B
Section 30 T153N R100W
McKenzie County, ND**

PRODUCTION LINER

Size	Interval	Weight	Grade	Coupling	I.D.	Drift	Estimated Torque
4-1/2"	10,179' – 20,729'	13.5	P-110	BTC	3.92"	3.795"	4,500

Interval	Description	Collapse	Burst	Tension
		(psi) a	(psi) b	(1000 lbs) c
10,179' – 20,729'	4-1/2", 13.5 lb, P-110, BTC, 8rd	10670 / 2.02	12410 / 1.28	443 / 1.98

API Rating & Safety Factor

- a) Based on full casing evacuation with 9.5 ppg fluid on backside @ 10,706' TVD.
Burst pressure based on 9000 psi treating pressure with 10.2 ppg internal fluid gradient and 9 ppg external
- b) fluid gradient @ 10,706' TVD.
- c) Based on string weight in 9.5 ppg fluid (Buoyed weight: 124k lbs.) plus 100k lbs overpull.

Oasis Petroleum does not use Diesel Fuel, as defined by the US EPA in the list below, in our hydraulic fracture operations.

68334-30-5 (Primary Name: Fuels, diesel)
68476-34-6 (Primary Name: Fuels, diesel, No. 2)
68476-30-2 (Primary Name: Fuel oil No. 2)
68476-31-3 (Primary Name: Fuel oil, No. 4)
8008-20-6 (Primary Name: Kerosene)

Oasis

Indian Hills

153N-100W-29/30

Wade Federal 5300 41-30 9B

Wade Federal 5300 41-30 9B

Plan: Plan #1

Standard Planning Report

27 June, 2014

Oasis Petroleum

Planning Report

Database:	OpenWellsCompass - EDM Prod	Local Co-ordinate Reference:	Well Wade Federal 5300 41-30 9B
Company:	Oasis	TVD Reference:	WELL @ 2070.0usft (Original Well Elev)
Project:	Indian Hills	MD Reference:	WELL @ 2070.0usft (Original Well Elev)
Site:	153N-100W-29/30	North Reference:	True
Well:	Wade Federal 5300 41-30 9B	Survey Calculation Method:	Minimum Curvature
Wellbore:	Wade Federal 5300 41-30 9B		
Design:	Plan #1		

Project	Indian Hills		
Map System:	US State Plane 1983	System Datum:	Mean Sea Level
Geo Datum:	North American Datum 1983		
Map Zone:	North Dakota Northern Zone		

Site	153N-100W-29/30		
Site Position:		Northing:	395,519.95 usft
From:	Lat/Long	Easting:	1,209,617.13 usft
Position Uncertainty:	0.0 usft	Slot Radius:	13-3/16 "
		Latitude:	48° 2' 32.580 N
		Longitude:	103° 36' 11.410 W
		Grid Convergence:	-2.31 °

Well	Wade Federal 5300 41-30 9B		
Well Position	+N/-S	-497.5 usft	Northing: 395,021.22 usft
	+E/-W	40.1 usft	Easting: 1,209,637.14 usft
Position Uncertainty	0.0 usft	Wellhead Elevation:	Latitude: 48° 2' 27.670 N
			Longitude: 103° 36' 10.820 W
			Ground Level: 2,045.0 usft

Wellbore	Wade Federal 5300 41-30 9B		
Magnetics	Model Name	Sample Date	Declination (°)
	IGRF200510	3/6/2014	8.17
			Dip Angle (°) 72.95
			Field Strength (nT) 56,478

Design	Plan #1		
Audit Notes:			
Version:	Phase:	PROTOTYPE	Tie On Depth: 0.0
Vertical Section:	Depth From (TVD)	+N/-S	+E/-W
	(usft)	(usft)	(usft)
	0.0	0.0	0.0
			Direction (°) 94.62

Plan Sections										
Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)	TFO (°)	Target
0.0	0.00	0.00	0.0	0.0	0.0	0.00	0.00	0.00	0.00	
10,225.0	0.50	180.00	10,224.8	-44.6	0.0	0.00	0.00	0.00	180.00	
10,229.0	0.00	0.00	10,228.8	-44.6	0.0	12.50	-12.50	0.00	180.00	
10,979.0	90.00	122.00	10,706.3	-297.6	404.9	12.00	12.00	0.00	122.00	
11,529.0	90.00	122.00	10,706.3	-589.1	871.3	0.00	0.00	0.00	0.00	
12,344.7	90.00	90.00	10,706.3	-811.0	1,645.3	3.92	0.00	-3.92	270.00	
20,729.4	90.00	90.00	10,706.3	-811.0	10,030.0	0.00	0.00	0.00	0.00	WADE 9B PBHL

Oasis Petroleum

Planning Report

Database:	OpenWellsCompass - EDM Prod	Local Co-ordinate Reference:	Well Wade Federal 5300 41-30 9B
Company:	Oasis	TVD Reference:	WELL @ 2070.0usft (Original Well Elev)
Project:	Indian Hills	MD Reference:	WELL @ 2070.0usft (Original Well Elev)
Site:	153N-100W-29/30	North Reference:	True
Well:	Wade Federal 5300 41-30 9B	Survey Calculation Method:	Minimum Curvature
Wellbore:	Wade Federal 5300 41-30 9B		
Design:	Plan #1		

Planned Survey

Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N-S (usft)	+E-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)
0.0	0.00	0.00	0.0	0.0	0.0	0.0	0.00	0.00	0.00
100.0	0.00	180.00	100.0	0.0	0.0	0.0	0.00	0.00	0.00
200.0	0.01	180.00	200.0	0.0	0.0	0.0	0.00	0.00	0.00
300.0	0.01	180.00	300.0	0.0	0.0	0.0	0.00	0.00	0.00
400.0	0.02	180.00	400.0	-0.1	0.0	0.0	0.00	0.00	0.00
500.0	0.02	180.00	500.0	-0.1	0.0	0.0	0.00	0.00	0.00
600.0	0.03	180.00	600.0	-0.2	0.0	0.0	0.00	0.00	0.00
700.0	0.03	180.00	700.0	-0.2	0.0	0.0	0.00	0.00	0.00
800.0	0.04	180.00	800.0	-0.3	0.0	0.0	0.00	0.00	0.00
900.0	0.04	180.00	900.0	-0.3	0.0	0.0	0.00	0.00	0.00
1,000.0	0.05	180.00	1,000.0	-0.4	0.0	0.0	0.00	0.00	0.00
1,100.0	0.05	180.00	1,100.0	-0.5	0.0	0.0	0.00	0.00	0.00
1,200.0	0.06	180.00	1,200.0	-0.6	0.0	0.0	0.00	0.00	0.00
1,300.0	0.06	180.00	1,300.0	-0.7	0.0	0.1	0.00	0.00	0.00
1,400.0	0.07	180.00	1,400.0	-0.8	0.0	0.1	0.00	0.00	0.00
1,500.0	0.07	180.00	1,500.0	-1.0	0.0	0.1	0.00	0.00	0.00
1,600.0	0.08	180.00	1,600.0	-1.1	0.0	0.1	0.00	0.00	0.00
1,700.0	0.08	180.00	1,700.0	-1.2	0.0	0.1	0.00	0.00	0.00
1,800.0	0.09	180.00	1,800.0	-1.4	0.0	0.1	0.00	0.00	0.00
1,900.0	0.09	180.00	1,900.0	-1.5	0.0	0.1	0.00	0.00	0.00
1,920.0	0.08	180.00	1,920.0	-1.6	0.0	0.1	0.05	-0.05	0.00
Pierre									
2,000.0	0.10	180.00	2,000.0	-1.7	0.0	0.1	0.02	0.02	0.00
2,050.0	0.10	180.00	2,050.0	-1.8	0.0	0.1	0.00	0.00	0.00
9 5/8"									
2,100.0	0.10	180.00	2,100.0	-1.9	0.0	0.2	0.00	0.00	0.00
2,200.0	0.11	180.00	2,200.0	-2.1	0.0	0.2	0.00	0.00	0.00
2,300.0	0.11	180.00	2,300.0	-2.3	0.0	0.2	0.00	0.00	0.00
2,400.0	0.12	180.00	2,400.0	-2.5	0.0	0.2	0.00	0.00	0.00
2,500.0	0.12	180.00	2,500.0	-2.7	0.0	0.2	0.00	0.00	0.00
2,600.0	0.13	180.00	2,600.0	-2.9	0.0	0.2	0.00	0.00	0.00
2,700.0	0.13	180.00	2,700.0	-3.1	0.0	0.3	0.00	0.00	0.00
2,800.0	0.14	180.00	2,800.0	-3.3	0.0	0.3	0.00	0.00	0.00
2,900.0	0.14	180.00	2,900.0	-3.6	0.0	0.3	0.00	0.00	0.00
3,000.0	0.15	180.00	3,000.0	-3.8	0.0	0.3	0.00	0.00	0.00
3,100.0	0.15	180.00	3,100.0	-4.1	0.0	0.3	0.00	0.00	0.00
3,200.0	0.16	180.00	3,200.0	-4.4	0.0	0.4	0.00	0.00	0.00
3,300.0	0.16	180.00	3,300.0	-4.6	0.0	0.4	0.00	0.00	0.00
3,400.0	0.17	180.00	3,400.0	-4.9	0.0	0.4	0.00	0.00	0.00
3,500.0	0.17	180.00	3,500.0	-5.2	0.0	0.4	0.00	0.00	0.00
3,600.0	0.18	180.00	3,600.0	-5.5	0.0	0.4	0.00	0.00	0.00
3,700.0	0.18	180.00	3,700.0	-5.8	0.0	0.5	0.00	0.00	0.00
3,800.0	0.19	180.00	3,800.0	-6.2	0.0	0.5	0.00	0.00	0.00
3,900.0	0.19	180.00	3,900.0	-6.5	0.0	0.5	0.00	0.00	0.00
4,000.0	0.20	180.00	4,000.0	-6.8	0.0	0.6	0.00	0.00	0.00
4,100.0	0.20	180.00	4,100.0	-7.2	0.0	0.6	0.00	0.00	0.00
4,200.0	0.21	180.00	4,200.0	-7.5	0.0	0.6	0.00	0.00	0.00
4,300.0	0.21	180.00	4,300.0	-7.9	0.0	0.6	0.00	0.00	0.00
4,400.0	0.22	180.00	4,400.0	-8.3	0.0	0.7	0.00	0.00	0.00
4,500.0	0.22	180.00	4,500.0	-8.6	0.0	0.7	0.00	0.00	0.00
4,566.0	0.22	180.00	4,566.0	-8.9	0.0	0.7	0.00	0.00	0.00
Greenhorn									
4,600.0	0.22	180.00	4,600.0	-9.0	0.0	0.7	0.01	0.01	0.00
4,700.0	0.23	180.00	4,700.0	-9.4	0.0	0.8	0.00	0.00	0.00
4,800.0	0.23	180.00	4,800.0	-9.8	0.0	0.8	0.00	0.00	0.00

Oasis Petroleum

Planning Report

Database:	OpenWellsCompass - EDM Prod	Local Co-ordinate Reference:	Well Wade Federal 5300 41-30 9B
Company:	Oasis	TVD Reference:	WELL @ 2070.0usft (Original Well Elev)
Project:	Indian Hills	MD Reference:	WELL @ 2070.0usft (Original Well Elev)
Site:	153N-100W-29/30	North Reference:	True
Well:	Wade Federal 5300 41-30 9B	Survey Calculation Method:	Minimum Curvature
Wellbore:	Wade Federal 5300 41-30 9B		
Design:	Plan #1		

Planned Survey

Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)
4,900.0	0.24	180.00	4,900.0	-10.2	0.0	0.8	0.00	0.00	0.00
4,969.0	0.24	180.00	4,969.0	-10.5	0.0	0.8	0.00	0.00	0.00
Mowry									
5,000.0	0.24	180.00	5,000.0	-10.7	0.0	0.9	0.02	0.02	0.00
5,100.0	0.25	180.00	5,100.0	-11.1	0.0	0.9	0.00	0.00	0.00
5,200.0	0.25	180.00	5,200.0	-11.5	0.0	0.9	0.00	0.00	0.00
5,300.0	0.26	180.00	5,300.0	-12.0	0.0	1.0	0.00	0.00	0.00
5,391.0	0.28	180.00	5,391.0	-12.4	0.0	1.0	0.03	0.03	0.00
Dakota									
5,400.0	0.26	180.00	5,400.0	-12.4	0.0	1.0	0.21	-0.21	0.00
5,500.0	0.27	180.00	5,500.0	-12.9	0.0	1.0	0.00	0.00	0.00
5,600.0	0.27	180.00	5,600.0	-13.4	0.0	1.1	0.00	0.00	0.00
5,700.0	0.28	180.00	5,700.0	-13.9	0.0	1.1	0.00	0.00	0.00
5,800.0	0.28	180.00	5,800.0	-14.4	0.0	1.2	0.00	0.00	0.00
5,900.0	0.29	180.00	5,900.0	-14.9	0.0	1.2	0.00	0.00	0.00
6,000.0	0.29	180.00	6,000.0	-15.4	0.0	1.2	0.00	0.00	0.00
6,100.0	0.30	180.00	6,100.0	-15.9	0.0	1.3	0.00	0.00	0.00
6,200.0	0.30	180.00	6,200.0	-16.4	0.0	1.3	0.00	0.00	0.00
6,300.0	0.31	180.00	6,300.0	-16.9	0.0	1.4	0.00	0.00	0.00
6,400.0	0.31	180.00	6,400.0	-17.5	0.0	1.4	0.00	0.00	0.00
6,407.0	0.27	180.00	6,407.0	-17.5	0.0	1.4	0.62	-0.62	0.00
Rierdon									
6,500.0	0.32	180.00	6,500.0	-18.0	0.0	1.5	0.05	0.05	0.00
6,600.0	0.32	180.00	6,600.0	-18.6	0.0	1.5	0.00	0.00	0.00
6,700.0	0.33	180.00	6,700.0	-19.2	0.0	1.5	0.00	0.00	0.00
6,800.0	0.33	180.00	6,800.0	-19.7	0.0	1.6	0.00	0.00	0.00
6,896.0	0.33	180.00	6,896.0	-20.3	0.0	1.6	0.00	0.00	0.00
Dunham Salt									
6,900.0	0.34	180.00	6,900.0	-20.3	0.0	1.6	0.08	0.08	0.00
6,942.0	0.33	180.00	6,942.0	-20.6	0.0	1.7	0.01	-0.01	0.00
Dunham Salt Base									
7,000.0	0.34	180.00	7,000.0	-20.9	0.0	1.7	0.02	0.02	0.00
7,100.0	0.35	180.00	7,100.0	-21.5	0.0	1.7	0.00	0.00	0.00
7,200.0	0.35	180.00	7,200.0	-22.1	0.0	1.8	0.00	0.00	0.00
7,205.0	0.29	180.00	7,205.0	-22.1	0.0	1.8	1.21	-1.21	0.00
Pine Salt									
7,229.0	0.34	180.00	7,229.0	-22.3	0.0	1.8	0.22	0.22	0.00
Pine Salt Base									
7,291.0	0.35	180.00	7,291.0	-22.7	0.0	1.8	0.01	0.01	0.00
Opeche Salt									
7,300.0	0.36	180.00	7,300.0	-22.7	0.0	1.8	0.05	0.05	0.00
7,371.0	0.38	180.00	7,371.0	-23.2	0.0	1.9	0.04	0.04	0.00
Opeche Salt Base									
7,400.0	0.36	180.00	7,400.0	-23.4	0.0	1.9	0.08	-0.08	0.00
7,500.0	0.37	180.00	7,499.9	-24.0	0.0	1.9	0.00	0.00	0.00
7,600.0	0.37	180.00	7,599.9	-24.6	0.0	2.0	0.00	0.00	0.00
7,615.0	0.35	180.00	7,615.0	-24.7	0.0	2.0	0.12	-0.12	0.00
Amsden									
7,700.0	0.38	180.00	7,699.9	-25.3	0.0	2.0	0.03	0.03	0.00
7,771.0	0.38	180.00	7,771.0	-25.8	0.0	2.1	0.00	0.00	0.00
Tyler									
7,800.0	0.38	180.00	7,799.9	-26.0	0.0	2.1	0.02	0.02	0.00
7,900.0	0.39	180.00	7,899.9	-26.6	0.0	2.1	0.00	0.00	0.00
7,994.0	0.39	180.00	7,994.0	-27.3	0.0	2.2	0.00	0.00	0.00

Oasis Petroleum

Planning Report

Database: OpenWellsCompass - EDM Prod
Company: Oasis
Project: Indian Hills
Site: 153N-100W-29/30
Well: Wade Federal 5300 41-30 9B
Wellbore: Wade Federal 5300 41-30 9B
Design: Plan #1

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TVD Reference: WELL @ 2070.0usft (Original Well Elev)
MD Reference: WELL @ 2070.0usft (Original Well Elev)
North Reference: True
Survey Calculation Method: Minimum Curvature

Planned Survey

Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)
Otter/Base Minnelusa									
8,000.0	0.39	180.00	7,999.9	-27.3	0.0	2.2	0.08	0.08	0.00
8,100.0	0.40	180.00	8,099.9	-28.0	0.0	2.3	0.00	0.00	0.00
8,200.0	0.40	180.00	8,199.9	-28.7	0.0	2.3	0.00	0.00	0.00
8,300.0	0.41	180.00	8,299.9	-29.4	0.0	2.4	0.00	0.00	0.00
8,336.1	0.40	180.00	8,336.0	-29.7	0.0	2.4	0.02	-0.02	0.00
Kibbey Lime									
8,400.0	0.41	180.00	8,399.9	-30.1	0.0	2.4	0.02	0.02	0.00
8,484.1	0.41	180.00	8,484.0	-30.7	0.0	2.5	0.00	0.00	0.00
Charles Salt									
8,500.0	0.42	180.00	8,499.9	-30.8	0.0	2.5	0.03	0.03	0.00
8,600.0	0.42	180.00	8,599.9	-31.6	0.0	2.5	0.00	0.00	0.00
8,700.0	0.43	180.00	8,699.9	-32.3	0.0	2.6	0.00	0.00	0.00
8,800.0	0.43	180.00	8,799.9	-33.0	0.0	2.7	0.00	0.00	0.00
8,900.0	0.44	180.00	8,899.9	-33.8	0.0	2.7	0.00	0.00	0.00
9,000.0	0.44	180.00	8,999.9	-34.6	0.0	2.8	0.00	0.00	0.00
9,100.0	0.44	180.00	9,099.9	-35.3	0.0	2.8	0.00	0.00	0.00
9,163.1	0.44	180.00	9,163.0	-35.8	0.0	2.9	0.00	0.00	0.00
Base Last Salt									
9,200.0	0.45	180.00	9,199.9	-36.1	0.0	2.9	0.02	0.02	0.00
9,300.0	0.45	180.00	9,299.9	-36.9	0.0	3.0	0.00	0.00	0.00
9,377.1	0.45	180.00	9,377.0	-37.5	0.0	3.0	0.00	0.00	0.00
Mission Canyon									
9,400.0	0.46	180.00	9,399.9	-37.7	0.0	3.0	0.02	0.02	0.00
9,500.0	0.46	180.00	9,499.9	-38.5	0.0	3.1	0.00	0.00	0.00
9,600.0	0.47	180.00	9,599.9	-39.3	0.0	3.2	0.00	0.00	0.00
9,700.0	0.47	180.00	9,699.9	-40.2	0.0	3.2	0.00	0.00	0.00
9,800.0	0.48	180.00	9,799.9	-41.0	0.0	3.3	0.00	0.00	0.00
9,900.0	0.48	180.00	9,899.9	-41.8	0.0	3.4	0.00	0.00	0.00
9,926.1	0.47	180.00	9,926.0	-42.0	0.0	3.4	0.04	-0.04	0.00
Lodgepole									
10,000.0	0.49	180.00	9,999.9	-42.7	0.0	3.4	0.02	0.02	0.00
10,100.0	0.49	180.00	10,099.9	-43.5	0.0	3.5	0.00	0.00	0.00
10,200.0	0.50	180.00	10,199.9	-44.4	0.0	3.6	0.00	0.00	0.00
10,225.0	0.50	180.00	10,224.8	-44.6	0.0	3.6	0.00	0.00	0.00
Start Drop -12.50									
10,229.0	0.00	0.00	10,228.8	-44.6	0.0	3.6	12.50	-12.50	0.00
Start Build 12.00									
10,250.0	2.52	122.00	10,249.9	-44.9	0.4	4.0	12.00	12.00	0.00
10,275.0	5.52	122.00	10,274.8	-45.8	1.9	5.6	12.00	12.00	0.00
10,300.0	8.52	122.00	10,299.6	-47.4	4.5	8.3	12.00	12.00	0.00
10,325.0	11.52	122.00	10,324.2	-49.7	8.2	12.1	12.00	12.00	0.00
10,350.0	14.52	122.00	10,348.6	-52.7	12.9	17.1	12.00	12.00	0.00
10,375.0	17.52	122.00	10,372.6	-56.4	18.8	23.3	12.00	12.00	0.00
10,400.0	20.52	122.00	10,396.2	-60.7	25.7	30.5	12.00	12.00	0.00
10,425.0	23.52	122.00	10,419.4	-65.7	33.6	38.8	12.00	12.00	0.00
10,450.0	26.52	122.00	10,442.1	-71.3	42.6	48.2	12.00	12.00	0.00
10,475.0	29.52	122.00	10,464.1	-77.5	52.6	58.6	12.00	12.00	0.00
10,500.0	32.52	122.00	10,485.5	-84.3	63.5	70.1	12.00	12.00	0.00
10,525.0	35.52	122.00	10,506.3	-91.7	75.4	82.5	12.00	12.00	0.00
10,550.0	38.52	122.00	10,526.2	-99.7	88.1	95.9	12.00	12.00	0.00
10,575.0	41.52	122.00	10,545.4	-108.2	101.8	110.1	12.00	12.00	0.00
10,600.0	44.52	122.00	10,563.6	-117.3	116.2	125.3	12.00	12.00	0.00
10,625.0	47.52	122.00	10,581.0	-126.8	131.5	141.3	12.00	12.00	0.00

Oasis Petroleum

Planning Report

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Design: Plan #1

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TVD Reference: WELL @ 2070.0usft (Original Well Elev)
MD Reference: WELL @ 2070.0usft (Original Well Elev)
North Reference: True
Survey Calculation Method: Minimum Curvature

Planned Survey

Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)
10,650.0	50.52	122.00	10,597.4	-136.8	147.5	158.0	12.00	12.00	0.00
10,675.0	53.52	122.00	10,612.8	-147.2	164.2	175.5	12.00	12.00	0.00
10,700.0	56.52	122.00	10,627.1	-158.1	181.6	193.7	12.00	12.00	0.00
10,725.0	59.52	122.00	10,640.3	-169.3	199.5	212.5	12.00	12.00	0.00
10,750.0	62.52	122.00	10,652.5	-180.9	218.1	232.0	12.00	12.00	0.00
10,757.8	63.45	122.00	10,656.0	-184.6	224.0	238.1	12.00	12.00	0.00
False Bakken									
10,775.0	65.52	122.00	10,663.4	-192.8	237.1	251.9	12.00	12.00	0.00
10,786.3	66.88	122.00	10,668.0	-198.3	245.9	261.1	12.00	12.00	0.00
Upper Bakken Shale									
10,800.0	68.52	122.00	10,673.2	-205.0	256.7	272.3	12.00	12.00	0.00
10,825.0	71.52	122.00	10,681.7	-217.5	276.6	293.2	12.00	12.00	0.00
10,850.0	74.52	122.00	10,689.0	-230.1	296.9	314.4	12.00	12.00	0.00
10,874.7	77.49	122.00	10,695.0	-242.8	317.2	335.7	12.00	12.00	0.00
Middle Bakken (Top of Target)									
10,875.0	77.52	122.00	10,695.0	-243.0	317.4	336.0	12.00	12.00	0.00
10,900.0	80.52	122.00	10,699.8	-256.0	338.2	357.8	12.00	12.00	0.00
10,925.0	83.52	122.00	10,703.3	-269.1	359.2	379.8	12.00	12.00	0.00
10,950.0	86.52	122.00	10,705.4	-282.3	380.4	401.9	12.00	12.00	0.00
10,961.1	87.85	122.00	10,706.0	-288.2	389.7	411.7	12.00	12.00	0.00
Middle Bakken (Base of target)									
10,975.0	89.52	122.00	10,706.3	-295.5	401.5	424.1	12.00	12.00	0.00
10,979.0	90.00	122.00	10,706.3	-297.6	404.9	427.6	12.00	12.00	0.00
Start 550.0 hold at 10979.0 MD									
11,000.0	90.00	122.00	10,706.3	-308.8	422.7	446.3	0.00	0.00	0.00
11,100.0	90.00	122.00	10,706.3	-361.8	507.5	535.1	0.00	0.00	0.00
11,200.0	90.00	122.00	10,706.3	-414.8	592.3	623.9	0.00	0.00	0.00
11,300.0	90.00	122.00	10,706.3	-467.8	677.2	712.7	0.00	0.00	0.00
11,400.0	90.00	122.00	10,706.3	-520.8	762.0	801.5	0.00	0.00	0.00
11,500.0	90.00	122.00	10,706.3	-573.7	846.8	890.3	0.00	0.00	0.00
11,529.0	90.00	122.00	10,706.3	-589.1	871.3	916.0	0.00	0.00	0.00
Start DLS 3.92 TFO 270.00 - 7"									
11,600.0	90.00	119.21	10,706.3	-625.3	932.5	979.8	3.92	0.00	-3.92
11,700.0	90.00	115.29	10,706.3	-671.0	1,021.3	1,072.1	3.92	0.00	-3.92
11,800.0	90.00	111.37	10,706.3	-710.6	1,113.1	1,166.8	3.92	0.00	-3.92
11,900.0	90.00	107.44	10,706.3	-743.9	1,207.4	1,263.5	3.92	0.00	-3.92
12,000.0	90.00	103.52	10,706.3	-770.5	1,303.8	1,361.7	3.92	0.00	-3.92
12,100.0	90.00	99.60	10,706.3	-790.6	1,401.8	1,460.9	3.92	0.00	-3.92
12,200.0	90.00	95.68	10,706.3	-803.9	1,500.8	1,560.8	3.92	0.00	-3.92
12,300.0	90.00	91.75	10,706.3	-810.3	1,600.6	1,660.7	3.92	0.00	-3.92
12,344.7	90.00	90.00	10,706.3	-811.0	1,645.3	1,705.3	3.92	0.00	-3.92
Start 8384.7 hold at 12344.7 MD									
12,400.0	90.00	90.00	10,706.3	-811.0	1,700.6	1,760.4	0.00	0.00	0.00
12,500.0	90.00	90.00	10,706.3	-811.0	1,800.6	1,860.1	0.00	0.00	0.00
12,600.0	90.00	90.00	10,706.3	-811.0	1,900.6	1,959.8	0.00	0.00	0.00
12,700.0	90.00	90.00	10,706.3	-811.0	2,000.6	2,059.5	0.00	0.00	0.00
12,800.0	90.00	90.00	10,706.3	-811.0	2,100.6	2,159.1	0.00	0.00	0.00
12,900.0	90.00	90.00	10,706.3	-811.0	2,200.6	2,258.8	0.00	0.00	0.00
13,000.0	90.00	90.00	10,706.3	-811.0	2,300.6	2,358.5	0.00	0.00	0.00
13,100.0	90.00	90.00	10,706.3	-811.0	2,400.6	2,458.2	0.00	0.00	0.00
13,200.0	90.00	90.00	10,706.3	-811.0	2,500.6	2,557.8	0.00	0.00	0.00
13,300.0	90.00	90.00	10,706.3	-811.0	2,600.6	2,657.5	0.00	0.00	0.00
13,400.0	90.00	90.00	10,706.3	-811.0	2,700.6	2,757.2	0.00	0.00	0.00
13,500.0	90.00	90.00	10,706.3	-811.0	2,800.6	2,856.9	0.00	0.00	0.00

Oasis Petroleum

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13,600.0	90.00	90.00	10,706.3	-811.0	2,900.6	2,956.5	0.00	0.00	0.00
13,700.0	90.00	90.00	10,706.3	-811.0	3,000.6	3,056.2	0.00	0.00	0.00
13,800.0	90.00	90.00	10,706.3	-811.0	3,100.6	3,155.9	0.00	0.00	0.00
13,900.0	90.00	90.00	10,706.3	-811.0	3,200.6	3,255.6	0.00	0.00	0.00
14,000.0	90.00	90.00	10,706.3	-811.0	3,300.6	3,355.2	0.00	0.00	0.00
14,100.0	90.00	90.00	10,706.3	-811.0	3,400.6	3,454.9	0.00	0.00	0.00
14,200.0	90.00	90.00	10,706.3	-811.0	3,500.6	3,554.6	0.00	0.00	0.00
14,300.0	90.00	90.00	10,706.3	-811.0	3,600.6	3,654.3	0.00	0.00	0.00
14,400.0	90.00	90.00	10,706.3	-811.0	3,700.6	3,753.9	0.00	0.00	0.00
14,500.0	90.00	90.00	10,706.3	-811.0	3,800.6	3,853.6	0.00	0.00	0.00
14,600.0	90.00	90.00	10,706.3	-811.0	3,900.6	3,953.3	0.00	0.00	0.00
14,700.0	90.00	90.00	10,706.3	-811.0	4,000.6	4,053.0	0.00	0.00	0.00
14,800.0	90.00	90.00	10,706.3	-811.0	4,100.6	4,152.6	0.00	0.00	0.00
14,900.0	90.00	90.00	10,706.3	-811.0	4,200.6	4,252.3	0.00	0.00	0.00
15,000.0	90.00	90.00	10,706.3	-811.0	4,300.6	4,352.0	0.00	0.00	0.00
15,100.0	90.00	90.00	10,706.3	-811.0	4,400.6	4,451.7	0.00	0.00	0.00
15,200.0	90.00	90.00	10,706.3	-811.0	4,500.6	4,551.3	0.00	0.00	0.00
15,300.0	90.00	90.00	10,706.3	-811.0	4,600.6	4,651.0	0.00	0.00	0.00
15,400.0	90.00	90.00	10,706.3	-811.0	4,700.6	4,750.7	0.00	0.00	0.00
15,500.0	90.00	90.00	10,706.3	-811.0	4,800.6	4,850.4	0.00	0.00	0.00
15,600.0	90.00	90.00	10,706.3	-811.0	4,900.6	4,950.0	0.00	0.00	0.00
15,700.0	90.00	90.00	10,706.3	-811.0	5,000.6	5,049.7	0.00	0.00	0.00
15,800.0	90.00	90.00	10,706.3	-811.0	5,100.6	5,149.4	0.00	0.00	0.00
15,900.0	90.00	90.00	10,706.3	-811.0	5,200.6	5,249.1	0.00	0.00	0.00
16,000.0	90.00	90.00	10,706.3	-811.0	5,300.6	5,348.7	0.00	0.00	0.00
16,100.0	90.00	90.00	10,706.3	-811.0	5,400.6	5,448.4	0.00	0.00	0.00
16,200.0	90.00	90.00	10,706.3	-811.0	5,500.6	5,548.1	0.00	0.00	0.00
16,300.0	90.00	90.00	10,706.3	-811.0	5,600.6	5,647.8	0.00	0.00	0.00
16,400.0	90.00	90.00	10,706.3	-811.0	5,700.6	5,747.4	0.00	0.00	0.00
16,500.0	90.00	90.00	10,706.3	-811.0	5,800.6	5,847.1	0.00	0.00	0.00
16,600.0	90.00	90.00	10,706.3	-811.0	5,900.6	5,946.8	0.00	0.00	0.00
16,700.0	90.00	90.00	10,706.3	-811.0	6,000.6	6,046.5	0.00	0.00	0.00
16,800.0	90.00	90.00	10,706.3	-811.0	6,100.6	6,146.1	0.00	0.00	0.00
16,900.0	90.00	90.00	10,706.3	-811.0	6,200.6	6,245.8	0.00	0.00	0.00
17,000.0	90.00	90.00	10,706.3	-811.0	6,300.6	6,345.5	0.00	0.00	0.00
17,100.0	90.00	90.00	10,706.3	-811.0	6,400.6	6,445.2	0.00	0.00	0.00
17,200.0	90.00	90.00	10,706.3	-811.0	6,500.6	6,544.8	0.00	0.00	0.00
17,300.0	90.00	90.00	10,706.3	-811.0	6,600.6	6,644.5	0.00	0.00	0.00
17,400.0	90.00	90.00	10,706.3	-811.0	6,700.6	6,744.2	0.00	0.00	0.00
17,500.0	90.00	90.00	10,706.3	-811.0	6,800.6	6,843.9	0.00	0.00	0.00
17,600.0	90.00	90.00	10,706.3	-811.0	6,900.6	6,943.5	0.00	0.00	0.00
17,700.0	90.00	90.00	10,706.3	-811.0	7,000.6	7,043.2	0.00	0.00	0.00
17,800.0	90.00	90.00	10,706.3	-811.0	7,100.6	7,142.9	0.00	0.00	0.00
17,900.0	90.00	90.00	10,706.3	-811.0	7,200.6	7,242.6	0.00	0.00	0.00
18,000.0	90.00	90.00	10,706.3	-811.0	7,300.6	7,342.2	0.00	0.00	0.00
18,100.0	90.00	90.00	10,706.3	-811.0	7,400.6	7,441.9	0.00	0.00	0.00
18,200.0	90.00	90.00	10,706.3	-811.0	7,500.6	7,541.6	0.00	0.00	0.00
18,300.0	90.00	90.00	10,706.3	-811.0	7,600.6	7,641.3	0.00	0.00	0.00
18,400.0	90.00	90.00	10,706.3	-811.0	7,700.6	7,740.9	0.00	0.00	0.00
18,500.0	90.00	90.00	10,706.3	-811.0	7,800.6	7,840.6	0.00	0.00	0.00
18,600.0	90.00	90.00	10,706.3	-811.0	7,900.6	7,940.3	0.00	0.00	0.00
18,700.0	90.00	90.00	10,706.3	-811.0	8,000.6	8,039.9	0.00	0.00	0.00
18,800.0	90.00	90.00	10,706.3	-811.0	8,100.6	8,139.6	0.00	0.00	0.00
18,900.0	90.00	90.00	10,706.3	-811.0	8,200.6	8,239.3	0.00	0.00	0.00
19,000.0	90.00	90.00	10,706.3	-811.0	8,300.6	8,339.0	0.00	0.00	0.00

Oasis Petroleum

Planning Report

Database:	OpenWellsCompass - EDM Prod	Local Co-ordinate Reference:	Well Wade Federal 5300 41-30 9B
Company:	Oasis	TVD Reference:	WELL @ 2070.0usft (Original Well Elev)
Project:	Indian Hills	MD Reference:	WELL @ 2070.0usft (Original Well Elev)
Site:	153N-100W-29/30	North Reference:	True
Well:	Wade Federal 5300 41-30 9B	Survey Calculation Method:	Minimum Curvature
Wellbore:	Wade Federal 5300 41-30 9B		
Design:	Plan #1		

Planned Survey									
Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)
19,100.0	90.00	90.00	10,706.3	-811.0	8,400.6	8,438.6	0.00	0.00	0.00
19,200.0	90.00	90.00	10,706.3	-811.0	8,500.6	8,538.3	0.00	0.00	0.00
19,300.0	90.00	90.00	10,706.3	-811.0	8,600.6	8,638.0	0.00	0.00	0.00
19,400.0	90.00	90.00	10,706.3	-811.0	8,700.6	8,737.7	0.00	0.00	0.00
19,500.0	90.00	90.00	10,706.3	-811.0	8,800.6	8,837.3	0.00	0.00	0.00
19,600.0	90.00	90.00	10,706.3	-811.0	8,900.6	8,937.0	0.00	0.00	0.00
19,700.0	90.00	90.00	10,706.3	-811.0	9,000.6	9,036.7	0.00	0.00	0.00
19,800.0	90.00	90.00	10,706.3	-811.0	9,100.6	9,136.4	0.00	0.00	0.00
19,900.0	90.00	90.00	10,706.3	-811.0	9,200.6	9,236.0	0.00	0.00	0.00
20,000.0	90.00	90.00	10,706.3	-811.0	9,300.6	9,335.7	0.00	0.00	0.00
20,100.0	90.00	90.00	10,706.3	-811.0	9,400.6	9,435.4	0.00	0.00	0.00
20,200.0	90.00	90.00	10,706.3	-811.0	9,500.6	9,535.1	0.00	0.00	0.00
20,300.0	90.00	90.00	10,706.3	-811.0	9,600.6	9,634.7	0.00	0.00	0.00
20,400.0	90.00	90.00	10,706.3	-811.0	9,700.6	9,734.4	0.00	0.00	0.00
20,500.0	90.00	90.00	10,706.3	-811.0	9,800.6	9,834.1	0.00	0.00	0.00
20,600.0	90.00	90.00	10,706.3	-811.0	9,900.6	9,933.8	0.00	0.00	0.00
20,700.0	90.00	90.00	10,706.3	-811.0	10,000.6	10,033.4	0.00	0.00	0.00
20,729.4	90.00	90.00	10,706.3	-811.0	10,030.0	10,062.7	0.00	0.00	0.00
TD at 20729.4									

Design Targets									
Target Name	Dip Angle (°)	Dip Dir. (°)	TVD (usft)	+N/-S (usft)	+E/-W (usft)	Northing (usft)	Easting (usft)	Latitude	Longitude
- hit/miss target									
- Shape									
WADE 9B PBHL	0.00	0.00	10,706.3	-811.0	10,030.0	393,806.78	1,219,626.30	48° 2' 19.640 N	103° 33' 43.220 W
- plan hits target center									
- Point									

Casing Points					
Measured Depth (usft)	Vertical Depth (usft)	Name	Casing Diameter (")	Hole Diameter (")	
2,050.0	2,050.0	13 3/8"	13-3/8	17 1/2	
11,529.0	10,706.3	7"	7	8-3/4	

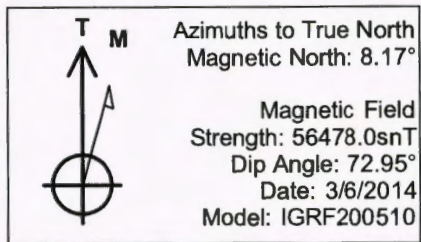
Oasis Petroleum

Planning Report

Database:	OpenWellsCompass - EDM Prod	Local Co-ordinate Reference:	Well Wade Federal 5300 41-30 9B
Company:	Oasis	TVD Reference:	WELL @ 2070.0usft (Original Well Elev)
Project:	Indian Hills	MD Reference:	WELL @ 2070.0usft (Original Well Elev)
Site:	153N-100W-29/30	North Reference:	True
Well:	Wade Federal 5300 41-30 9B	Survey Calculation Method:	Minimum Curvature
Wellbore:	Wade Federal 5300 41-30 9B		
Design:	Plan #1		

Formations						
Measured Depth (usft)	Vertical Depth (usft)	Name	Lithology	Dip (°)	Dip Direction (°)	
1,920.0	1,920.0	Pierre				
4,566.0	4,566.0	Greenhorn				
4,969.0	4,969.0	Mowry				
5,391.0	5,391.0	Dakota				
6,407.0	6,407.0	Rierdon				
6,896.0	6,896.0	Dunham Salt				
6,942.0	6,942.0	Dunham Salt Base				
7,205.0	7,205.0	Pine Salt				
7,229.0	7,229.0	Pine Salt Base				
7,291.0	7,291.0	Opeche Salt				
7,371.0	7,371.0	Opeche Salt Base				
7,615.0	7,615.0	Amsden				
7,771.0	7,771.0	Tyler				
7,994.0	7,994.0	Otter/Base Minnelusa				
8,336.1	8,336.0	Kibbey Lime				
8,484.1	8,484.0	Charles Salt				
9,163.1	9,163.0	Base Last Salt				
9,377.1	9,377.0	Mission Canyon				
9,926.1	9,926.0	Lodgepole				
10,757.8	10,656.0	False Bakken				
10,786.3	10,668.0	Upper Bakken Shale				
10,874.7	10,695.0	Middle Bakken (Top of Target)				
10,961.1	10,706.0	Middle Bakken (Base of target)				

Plan Annotations					
Measured Depth (usft)	Vertical Depth (usft)	Local Coordinates			
		+N/-S (usft)	+E/-W (usft)	Comment	
10,225.0	10,224.8	-44.6	0.0	Start Drop -12.50	
10,229.0	10,228.8	-44.6	0.0	Start Build 12.00	
10,979.0	10,706.3	-297.6	404.9	Start 550.0 hold at 10979.0 MD	
11,529.0	10,706.3	-589.1	871.3	Start DLS 3.92 TFO 270.00	
12,344.7	10,706.3	-811.0	1,645.3	Start 8384.7 hold at 12344.7 MD	
20,729.4	10,706.3	-811.0	10,030.0	TD at 20729.4	

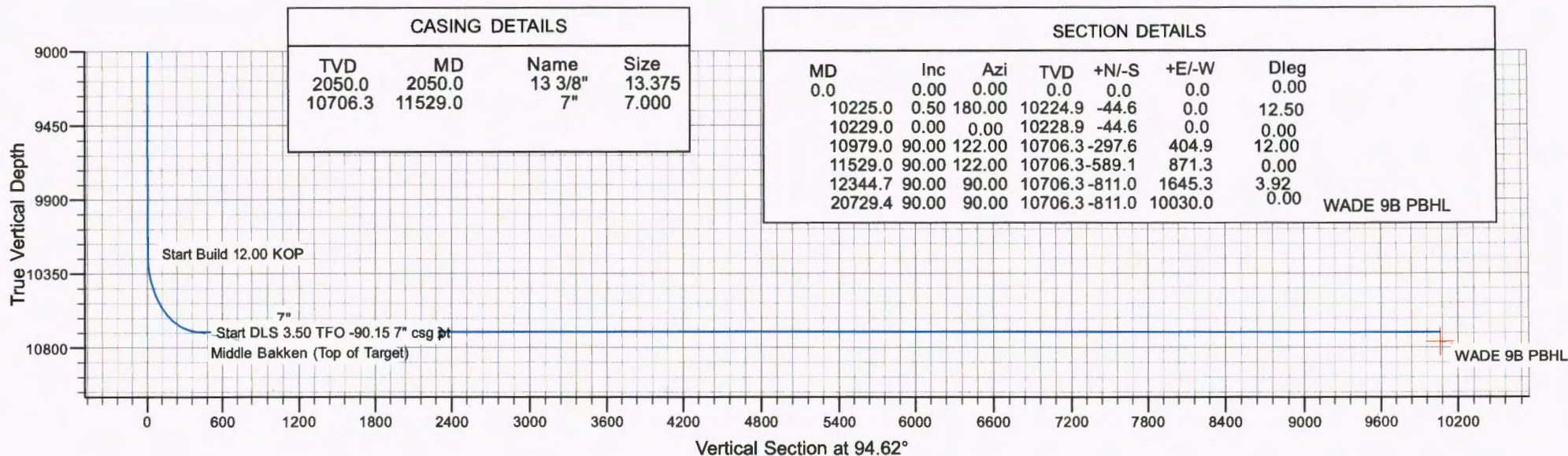
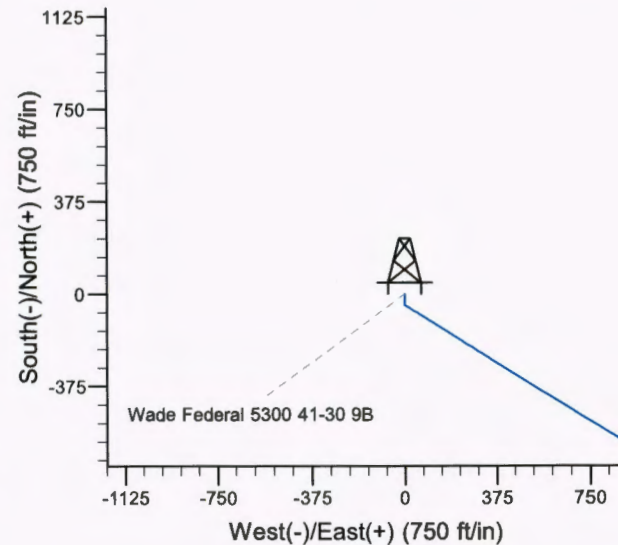
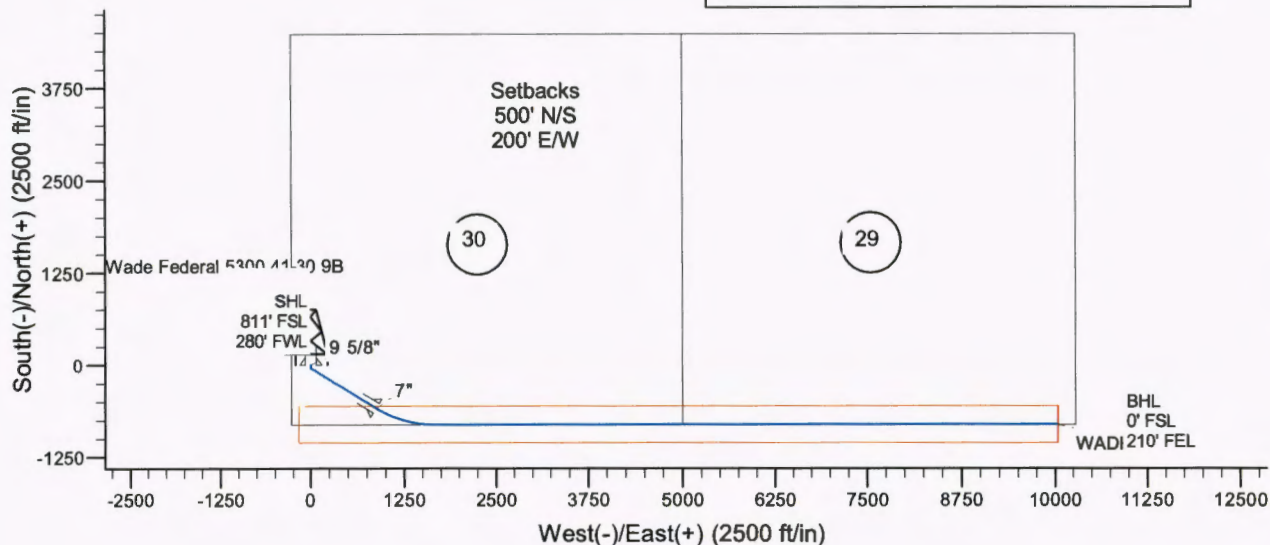


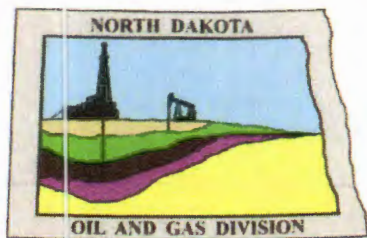
Project: Indian Hills
Site: 153N-100W-29/30
Well: Wade Federal 5300 41-30 9B
Wellbore: Wade Federal 5300 41-30 9B
Design: Plan #1

SITE DETAILS: 153N-100W-29/30

Site Centre Latitude: 48° 2' 27.670 N
Longitude: 103° 36' 10.820 W

Positional Uncertainty: 0.0
Convergence: -2.31
Local North: True





Oil and Gas Division

Lynn D. Helms - Director

Bruce E. Hicks - Assistant Director

Department of Mineral Resources

Lynn D. Helms - Director

North Dakota Industrial Commission

www.dmr.nd.gov/oilgas

BRANDI TERRY
OASIS PETROLEUM NORTH AMERICA LLC
1001 FANNIN STE 1500
HOUSTON, TX 77002 USA

Date: 6/30/2014

RE: CORES AND SAMPLES

Well Name: **WADE FEDERAL 5300 41-30 9B** Well File No.: **28744**
Location: **LOT4 30-153-100** County: **MCKENZIE**
Permit Type: **Development - HORIZONTAL**
Field: **BAKER** Target Horizon: **BAKKEN**

Dear BRANDI TERRY:

North Dakota Century Code Section 38-08-04 provides for the preservation of cores and samples and their shipment to the State Geologist when requested. The following is required on the above referenced well:

- 1) All cores, core chips and samples must be submitted to the State Geologist as provided for under North Dakota Century Code: Section 38-08-04 and North Dakota Administrative Code: Section 43-02-03-38.1.
- 2) Samples: The Operator is to begin collecting sample drill cuttings no lower than the:
Base of the Last Charles Salt
 - Sample cuttings shall be collected at:
 - o 30' maximum intervals through all vertical and build sections.
 - o 100' maximum intervals through any horizontal sections.
 - Samples must be washed, dried, placed in standard sample envelopes (3" x 4.5"), packed in the correct order into standard sample boxes (3.5" x 5.25" x 15.25").
 - Samples boxes are to be carefully identified with a label that indicates the operator, well name, well file number, American Petroleum Institute (API) number, location and depth of samples; and forwarded in to the state core and sample library within 30 days of the completion of drilling operations.
- 3) Cores: Any cores cut shall be preserved in correct order, boxed in standard core boxes (4.5", 4.5", 35.75"), and the entire core forwarded to the state core and samples library within 180 days of completion of drilling operations. Any extension of time must have approval on a Form 4 Sundry Notice.

All cores, core chips, and samples must be shipped, prepaid, to the state core and samples library at the following address:

**ND Geological Survey Core Library
2835 Campus Road, Stop 8156
Grand Forks, ND 58202**

North Dakota Century Code Section 38-08-16 allows for a civil penalty for any violation of Chapter 38 08 not to exceed \$12,500 for each offense, and each day's violation is a separate offense.

Sincerely

Stephen Fried
Geologist

**SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4**

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2008)



Well File No.

28744

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.

PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☒ Notice of Intent	Approximate Start Date October 1, 2014
☐ Report of Work Done	Date Work Completed
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	Approximate Start Date

☐ Drilling Prognosis	☐ Spill Report
☐ Redrilling or Repair	☐ Shooting
☐ Casing or Liner	☐ Acidizing
☐ Plug Well	☐ Fracture Treatment
☐ Supplemental History	☐ Change Production Method
☐ Temporarily Abandon	☐ Reclamation
☒ Other	Waiver to rule Rule 43-02-03-31

Well Name and Number Wade Federal 5300 41-30 9B					
Footages		Qtr-Qtr	Section	Township	Range
811 F S L 280 F W L		SWSW	30	153 N	100 W
Field		Pool		County	
		Bakken		McKenzie	

24-HOUR PRODUCTION RATE			
Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

Oasis Petroleum respectfully requests a waiver to Rule 43-02-03-31 in regards to running open hole logs for the above referenced well. Justification for this request is as follows:

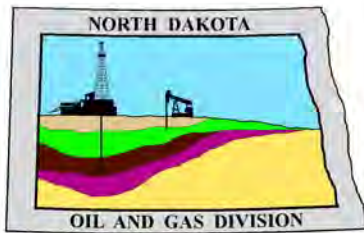
The Oasis Petroleum Wade Federal 5300 21-30H located within a mile of subject location

20197

If this exception is approved, Oasis Petroleum will run a CBL on the intermediate string, and we will also run GR to surface. Oasis Petroleum will also submit two digital copies of each cased hole log and a copy of the mud log containing MWD gamma ray.

Company Oasis Petroleum North America LLC		Telephone Number 281-404-9491	
Address 1001 Fannin, Suite 1500			
City Houston		State TX	Zip Code 77002
Signature 		Printed Name Brandi Terry	
Title Regulatory Specialist		Date March 31, 2014	
Email Address bterry@oasispetroleum.com			

FOR STATE USE ONLY	
☐ Received	☒ Approved
Date 6-27-2014	
By 	
Title Stephen Fried Geologist	



Oil and Gas Division

Lynn D. Helms - Director

Bruce E. Hicks - Assistant Director

Department of Mineral Resources

Lynn D. Helms - Director

North Dakota Industrial Commission

www.oilgas.nd.gov

June 27, 2014

Brandi Terry
REGULATORY SPECIALIST
OASIS PETROLEUM NORTH AMERICA LLC
1001 Fannin Street, Suite 1500
Houston, TX 77002

RE: HORIZONTAL WELL
WADE FEDERAL 5300 41-30 9B
LOT4 Section 30-153N-100W
McKenzie County
Well File # 28744

Dear Brandi:

Pursuant to Commission Order No. 23752, approval to drill the above captioned well is hereby given. The approval is granted on the condition that all portions of the well bore not isolated by cement, be located approximately down the east-west axis (500' drilling corridor, per Commission policy) and no closer than 200 feet to the east or west boundary within the 2560 acre spacing unit consisting of Sections 29, 30, 31 & 32 T153N R100W. **Tool error is not required pursuant to order.**

PERMIT STIPULATIONS: Effective June 1, 2014, a covered leak-proof container (with placard) for filter sock disposal must be maintained on the well site beginning when the well is spud, and must remain on-site during clean-out, completion, and flow-back whenever filtration operations are conducted. Due to drainage adjacent to the well site, a dike is required surrounding the entire location. OASIS PETROLEUM NORTH AMERICA LLC must take into consideration NDAC 43-02-03-28 (Safety Regulation) when contemplating simultaneous operations on the above captioned location. Pursuant to NDAC 43-02-03-28 (Safety Regulation) "No boiler, portable electric lighting generator, or treater shall be placed nearer than 150 feet to any producing well or oil tank." OASIS PETRO NO AMER must contact NDIC Field Inspector Richard Dunn at 701-770-3554 prior to location construction.

Drilling pit

NDAC 43-02-03-19.4 states that "a pit may be utilized to bury drill cuttings and solids generated during well drilling and completion operations, providing the pit can be constructed, used and reclaimed in a manner that will prevent pollution of the land surface and freshwaters. Reserve and circulation of mud system through earthen pits are prohibited. All pits shall be inspected by an authorized representative of the director prior to lining and use. Drill cuttings and solids must be stabilized in a manner approved by the director prior to placement in a cuttings pit."

Form 1 Changes & Hard Lines

Any changes, shortening of casing point or lengthening at Total Depth must have prior approval by the NDIC. The proposed directional plan is at a legal location. The minimum legal coordinate from the well head at casing point is: 561°S. Also, based on the azimuth of the proposed lateral the maximum legal coordinate from the well head is: 1061°S and 10042°E.

Location Construction Commencement (Three Day Waiting Period)

Operators shall not commence operations on a drill site until the 3rd business day following publication of the approved drilling permit on the NDIC - OGD Daily Activity Report. If circumstances require operations to commence before the 3rd business day following publication on the Daily Activity Report, the waiting period may be waived by the Director. Application for a waiver must be by sworn affidavit providing the information necessary to evaluate the extenuating circumstances, the factors of NDAC 43-02-03-16.2 (1), (a)-(f), and any other information that would allow the Director to conclude that in the event another owner seeks revocation of the drilling permit, the applicant should retain the permit.

Permit Fee & Notification

Payment was received in the amount of \$100 via credit card. The permit fee has been received. It is requested that notification be given immediately upon the spudding of the well. This information should be relayed to the Oil & Gas Division, Bismarck, via telephone. The following information must be included: Well name, legal location, permit number, drilling contractor, company representative, date and time of spudding. Office hours are 8:00 a.m. to 12:00 p.m. and 1:00 p.m. to 5:00 p.m. Central Time. Our telephone number is (701) 328-8020, leave a message if after hours or on the weekend.

Survey Requirements for Horizontal, Horizontal Re-entry, and Directional Wells

NDAC Section 43-02-03-25 (Deviation Tests and Directional Surveys) states in part (that) the survey contractor shall file a certified copy of all surveys with the director free of charge within thirty days of completion. Surveys must be submitted as one electronic copy, or in a form approved by the director. However, the director may require the directional survey to be filed immediately after completion if the survey is needed to conduct the operation of the director's office in a timely manner. Certified surveys must be submitted via email in one adobe document, with a certification cover page to certsurvey@nd.gov.

Survey points shall be of such frequency to accurately determine the entire location of the well bore.

Specifically, the Horizontal and Directional well survey frequency is 100 feet in the vertical, 30 feet in the curve (or when sliding) and 90 feet in the lateral.

Surface casing cement

Tail cement utilized on surface casing must have a minimum compressive strength of 500 psi within 12 hours, and tail cement utilized on production casing must have a minimum compressive strength of 500 psi before drilling the plug or initiating tests.

Logs

NDAC Section 43-02-03-31 requires the running of (1) a suite of open hole logs from which formation tops and porosity zones can be determined, (2) a Gamma Ray Log run from total depth to ground level elevation of the well bore, and (3) a log from which the presence and quality of cement can be determined (Standard CBL or Ultrasonic cement evaluation log) in every well in which production or intermediate casing has been set, this log must be run prior to completing the well. All logs run must be submitted free of charge, as one digital TIFF (tagged image file format) copy and one digital LAS (log ASCII) formatted copy. Digital logs may be submitted on a standard CD, DVD, or attached to an email sent to digitallogs@nd.gov

Thank you for your cooperation.

Sincerely,

Alice Webber
Engineering Tech



APPLICATION FOR PERMIT TO DRILL HORIZONTAL WELL - FORM 1H

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 54269 (08-2005)

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

Type of Work New Location	Type of Well Oil & Gas	Approximate Date Work Will Start 10 / 01 / 2014	Confidential Status No
Operator OASIS PETROLEUM NORTH AMERICA LLC			Telephone Number 281-404-9491
Address 1001 Fannin Street, Suite 1500		City Houston	State TX Zip Code 77002

☒ Notice has been provided to the owner of any permanently occupied dwelling within 1,320 feet.

☒ This well is not located within five hundred feet of an occupied dwelling.

WELL INFORMATION (If more than one lateral proposed, enter data for additional laterals on page 2)

Well Name WADE FEDERAL				Well Number 5300 41-30 9B			
Surface Footages 811 F S L 280 F W L		Qtr-Qtr LOT4	Section 30	Township 153 N	Range 100 W	County McKenzie	
Longstring Casing Point Footages 222 F S L 1151 F W L		Qtr-Qtr LOT4	Section 30	Township 153 N	Range 100 W	County McKenzie	
Longstring Casing Point Coordinates From Well Head 589 S From WH 871 E From WH		Azimuth 122 °	Longstring Total Depth 11529 Feet MD 10706 Feet TVD				
Bottom Hole Footages From Nearest Section Line 0 F S L 212 F E L		Qtr-Qtr LOT6	Section 29	Township 153 N	Range 100 W	County Williams	
Bottom Hole Coordinates From Well Head 811 S From WH 10030 E From WH		KOP Lateral 1 10229 Feet MD	Azimuth Lateral 1 90 °		Estimated Total Depth Lateral 1 20729 Feet MD 10706 Feet TVD		
Latitude of Well Head 48 ° 02 ' 27.67 "		Longitude of Well Head -103 ° 36 ' 10.82 "		NAD Reference NAD83	Description of (Subject to NDIC Approval) Spacing Unit: Secs 29,30,31&32 T153N R100W		
Ground Elevation 2058 Feet Above S.L.	Acres in Spacing/Drilling Unit 2560	Spacing/Drilling Unit Setback Requirement 500 Feet N/S 200 Feet E/W			Industrial Commission Order 23752		
North Line of Spacing/Drilling Unit 10513 Feet	South Line of Spacing/Drilling Unit 10535 Feet	East Line of Spacing/Drilling Unit 10362 Feet			West Line of Spacing/Drilling Unit 10485 Feet		
Objective Horizons BAKKEN						Pierre Shale Top 1920	
Proposed Surface Casing	Size 9 - 5/8 "	Weight 36 Lb./Ft.	Depth 2050 Feet	Cement Volume 735 Sacks	NOTE: Surface hole must be drilled with fresh water and surface casing must be cemented back to surface.		
Proposed Longstring Casing	Size 7 - "	Weight(s) 29/32 Lb./Ft.	Longstring Total Depth 11529 Feet MD 10706 Feet TVD		Cement Volume 781 Sacks	Cement Top 4891 Feet	Top Dakota Sand 5391 Feet
Base Last Charles Salt (If Applicable) 9163 Feet		NOTE: Intermediate or longstring casing string must be cemented above the top Dakota Group Sand.					
Proposed Logs TRIPLE COMBO: KOP TO KIBBEY GR/RES TO BSC GR TO SURF CND THROUGH THE DAKOTA							
Drilling Mud Type (Vertical Hole - Below Surface Casing) Invert				Drilling Mud Type (Lateral) Salt Water Gel			
Survey Type in Vertical Portion of Well MWD Every 100 Feet		Survey Frequency: Build Section 30 Feet		Survey Frequency: Lateral 90 Feet		Survey Contractor RYAN	

NOTE: A Gamma Ray log must be run to ground surface and a CBL must be run on intermediate or longstring casing string if set.

Surveys are required at least every 30 feet in the build section and every 90 feet in the lateral section of a horizontal well. Measurement inaccuracies are not considered when determining compliance with the spacing/drilling unit boundary setback requirement except in the following scenarios: 1) When the angle between the well bore and the respective boundary is 10 degrees or less; or 2) If Industry standard methods and equipment are not utilized. Consult the applicable field order for exceptions.

If measurement inaccuracies are required to be considered, a 2° MWD measurement inaccuracy will be applied to the horizontal portion of the well bore. This measurement inaccuracy is applied to the well bore from KOP to TD.

REQUIRED ATTACHMENTS: Certified surveyor's plat, horizontal section plat, estimated geological tops, proposed mud/cementing plan, directional plot/plan, \$100 fee.
See Page 2 for Comments section and signature block.

COMMENTS, ADDITIONAL INFORMATION, AND/OR LIST OF ATTACHMENTS

--

Lateral 2

KOP Lateral 2 Feet MD	Azimuth Lateral 2 °	Estimated Total Depth Lateral 2 Feet MD Feet TVD	KOP Coordinates From Well Head From WH From WH		
Formation Entry Point Coordinates From Well Head From WH From WH		Bottom Hole Coordinates From Well Head From WH From WH			
KOP Footages From Nearest Section Line F L F L		Qtr-Qtr	Section	Township N	Range W County
Bottom Hole Footages From Nearest Section Line F L F L		Qtr-Qtr	Section	Township N	Range W County

Lateral 3

KOP Lateral 3 Feet MD	Azimuth Lateral 3 °	Estimated Total Depth Lateral 3 Feet MD Feet TVD	KOP Coordinates From Well Head From WH From WH		
Formation Entry Point Coordinates From Well Head From WH From WH		Bottom Hole Coordinates From Well Head From WH From WH			
KOP Footages From Nearest Section Line F L F L		Qtr-Qtr	Section	Township N	Range W County
Bottom Hole Footages From Nearest Section Line F L F L		Qtr-Qtr	Section	Township N	Range W County

Lateral 4

KOP Lateral 4 Feet MD	Azimuth Lateral 4 °	Estimated Total Depth Lateral 4 Feet MD Feet TVD	KOP Coordinates From Well Head From WH From WH		
Formation Entry Point Coordinates From Well Head From WH From WH		Bottom Hole Coordinates From Well Head From WH From WH			
KOP Footages From Nearest Section Line F L F L		Qtr-Qtr	Section	Township N	Range W County
Bottom Hole Footages From Nearest Section Line F L F L		Qtr-Qtr	Section	Township N	Range W County

Lateral 5

KOP Lateral 5 Feet MD	Azimuth Lateral 5 °	Estimated Total Depth Lateral 5 Feet MD Feet TVD	KOP Coordinates From Well Head From WH From WH		
Formation Entry Point Coordinates From Well Head From WH From WH		Bottom Hole Coordinates From Well Head From WH From WH			
KOP Footages From Nearest Section Line F L F L		Qtr-Qtr	Section	Township N	Range W County
Bottom Hole Footages From Nearest Section Line F L F L		Qtr-Qtr	Section	Township N	Range W County

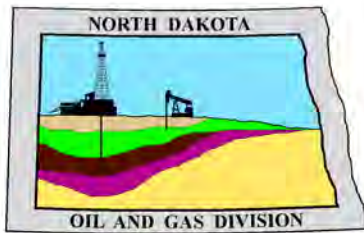
I hereby swear or affirm the information provided is true, complete and correct as determined from all available records.		Date 03 / 31 / 2014
ePermit	Printed Name Brandi Terry	Title REGULATORY SPECIALIST

FOR STATE USE ONLY

Permit and File Number 28744	API Number 33 - 053 - 06051
Field BAKER	
Pool BAKKEN	Permit Type DEVELOPMENT

FOR STATE USE ONLY

Date Approved 6 / 27 / 2014
By Alice Webber
Title Engineering Tech



Oil and Gas Division

Lynn D. Helms - Director

Bruce E. Hicks - Assistant Director

Department of Mineral Resources

Lynn D. Helms - Director

North Dakota Industrial Commission

www.oilgas.nd.gov

April 9, 2014

**RE: Filter Socks and Other Filter Media
Leakproof Container Required
Oil and Gas Wells**

Dear Operator,

North Dakota Administrative Code Section 43-02-03-19.2 states in part that all waste material associated with exploration or production of oil and gas must be properly disposed of in an authorized facility in accord with all applicable local, state, and federal laws and regulations.

Filtration systems are commonly used during oil and gas operations in North Dakota. The Commission is very concerned about the proper disposal of used filters (including filter socks) used by the oil and gas industry.

Effective June 1, 2014, a container must be maintained on each well drilled in North Dakota beginning when the well is spud and must remain on-site during clean-out, completion, and flow-back whenever filtration operations are conducted. The on-site container must be used to store filters until they can be properly disposed of in an authorized facility. Such containers must be:

- leakproof to prevent any fluids from escaping the container
- covered to prevent precipitation from entering the container
- placard to indicate only filters are to be placed in the container

If the operator will not utilize a filtration system, a waiver to the container requirement will be considered, but only upon the operator submitting a Sundry Notice (Form 4) justifying their request.

As previously stated in our March 13, 2014 letter, North Dakota Administrative Code Section 33-20-02.1-01 states in part that every person who transports solid waste (which includes oil and gas exploration and production wastes) is required to have a valid permit issued by the North Dakota Department of Health, Division of Waste Management. Please contact the Division of Waste Management at (701) 328-5166 with any questions on the solid waste program. Note oil and gas exploration and production wastes include produced water, drilling mud, invert mud, tank bottom sediment, pipe scale, filters, and fly ash.

Thank you for your cooperation.

Sincerely,

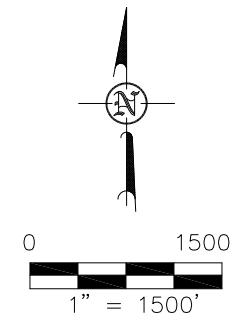
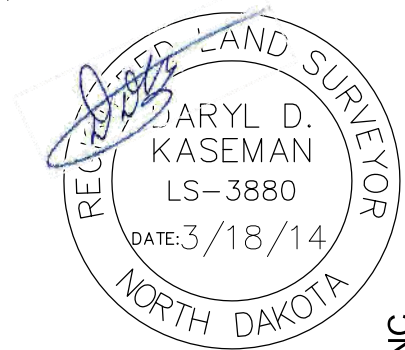
Bruce E. Hicks

Assistant Director

STAKED ON 1/10/14
VERTICAL CONTROL DATUM WAS BASED UPON
CONTROL POINT 705 WITH AN ELEVATION OF 2158.3'

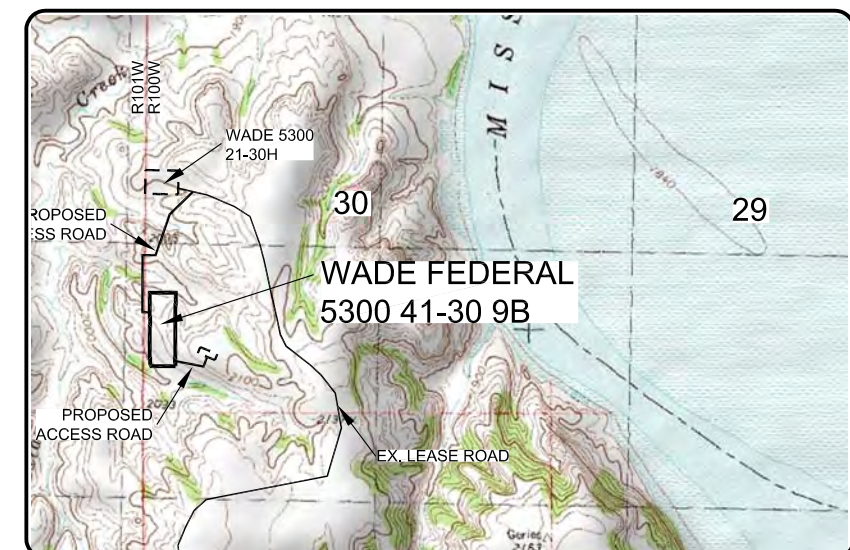

DARYL D. KASEMAN LS-3880

THIS DOCUMENT WAS ORIGINALLY
ISSUED AND SEALED BY DARYL D.
KASEMAN, PLS., REGISTRATION NUMBER
3880 ON 3/18/14 AND THE
ORIGINAL DOCUMENTS ARE STORED AT
THE OFFICES OF INTERSTATE
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Professionals you need, people you trust

1/8 SHEET NO.

[illegible]

OASIS PETROLEUM NORTH AMERICA, LLC
WELL LOCATION PLAT
SECTION 30, T153N, R100W

MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: <u>B.H.H.</u>	Project No.: <u>S13-09S-361.08</u>	
Checked By: <u>D.J.K.</u>	Date: <u>JAN, 2014</u>	

Interstate Engineering, Inc.
P.O. Box 648
425 East Main Street
Sidney, Montana 59270
Ph (406) 433-5617
Fax (406) 433-5618
www.interstateeng.com

Other offices in Minnesota, North Dakota and South Dakota

DRILLING PLAN									
OPERATOR	Oasis Petroleum				COUNTY/STATE	McKenzie Co. , ND			
WELL NAME	Wade Federal 5300 41-30 9B				RIG	B22			
WELL TYPE	Horizontal Middle Bakken								
LOCATION	SW SW 30-153N-100W				Surface Location (survey plat): 811' FSL		280' FWL		
EST. T.D.	20,729'				GROUND ELEV:		2,045'		Sub Height: 25'
TOTAL LATERAL:		9,200'		KB ELEV:		2,070'			
MARKER		TVD	Subsea TVD	LOGS:	Type	Interval			
Pierre	NDIC MAP	1,920	150	OH Logs: Request Log waiver based on the Wade Federal 5300 21-30H 2,150' N of surface location					
Greenhorn		4,566	-2,496	CBL/GR: Above top of cement/GR to base of casing					
Mowry		4,969	-2,899	MWD GR: KOP to lateral TD					
Dakota		5,391	-3,321	DEVIATION:	Surf:	3 deg. max., 1 deg / 100'; srvy every 500'			
Rierdon		6,407	-4,337		Prod:	5 deg. max., 1 deg / 100'; srvy every 100'			
Dunham Salt		6,896	-4,826						
Dunham Salt Base		6,942	-4,872						
Pine Salt		7,205	-5,135						
Pine Salt Base		7,229	-5,159						
Opeche Salt		7,291	-5,221						
Opeche Salt Base		7,371	-5,301						
Amsden		7,615	-5,545						
Tyler		7,771	-5,701						
Otter/Base Minnelusa		7,994	-5,924	DST'S:	None planned				
Kibbey Lime		8,336	-6,266	CORES:	None planned				
Charles Salt		8,484	-6,414						
Base Last Salt		9,163	-7,093						
Mission Canyon		9,377	-7,307						
Lodgepole		9,926	-7,856						
False Bakken		10,656	-8,586	MUDLOGGING:	Two-Man:	Begin 200' above Kibbey			
Upper Bakken Shale		10,668	-8,598		30' samples in curve and lateral				
Middle Bakken (Top of Target)		10,695	-8,625						
Middle Bakken (Base of target)		10,706	-8,636						
Lower Bakken Shale		10,719	-8,649						
Threeforks		10,744	-8,674						
Est. Dip Rate: -0.35									
Max. Anticipated BHP: 4645				Surface Formation: Glacial till					
MUD:	Interval	Type	WT	Vis	WL	Remarks			
Surface:	0' -	2,050'	FW/Gel - Lime Sweeps	8.4-9.0	28-32	NC	Circ Mud Tanks		
Intermediate:	2,050' -	11,529'	Invert	9.5-10.4	40-50	30+HTHp	Circ Mud Tanks		
Lateral:	11,529' -	20,729'	Salt Water	9.8-10.2	28-32	NC	Circ Mud Tanks		
CASING:	Size	Wt ppf	Hole	Depth	Cement	WOC	Remarks		
Surface:	9-5/8"	36#	13-1/2"	2,050'	To Surface	12	100' into Pierre		
Intermediate:	7"	29/32#	8-3/4"	11,529'	4891	24	500' above Dakota		
Production Liner:	4.5"	11.6#	6"	20,729'	TOL @ 10,179'			50' above KOP	
PROBABLE PLUGS, IF REQ'D:									
OTHER:		MD	TVD	FNL/FSL	FEL/FWL	S-T-R	AZI		
Surface:		2,020	2,020	811' FSL	280' FWL	S 30 T153N R100W	Survey Company:		
KOP:		10,229'	10,229'	766' FSL	280' FWL	S 30 T153N R100W	Build Rate:	12 deg /100'	
EOC:		10,979'	10,706'	513' FSL	685' FWL	S 30 T153N R100W			
Casing Point:		11,529'	10,706'	222' FSL	1151' FWL	S 30 T153N R100W			
Middle Bakken Lateral TD:		20,729'	10,706'	0' FSL	210' FEL	S 29 T153N R100W			
Comments:									
Request Log waiver based on the Wade Federal 5300 21-30H 2,150' N of surface location									
Oasis Petroleum does not use Diesel Fuel, as defined by the US EPA in the list below, in our hydraulic fracture operations.									
68334-30-5 (Primary Name: Fuels, diesel), 68476-34-6 (Primary Name: Fuels, diesel, No. 2); 68476-30-2 (Primary Name: Fuel oil No. 2); 68476-31-3 (Primary Name: Fuel oil, No. 4)									
8008-20-6 (Primary Name: Kerosene)									
		35 Packers;35 Sleeves							
		No Frac String Planned							
Geology: N. Gabelman		1/20/2014		Engineering: DAD 5/19/14					

**Oasis Petroleum
Well Summary
Wade Federal 5300 4130 9B
Sec. 30 T153N R100W
McKenzie County, North Dakota**

SURFACE CASING AND CEMENT DESIGN

							Make-up Torque (ft-lbs)		
Size	Interval	Weight	Grade	Coupling	I.D.	Drift	Minimum	Optimum	Max
9-5/8"	0' to 2,050'	36	J-55	LTC	8.921"	8.765"	3400	4530	5660

Interval	Description	Collapse	Burst	Tension	Cost per ft
		(psi) a	(psi) b	(1000 lbs) c	
0' to 2,050'	9-5/8", 36#, J-55, LTC, 8rd	2020 / 2.13	3520 / 3.72	453 / 2.78	

API Rating & Safety Factor

- a) Based on full casing evacuation with 9 ppg fluid on backside (2,050' setting depth).
- b) Burst pressure based on 9 ppg fluid with no fluid on backside (2050' setting depth).
- c) Based on string weight in 9 ppg fluid at 2050' TVD plus 100k# overpull. (Buoyed weight equals 63k lbs.)

Cement volumes are based on 9-5/8" casing set in 13-1/2" hole with 60% excess to circulate cement back to surface. Mix and pump the following slurry.

Pre-flush (Spacer): **20 bbls** fresh water

Lead Slurry: **435 sks** (225 bbls) 2.9 yield conventional system with 94 lb/sk cement, .25 lb/sk D130 Lost Circulation Control Agent, 2% CaCL₂, 4% D079 Extender, and 2% D053 Expanding Agent.

Tail Slurry: **300 sks** (62 bbls) 1.16 yield conventional system with 94 lb/sk cement, .25 lb/sk Lost Circulation Control Agent, and .25% CaCL₂.

Displacement: **153 bbl** OBM

**Oasis Petroleum
Well Summary
Wade Federal 5300 4130 9B
Sec. 30 T153N R100W
McKenzie County, North Dakota**

INTERMEDIATE CASING AND CEMENT DESIGN

Size	Interval	Weight	Grade	Coupling	I.D.	Drift	Make-up Torque (ft-lbs)		
							Minimum	Optimum	Max
7"	0' - 6896'	29	P-110	LTC	6.184"	6.059"	5,980	7,970	8,770
7"	6896' - 10229'	32	HCP-110	LTC	6.094"	6.000"	6,730	8,970	9,870
7"	10229' - 11529'	29	P-110	LTC	6.184"	6.059"	5,980	7,970	8,770

****Special Drift**

Interval	Description	Collapse (psi) a	Burst (psi) b	Tension (1000 lbs) c
0' - 6896'	7", 29#, HCP-110, LTC, 8rd	8530 / 2.37*	11220 / 1.18	797 / 2.14
6896' - 10229'	7", 32#, HCP-110, LTC, 8rd	11820 / 2.22*	12460 / 1.29	
6896' - 10229'	7", 32#, HCP-110, LTC, 8rd	11820 / 1.07**	12460 / 1.29	
10229' - 11529'	7", 32#, HCP-110, LTC, 8rd	8530 / 1.53*	11220 / 1.16	

API Rating & Safety Factor

- a. *Assume full casing evacuation with 10 ppg fluid on backside. **Assume full casing evacuation with 1.2 psi/ft equivalent fluid gradient across salt intervals.
- b. Burst pressure based on 9,000 psig max press for stimulation plus 10.2 ppg fluid in casing and 9.0 ppg fluid on backside—to 10,706' TVD.
- c. Based on string weight in 10 ppg fluid, (271k lbs buoyed weight) plus 100k lbs overpull.

Cement volumes are estimates based on 7" casing set in an 8-3/4" hole with 30% excess.

Pre-flush (Spacer): **50 bbls** Saltwater
40 bbls Weighted MudPush Express

Lead Slurry: **189 sks** (87 bbls) 2.59 yield conventional system with 61 lb/sk cement, 23 lb/sk D035 Extender, .250 lb/sk D130 Lost Circulation Control Agent, 10% NaCl, .15% D208 Viscosifier, .20% D046 Anti Foamer, .8% D112 Fluid Loss Agent, and 6% D154 Extender.

Tail Slurry: **592 sks** (184 bbls) 1.55 yield conventional system with 94 lb/sk cement, .250 lb/sk D130 Lost Circulation Control Agent, 3% KCL, 35% Silica, .2% D167 Fluid Loss Agent, .5% D065 Dispersant, .2%D198 Retarder, .2% D046 Anti Foamer, and .25% D153 Anti-Settling Agent.

Displacement: **401 bbl** brine

**Oasis Petroleum
Well Summary
Wade Federal 5300 4130 9B
Sec. 30 T153N R100W
McKenzie County, North Dakota**

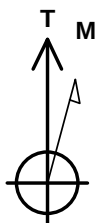
PRODUCTION LINER

							Make-up Torque (ft-lbs)		
Size	Interval	Weight	Grade	Coupling	I.D.	Drift	Minimum	Optimum	Max
4-1/2"	10179' - 20729'	11.6	P-110	BTC	4.000"	3.875"	2,270	3,020	3,780

Interval	Length	Description	Collapse	Burst	Tension	Condition
			(psi) a	(psi) b	(1000 lbs) c	
10179' - 20729'	10,550'	4-1/2", 11.6 lb, P-110, BTC, 8rd	7,560/1.53	10,690/1.11	385/1.83	New

API Rating & Safety Factor

- a) Based on full casing evacuation with 9.5 ppg fluid on backside @ 9,950' TVD.
- b) Burst pressure based on 9,000 psi treating pressure with 10.2 ppg internal fluid gradient and 9.0 ppg external fluid gradient @ 10,763' TVD.
- c) Based on string weight in 9.5 ppg fluid (Buoyed weight: 105k lbs.) plus 100k lbs overpull.



Azimuths to True North
Magnetic North: 8.17°

Magnetic Field
Strength: 56478.0snT
Dip Angle: 72.95°
Date: 3/6/2014
Model: IGRF200510

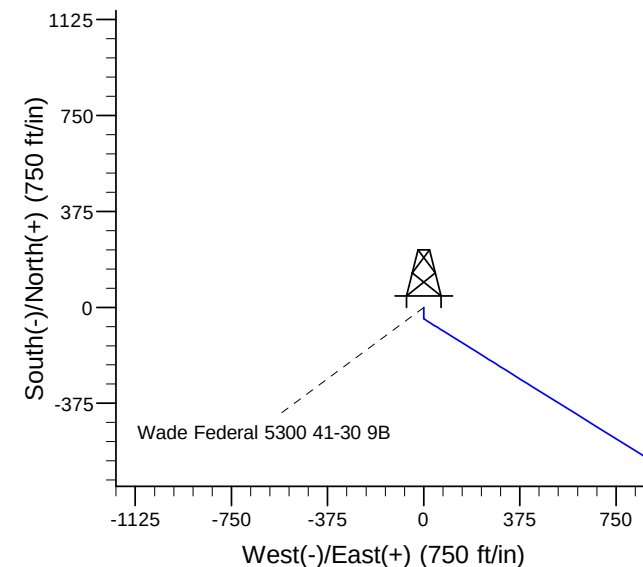
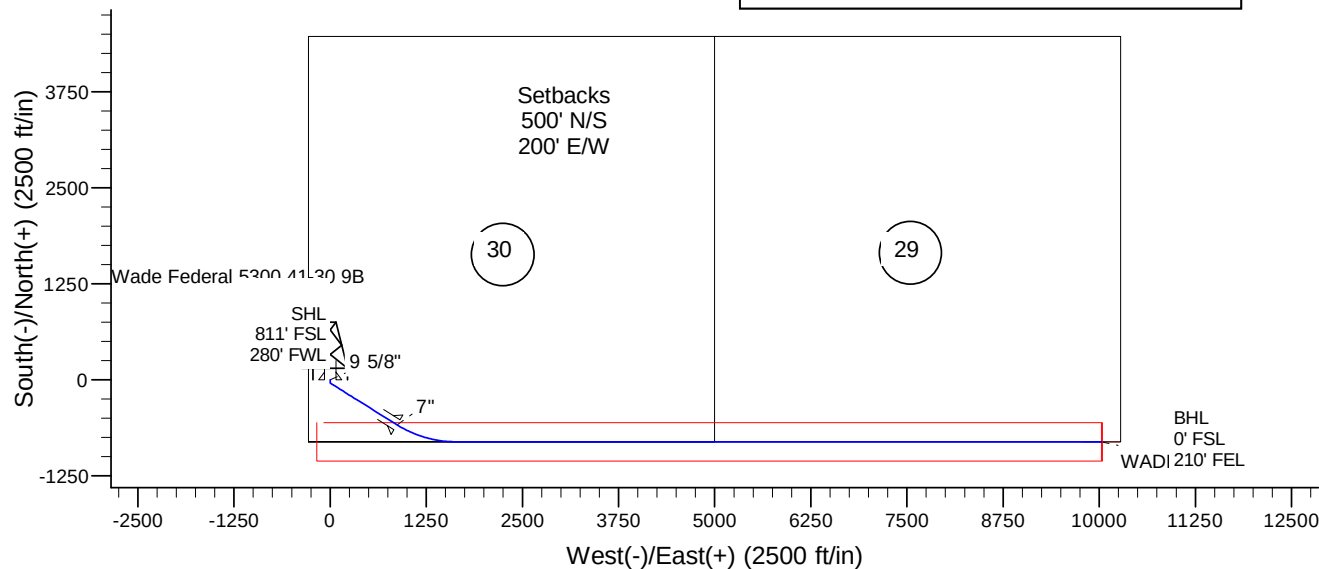


Project: Indian Hills
Site: 153N-100W-29/30
Well: Wade Federal 5300 41-30 9B
Wellbore: Wade Federal 5300 41-30 9B
Design: Plan #1

SITE DETAILS: 153N-100W-29/30

Site Centre Latitude: 48° 2' 27.670 N
Longitude: 103° 36' 10.820 W

Positional Uncertainty: 0.0
Convergence: -2.31
Local North: True



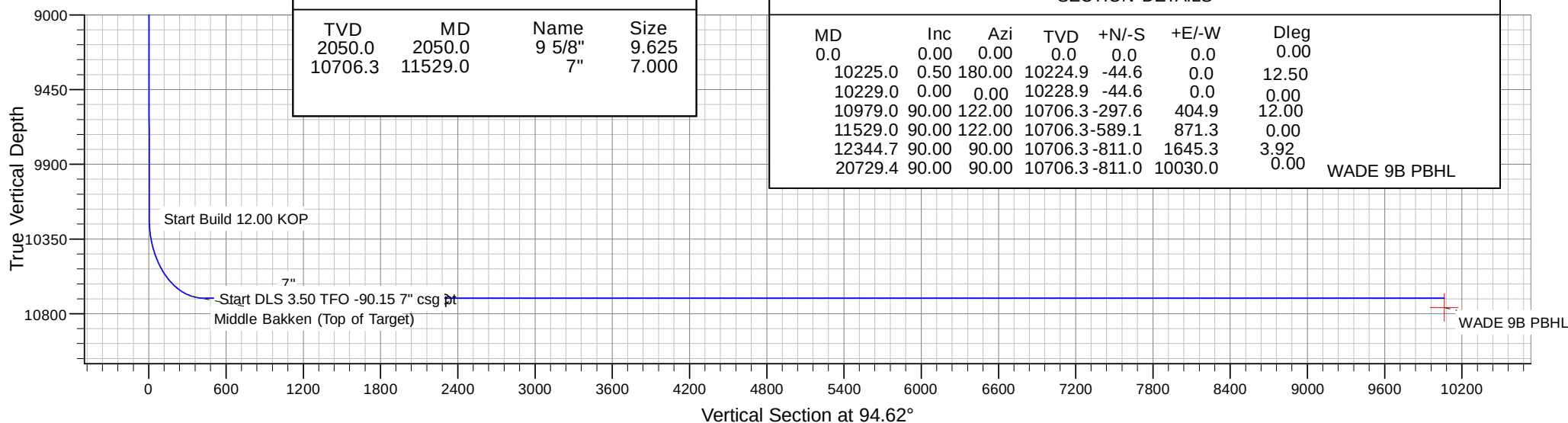
CASING DETAILS

TVD	MD	Name	Size
2050.0	2050.0	9 5/8"	9.625
10706.3	11529.0	7"	7.000

SECTION DETAILS

MD	Inc	Azi	TVD	+N/-S	+E/-W	Dleg
0.0	0.00	0.00	0.0	0.0	0.0	0.00
10225.0	0.50	180.00	10224.9	-44.6	0.0	12.50
10229.0	0.00	0.00	10228.9	-44.6	0.0	0.00
10979.0	90.00	122.00	10706.3	-297.6	404.9	12.00
11529.0	90.00	122.00	10706.3	-589.1	871.3	0.00
12344.7	90.00	90.00	10706.3	-811.0	1645.3	3.92
20729.4	90.00	90.00	10706.3	-811.0	10030.0	0.00

WADE 9B PBHL



Oasis

Indian Hills

153N-100W-29/30

Wade Federal 5300 41-30 9B

Wade Federal 5300 41-30 9B

Plan: Plan #1

Standard Planning Report

27 June, 2014

Oasis Petroleum

Planning Report

Database:	OpenWellsCompass - EDM Prod	Local Co-ordinate Reference:	Well Wade Federal 5300 41-30 9B
Company:	Oasis	TVD Reference:	WELL @ 2070.0usft (Original Well Elev)
Project:	Indian Hills	MD Reference:	WELL @ 2070.0usft (Original Well Elev)
Site:	153N-100W-29/30	North Reference:	True
Well:	Wade Federal 5300 41-30 9B	Survey Calculation Method:	Minimum Curvature
Wellbore:	Wade Federal 5300 41-30 9B		
Design:	Plan #1		

Project	Indian Hills		
Map System:	US State Plane 1983	System Datum:	Mean Sea Level
Geo Datum:	North American Datum 1983		
Map Zone:	North Dakota Northern Zone		

Site		153N-100W-29/30				
Site Position:		Northing:	395,519.95 usft	Latitude:	48° 2' 32.580 N	
From:	Lat/Long	Easting:	1,209,617.13 usft	Longitude:	103° 36' 11.410 W	
Position Uncertainty:		0.0 usft	Slot Radius:	13-3/16 "	Grid Convergence:	-2.31 °

Well	Wade Federal 5300 41-30 9B					
Well Position	+N/-S	-497.5 usft	Northing:	395,021.22 usft	Latitude:	48° 2' 27.670 N
	+E/-W	40.1 usft	Easting:	1,209,637.14 usft	Longitude:	103° 36' 10.820 W
Position Uncertainty		0.0 usft	Wellhead Elevation:		Ground Level:	2,045.0 usft

Wellbore	Wade Federal 5300 41-30 9B				
Magnetics	Model Name	Sample Date	Declination (°)	Dip Angle (°)	Field Strength (nT)
	IGRF200510	3/6/2014	8.17	72.95	56,478

Design	Plan #1			
Audit Notes:				
Version:	Phase:	PROTOTYPE	Tie On Depth:	0.0
Vertical Section:	Depth From (TVD) (usft)	+N/-S (usft)	+E/-W (usft)	Direction (°)
	0.0	0.0	0.0	94.62

Plan Sections										
Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)	TFO (°)	Target
0.0	0.00	0.00	0.0	0.0	0.0	0.00	0.00	0.00	0.00	
10,225.0	0.50	180.00	10,224.8	-44.6	0.0	0.00	0.00	0.00	180.00	
10,229.0	0.00	0.00	10,228.8	-44.6	0.0	12.50	-12.50	0.00	180.00	
10,979.0	90.00	122.00	10,706.3	-297.6	404.9	12.00	12.00	0.00	122.00	
11,529.0	90.00	122.00	10,706.3	-589.1	871.3	0.00	0.00	0.00	0.00	
12,344.7	90.00	90.00	10,706.3	-811.0	1,645.3	3.92	0.00	-3.92	270.00	
20,729.4	90.00	90.00	10,706.3	-811.0	10,030.0	0.00	0.00	0.00	0.00	WADE 9B PBHL

Oasis Petroleum

Planning Report

Database:	OpenWellsCompass - EDM Prod	Local Co-ordinate Reference:	Well Wade Federal 5300 41-30 9B
Company:	Oasis	TVD Reference:	WELL @ 2070.0usft (Original Well Elev)
Project:	Indian Hills	MD Reference:	WELL @ 2070.0usft (Original Well Elev)
Site:	153N-100W-29/30	North Reference:	True
Well:	Wade Federal 5300 41-30 9B	Survey Calculation Method:	Minimum Curvature
Wellbore:	Wade Federal 5300 41-30 9B		
Design:	Plan #1		

Planned Survey									
Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)
0.0	0.00	0.00	0.0	0.0	0.0	0.0	0.00	0.00	0.00
100.0	0.00	180.00	100.0	0.0	0.0	0.0	0.00	0.00	0.00
200.0	0.01	180.00	200.0	0.0	0.0	0.0	0.00	0.00	0.00
300.0	0.01	180.00	300.0	0.0	0.0	0.0	0.00	0.00	0.00
400.0	0.02	180.00	400.0	-0.1	0.0	0.0	0.00	0.00	0.00
500.0	0.02	180.00	500.0	-0.1	0.0	0.0	0.00	0.00	0.00
600.0	0.03	180.00	600.0	-0.2	0.0	0.0	0.00	0.00	0.00
700.0	0.03	180.00	700.0	-0.2	0.0	0.0	0.00	0.00	0.00
800.0	0.04	180.00	800.0	-0.3	0.0	0.0	0.00	0.00	0.00
900.0	0.04	180.00	900.0	-0.3	0.0	0.0	0.00	0.00	0.00
1,000.0	0.05	180.00	1,000.0	-0.4	0.0	0.0	0.00	0.00	0.00
1,100.0	0.05	180.00	1,100.0	-0.5	0.0	0.0	0.00	0.00	0.00
1,200.0	0.06	180.00	1,200.0	-0.6	0.0	0.0	0.00	0.00	0.00
1,300.0	0.06	180.00	1,300.0	-0.7	0.0	0.1	0.00	0.00	0.00
1,400.0	0.07	180.00	1,400.0	-0.8	0.0	0.1	0.00	0.00	0.00
1,500.0	0.07	180.00	1,500.0	-1.0	0.0	0.1	0.00	0.00	0.00
1,600.0	0.08	180.00	1,600.0	-1.1	0.0	0.1	0.00	0.00	0.00
1,700.0	0.08	180.00	1,700.0	-1.2	0.0	0.1	0.00	0.00	0.00
1,800.0	0.09	180.00	1,800.0	-1.4	0.0	0.1	0.00	0.00	0.00
1,900.0	0.09	180.00	1,900.0	-1.5	0.0	0.1	0.00	0.00	0.00
1,920.0	0.08	180.00	1,920.0	-1.6	0.0	0.1	0.05	-0.05	0.00
Pierre									
2,000.0	0.10	180.00	2,000.0	-1.7	0.0	0.1	0.02	0.02	0.00
2,050.0	0.10	180.00	2,050.0	-1.8	0.0	0.1	0.00	0.00	0.00
9 5/8"									
2,100.0	0.10	180.00	2,100.0	-1.9	0.0	0.2	0.00	0.00	0.00
2,200.0	0.11	180.00	2,200.0	-2.1	0.0	0.2	0.00	0.00	0.00
2,300.0	0.11	180.00	2,300.0	-2.3	0.0	0.2	0.00	0.00	0.00
2,400.0	0.12	180.00	2,400.0	-2.5	0.0	0.2	0.00	0.00	0.00
2,500.0	0.12	180.00	2,500.0	-2.7	0.0	0.2	0.00	0.00	0.00
2,600.0	0.13	180.00	2,600.0	-2.9	0.0	0.2	0.00	0.00	0.00
2,700.0	0.13	180.00	2,700.0	-3.1	0.0	0.3	0.00	0.00	0.00
2,800.0	0.14	180.00	2,800.0	-3.3	0.0	0.3	0.00	0.00	0.00
2,900.0	0.14	180.00	2,900.0	-3.6	0.0	0.3	0.00	0.00	0.00
3,000.0	0.15	180.00	3,000.0	-3.8	0.0	0.3	0.00	0.00	0.00
3,100.0	0.15	180.00	3,100.0	-4.1	0.0	0.3	0.00	0.00	0.00
3,200.0	0.16	180.00	3,200.0	-4.4	0.0	0.4	0.00	0.00	0.00
3,300.0	0.16	180.00	3,300.0	-4.6	0.0	0.4	0.00	0.00	0.00
3,400.0	0.17	180.00	3,400.0	-4.9	0.0	0.4	0.00	0.00	0.00
3,500.0	0.17	180.00	3,500.0	-5.2	0.0	0.4	0.00	0.00	0.00
3,600.0	0.18	180.00	3,600.0	-5.5	0.0	0.4	0.00	0.00	0.00
3,700.0	0.18	180.00	3,700.0	-5.8	0.0	0.5	0.00	0.00	0.00
3,800.0	0.19	180.00	3,800.0	-6.2	0.0	0.5	0.00	0.00	0.00
3,900.0	0.19	180.00	3,900.0	-6.5	0.0	0.5	0.00	0.00	0.00
4,000.0	0.20	180.00	4,000.0	-6.8	0.0	0.6	0.00	0.00	0.00
4,100.0	0.20	180.00	4,100.0	-7.2	0.0	0.6	0.00	0.00	0.00
4,200.0	0.21	180.00	4,200.0	-7.5	0.0	0.6	0.00	0.00	0.00
4,300.0	0.21	180.00	4,300.0	-7.9	0.0	0.6	0.00	0.00	0.00
4,400.0	0.22	180.00	4,400.0	-8.3	0.0	0.7	0.00	0.00	0.00
4,500.0	0.22	180.00	4,500.0	-8.6	0.0	0.7	0.00	0.00	0.00
4,566.0	0.22	180.00	4,566.0	-8.9	0.0	0.7	0.00	0.00	0.00
Greenhorn									
4,600.0	0.22	180.00	4,600.0	-9.0	0.0	0.7	0.01	0.01	0.00
4,700.0	0.23	180.00	4,700.0	-9.4	0.0	0.8	0.00	0.00	0.00
4,800.0	0.23	180.00	4,800.0	-9.8	0.0	0.8	0.00	0.00	0.00

Oasis Petroleum

Planning Report

Database:	OpenWellsCompass - EDM Prod	Local Co-ordinate Reference:	Well Wade Federal 5300 41-30 9B
Company:	Oasis	TVD Reference:	WELL @ 2070.0usft (Original Well Elev)
Project:	Indian Hills	MD Reference:	WELL @ 2070.0usft (Original Well Elev)
Site:	153N-100W-29/30	North Reference:	True
Well:	Wade Federal 5300 41-30 9B	Survey Calculation Method:	Minimum Curvature
Wellbore:	Wade Federal 5300 41-30 9B		
Design:	Plan #1		

Planned Survey										
Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)	
4,900.0	0.24	180.00	4,900.0	-10.2	0.0	0.8	0.00	0.00	0.00	
4,969.0	0.24	180.00	4,969.0	-10.5	0.0	0.8	0.00	0.00	0.00	
Mowry										
5,000.0	0.24	180.00	5,000.0	-10.7	0.0	0.9	0.02	0.02	0.00	
5,100.0	0.25	180.00	5,100.0	-11.1	0.0	0.9	0.00	0.00	0.00	
5,200.0	0.25	180.00	5,200.0	-11.5	0.0	0.9	0.00	0.00	0.00	
5,300.0	0.26	180.00	5,300.0	-12.0	0.0	1.0	0.00	0.00	0.00	
5,391.0	0.28	180.00	5,391.0	-12.4	0.0	1.0	0.03	0.03	0.00	
Dakota										
5,400.0	0.26	180.00	5,400.0	-12.4	0.0	1.0	0.21	-0.21	0.00	
5,500.0	0.27	180.00	5,500.0	-12.9	0.0	1.0	0.00	0.00	0.00	
5,600.0	0.27	180.00	5,600.0	-13.4	0.0	1.1	0.00	0.00	0.00	
5,700.0	0.28	180.00	5,700.0	-13.9	0.0	1.1	0.00	0.00	0.00	
5,800.0	0.28	180.00	5,800.0	-14.4	0.0	1.2	0.00	0.00	0.00	
5,900.0	0.29	180.00	5,900.0	-14.9	0.0	1.2	0.00	0.00	0.00	
6,000.0	0.29	180.00	6,000.0	-15.4	0.0	1.2	0.00	0.00	0.00	
6,100.0	0.30	180.00	6,100.0	-15.9	0.0	1.3	0.00	0.00	0.00	
6,200.0	0.30	180.00	6,200.0	-16.4	0.0	1.3	0.00	0.00	0.00	
6,300.0	0.31	180.00	6,300.0	-16.9	0.0	1.4	0.00	0.00	0.00	
6,400.0	0.31	180.00	6,400.0	-17.5	0.0	1.4	0.00	0.00	0.00	
6,407.0	0.27	180.00	6,407.0	-17.5	0.0	1.4	0.62	-0.62	0.00	
Rierdon										
6,500.0	0.32	180.00	6,500.0	-18.0	0.0	1.5	0.05	0.05	0.00	
6,600.0	0.32	180.00	6,600.0	-18.6	0.0	1.5	0.00	0.00	0.00	
6,700.0	0.33	180.00	6,700.0	-19.2	0.0	1.5	0.00	0.00	0.00	
6,800.0	0.33	180.00	6,800.0	-19.7	0.0	1.6	0.00	0.00	0.00	
6,896.0	0.33	180.00	6,896.0	-20.3	0.0	1.6	0.00	0.00	0.00	
Dunham Salt										
6,900.0	0.34	180.00	6,900.0	-20.3	0.0	1.6	0.08	0.08	0.00	
6,942.0	0.33	180.00	6,942.0	-20.6	0.0	1.7	0.01	-0.01	0.00	
Dunham Salt Base										
7,000.0	0.34	180.00	7,000.0	-20.9	0.0	1.7	0.02	0.02	0.00	
7,100.0	0.35	180.00	7,100.0	-21.5	0.0	1.7	0.00	0.00	0.00	
7,200.0	0.35	180.00	7,200.0	-22.1	0.0	1.8	0.00	0.00	0.00	
7,205.0	0.29	180.00	7,205.0	-22.1	0.0	1.8	1.21	-1.21	0.00	
Pine Salt										
7,229.0	0.34	180.00	7,229.0	-22.3	0.0	1.8	0.22	0.22	0.00	
Pine Salt Base										
7,291.0	0.35	180.00	7,291.0	-22.7	0.0	1.8	0.01	0.01	0.00	
Opeche Salt										
7,300.0	0.36	180.00	7,300.0	-22.7	0.0	1.8	0.05	0.05	0.00	
7,371.0	0.38	180.00	7,371.0	-23.2	0.0	1.9	0.04	0.04	0.00	
Opeche Salt Base										
7,400.0	0.36	180.00	7,400.0	-23.4	0.0	1.9	0.08	-0.08	0.00	
7,500.0	0.37	180.00	7,499.9	-24.0	0.0	1.9	0.00	0.00	0.00	
7,600.0	0.37	180.00	7,599.9	-24.6	0.0	2.0	0.00	0.00	0.00	
7,615.0	0.35	180.00	7,615.0	-24.7	0.0	2.0	0.12	-0.12	0.00	
Amsden										
7,700.0	0.38	180.00	7,699.9	-25.3	0.0	2.0	0.03	0.03	0.00	
7,771.0	0.38	180.00	7,771.0	-25.8	0.0	2.1	0.00	0.00	0.00	
Tyler										
7,800.0	0.38	180.00	7,799.9	-26.0	0.0	2.1	0.02	0.02	0.00	
7,900.0	0.39	180.00	7,899.9	-26.6	0.0	2.1	0.00	0.00	0.00	
7,994.0	0.39	180.00	7,994.0	-27.3	0.0	2.2	0.00	0.00	0.00	

Oasis Petroleum

Planning Report

Database:	OpenWellsCompass - EDM Prod	Local Co-ordinate Reference:	Well Wade Federal 5300 41-30 9B
Company:	Oasis	TVD Reference:	WELL @ 2070.0usft (Original Well Elev)
Project:	Indian Hills	MD Reference:	WELL @ 2070.0usft (Original Well Elev)
Site:	153N-100W-29/30	North Reference:	True
Well:	Wade Federal 5300 41-30 9B	Survey Calculation Method:	Minimum Curvature
Wellbore:	Wade Federal 5300 41-30 9B		
Design:	Plan #1		

Planned Survey									
Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)
Otter/Base Minnelusa									
8,000.0	0.39	180.00	7,999.9	-27.3	0.0	2.2	0.08	0.08	0.00
8,100.0	0.40	180.00	8,099.9	-28.0	0.0	2.3	0.00	0.00	0.00
8,200.0	0.40	180.00	8,199.9	-28.7	0.0	2.3	0.00	0.00	0.00
8,300.0	0.41	180.00	8,299.9	-29.4	0.0	2.4	0.00	0.00	0.00
8,336.1	0.40	180.00	8,336.0	-29.7	0.0	2.4	0.02	-0.02	0.00
Kibbey Lime									
8,400.0	0.41	180.00	8,399.9	-30.1	0.0	2.4	0.02	0.02	0.00
8,484.1	0.41	180.00	8,484.0	-30.7	0.0	2.5	0.00	0.00	0.00
Charles Salt									
8,500.0	0.42	180.00	8,499.9	-30.8	0.0	2.5	0.03	0.03	0.00
8,600.0	0.42	180.00	8,599.9	-31.6	0.0	2.5	0.00	0.00	0.00
8,700.0	0.43	180.00	8,699.9	-32.3	0.0	2.6	0.00	0.00	0.00
8,800.0	0.43	180.00	8,799.9	-33.0	0.0	2.7	0.00	0.00	0.00
8,900.0	0.44	180.00	8,899.9	-33.8	0.0	2.7	0.00	0.00	0.00
9,000.0	0.44	180.00	8,999.9	-34.6	0.0	2.8	0.00	0.00	0.00
9,100.0	0.44	180.00	9,099.9	-35.3	0.0	2.8	0.00	0.00	0.00
9,163.1	0.44	180.00	9,163.0	-35.8	0.0	2.9	0.00	0.00	0.00
Base Last Salt									
9,200.0	0.45	180.00	9,199.9	-36.1	0.0	2.9	0.02	0.02	0.00
9,300.0	0.45	180.00	9,299.9	-36.9	0.0	3.0	0.00	0.00	0.00
9,377.1	0.45	180.00	9,377.0	-37.5	0.0	3.0	0.00	0.00	0.00
Mission Canyon									
9,400.0	0.46	180.00	9,399.9	-37.7	0.0	3.0	0.02	0.02	0.00
9,500.0	0.46	180.00	9,499.9	-38.5	0.0	3.1	0.00	0.00	0.00
9,600.0	0.47	180.00	9,599.9	-39.3	0.0	3.2	0.00	0.00	0.00
9,700.0	0.47	180.00	9,699.9	-40.2	0.0	3.2	0.00	0.00	0.00
9,800.0	0.48	180.00	9,799.9	-41.0	0.0	3.3	0.00	0.00	0.00
9,900.0	0.48	180.00	9,899.9	-41.8	0.0	3.4	0.00	0.00	0.00
9,926.1	0.47	180.00	9,926.0	-42.0	0.0	3.4	0.04	-0.04	0.00
Lodgepole									
10,000.0	0.49	180.00	9,999.9	-42.7	0.0	3.4	0.02	0.02	0.00
10,100.0	0.49	180.00	10,099.9	-43.5	0.0	3.5	0.00	0.00	0.00
10,200.0	0.50	180.00	10,199.9	-44.4	0.0	3.6	0.00	0.00	0.00
10,225.0	0.50	180.00	10,224.8	-44.6	0.0	3.6	0.00	0.00	0.00
Start Drop -12.50									
10,229.0	0.00	0.00	10,228.8	-44.6	0.0	3.6	12.50	-12.50	0.00
Start Build 12.00									
10,250.0	2.52	122.00	10,249.9	-44.9	0.4	4.0	12.00	12.00	0.00
10,275.0	5.52	122.00	10,274.8	-45.8	1.9	5.6	12.00	12.00	0.00
10,300.0	8.52	122.00	10,299.6	-47.4	4.5	8.3	12.00	12.00	0.00
10,325.0	11.52	122.00	10,324.2	-49.7	8.2	12.1	12.00	12.00	0.00
10,350.0	14.52	122.00	10,348.6	-52.7	12.9	17.1	12.00	12.00	0.00
10,375.0	17.52	122.00	10,372.6	-56.4	18.8	23.3	12.00	12.00	0.00
10,400.0	20.52	122.00	10,396.2	-60.7	25.7	30.5	12.00	12.00	0.00
10,425.0	23.52	122.00	10,419.4	-65.7	33.6	38.8	12.00	12.00	0.00
10,450.0	26.52	122.00	10,442.1	-71.3	42.6	48.2	12.00	12.00	0.00
10,475.0	29.52	122.00	10,464.1	-77.5	52.6	58.6	12.00	12.00	0.00
10,500.0	32.52	122.00	10,485.5	-84.3	63.5	70.1	12.00	12.00	0.00
10,525.0	35.52	122.00	10,506.3	-91.7	75.4	82.5	12.00	12.00	0.00
10,550.0	38.52	122.00	10,526.2	-99.7	88.1	95.9	12.00	12.00	0.00
10,575.0	41.52	122.00	10,545.4	-108.2	101.8	110.1	12.00	12.00	0.00
10,600.0	44.52	122.00	10,563.6	-117.3	116.2	125.3	12.00	12.00	0.00
10,625.0	47.52	122.00	10,581.0	-126.8	131.5	141.3	12.00	12.00	0.00

Oasis Petroleum

Planning Report

Database:	OpenWellsCompass - EDM Prod	Local Co-ordinate Reference:	Well Wade Federal 5300 41-30 9B
Company:	Oasis	TVD Reference:	WELL @ 2070.0usft (Original Well Elev)
Project:	Indian Hills	MD Reference:	WELL @ 2070.0usft (Original Well Elev)
Site:	153N-100W-29/30	North Reference:	True
Well:	Wade Federal 5300 41-30 9B	Survey Calculation Method:	Minimum Curvature
Wellbore:	Wade Federal 5300 41-30 9B		
Design:	Plan #1		

Planned Survey									
Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)
10,650.0	50.52	122.00	10,597.4	-136.8	147.5	158.0	12.00	12.00	0.00
10,675.0	53.52	122.00	10,612.8	-147.2	164.2	175.5	12.00	12.00	0.00
10,700.0	56.52	122.00	10,627.1	-158.1	181.6	193.7	12.00	12.00	0.00
10,725.0	59.52	122.00	10,640.3	-169.3	199.5	212.5	12.00	12.00	0.00
10,750.0	62.52	122.00	10,652.5	-180.9	218.1	232.0	12.00	12.00	0.00
10,757.8	63.45	122.00	10,656.0	-184.6	224.0	238.1	12.00	12.00	0.00
False Bakken									
10,775.0	65.52	122.00	10,663.4	-192.8	237.1	251.9	12.00	12.00	0.00
10,786.3	66.88	122.00	10,668.0	-198.3	245.9	261.1	12.00	12.00	0.00
Upper Bakken Shale									
10,800.0	68.52	122.00	10,673.2	-205.0	256.7	272.3	12.00	12.00	0.00
10,825.0	71.52	122.00	10,681.7	-217.5	276.6	293.2	12.00	12.00	0.00
10,850.0	74.52	122.00	10,689.0	-230.1	296.9	314.4	12.00	12.00	0.00
10,874.7	77.49	122.00	10,695.0	-242.8	317.2	335.7	12.00	12.00	0.00
Middle Bakken (Top of Target)									
10,875.0	77.52	122.00	10,695.0	-243.0	317.4	336.0	12.00	12.00	0.00
10,900.0	80.52	122.00	10,699.8	-256.0	338.2	357.8	12.00	12.00	0.00
10,925.0	83.52	122.00	10,703.3	-269.1	359.2	379.8	12.00	12.00	0.00
10,950.0	86.52	122.00	10,705.4	-282.3	380.4	401.9	12.00	12.00	0.00
10,961.1	87.85	122.00	10,706.0	-288.2	389.7	411.7	12.00	12.00	0.00
Middle Bakken (Base of target)									
10,975.0	89.52	122.00	10,706.3	-295.5	401.5	424.1	12.00	12.00	0.00
10,979.0	90.00	122.00	10,706.3	-297.6	404.9	427.6	12.00	12.00	0.00
Start 550.0 hold at 10979.0 MD									
11,000.0	90.00	122.00	10,706.3	-308.8	422.7	446.3	0.00	0.00	0.00
11,100.0	90.00	122.00	10,706.3	-361.8	507.5	535.1	0.00	0.00	0.00
11,200.0	90.00	122.00	10,706.3	-414.8	592.3	623.9	0.00	0.00	0.00
11,300.0	90.00	122.00	10,706.3	-467.8	677.2	712.7	0.00	0.00	0.00
11,400.0	90.00	122.00	10,706.3	-520.8	762.0	801.5	0.00	0.00	0.00
11,500.0	90.00	122.00	10,706.3	-573.7	846.8	890.3	0.00	0.00	0.00
11,529.0	90.00	122.00	10,706.3	-589.1	871.3	916.0	0.00	0.00	0.00
Start DLS 3.92 TFO 270.00 - 7"									
11,600.0	90.00	119.21	10,706.3	-625.3	932.5	979.8	3.92	0.00	-3.92
11,700.0	90.00	115.29	10,706.3	-671.0	1,021.3	1,072.1	3.92	0.00	-3.92
11,800.0	90.00	111.37	10,706.3	-710.6	1,113.1	1,166.8	3.92	0.00	-3.92
11,900.0	90.00	107.44	10,706.3	-743.9	1,207.4	1,263.5	3.92	0.00	-3.92
12,000.0	90.00	103.52	10,706.3	-770.5	1,303.8	1,361.7	3.92	0.00	-3.92
12,100.0	90.00	99.60	10,706.3	-790.6	1,401.8	1,460.9	3.92	0.00	-3.92
12,200.0	90.00	95.68	10,706.3	-803.9	1,500.8	1,560.8	3.92	0.00	-3.92
12,300.0	90.00	91.75	10,706.3	-810.3	1,600.6	1,660.7	3.92	0.00	-3.92
12,344.7	90.00	90.00	10,706.3	-811.0	1,645.3	1,705.3	3.92	0.00	-3.92
Start 8384.7 hold at 12344.7 MD									
12,400.0	90.00	90.00	10,706.3	-811.0	1,700.6	1,760.4	0.00	0.00	0.00
12,500.0	90.00	90.00	10,706.3	-811.0	1,800.6	1,860.1	0.00	0.00	0.00
12,600.0	90.00	90.00	10,706.3	-811.0	1,900.6	1,959.8	0.00	0.00	0.00
12,700.0	90.00	90.00	10,706.3	-811.0	2,000.6	2,059.5	0.00	0.00	0.00
12,800.0	90.00	90.00	10,706.3	-811.0	2,100.6	2,159.1	0.00	0.00	0.00
12,900.0	90.00	90.00	10,706.3	-811.0	2,200.6	2,258.8	0.00	0.00	0.00
13,000.0	90.00	90.00	10,706.3	-811.0	2,300.6	2,358.5	0.00	0.00	0.00
13,100.0	90.00	90.00	10,706.3	-811.0	2,400.6	2,458.2	0.00	0.00	0.00
13,200.0	90.00	90.00	10,706.3	-811.0	2,500.6	2,557.8	0.00	0.00	0.00
13,300.0	90.00	90.00	10,706.3	-811.0	2,600.6	2,657.5	0.00	0.00	0.00
13,400.0	90.00	90.00	10,706.3	-811.0	2,700.6	2,757.2	0.00	0.00	0.00
13,500.0	90.00	90.00	10,706.3	-811.0	2,800.6	2,856.9	0.00	0.00	0.00

Oasis Petroleum

Planning Report

Database:	OpenWellsCompass - EDM Prod	Local Co-ordinate Reference:	Well Wade Federal 5300 41-30 9B
Company:	Oasis	TVD Reference:	WELL @ 2070.0usft (Original Well Elev)
Project:	Indian Hills	MD Reference:	WELL @ 2070.0usft (Original Well Elev)
Site:	153N-100W-29/30	North Reference:	True
Well:	Wade Federal 5300 41-30 9B	Survey Calculation Method:	Minimum Curvature
Wellbore:	Wade Federal 5300 41-30 9B		
Design:	Plan #1		

Planned Survey										
Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)	
13,600.0	90.00	90.00	10,706.3	-811.0	2,900.6	2,956.5	0.00	0.00	0.00	
13,700.0	90.00	90.00	10,706.3	-811.0	3,000.6	3,056.2	0.00	0.00	0.00	
13,800.0	90.00	90.00	10,706.3	-811.0	3,100.6	3,155.9	0.00	0.00	0.00	
13,900.0	90.00	90.00	10,706.3	-811.0	3,200.6	3,255.6	0.00	0.00	0.00	
14,000.0	90.00	90.00	10,706.3	-811.0	3,300.6	3,355.2	0.00	0.00	0.00	
14,100.0	90.00	90.00	10,706.3	-811.0	3,400.6	3,454.9	0.00	0.00	0.00	
14,200.0	90.00	90.00	10,706.3	-811.0	3,500.6	3,554.6	0.00	0.00	0.00	
14,300.0	90.00	90.00	10,706.3	-811.0	3,600.6	3,654.3	0.00	0.00	0.00	
14,400.0	90.00	90.00	10,706.3	-811.0	3,700.6	3,753.9	0.00	0.00	0.00	
14,500.0	90.00	90.00	10,706.3	-811.0	3,800.6	3,853.6	0.00	0.00	0.00	
14,600.0	90.00	90.00	10,706.3	-811.0	3,900.6	3,953.3	0.00	0.00	0.00	
14,700.0	90.00	90.00	10,706.3	-811.0	4,000.6	4,053.0	0.00	0.00	0.00	
14,800.0	90.00	90.00	10,706.3	-811.0	4,100.6	4,152.6	0.00	0.00	0.00	
14,900.0	90.00	90.00	10,706.3	-811.0	4,200.6	4,252.3	0.00	0.00	0.00	
15,000.0	90.00	90.00	10,706.3	-811.0	4,300.6	4,352.0	0.00	0.00	0.00	
15,100.0	90.00	90.00	10,706.3	-811.0	4,400.6	4,451.7	0.00	0.00	0.00	
15,200.0	90.00	90.00	10,706.3	-811.0	4,500.6	4,551.3	0.00	0.00	0.00	
15,300.0	90.00	90.00	10,706.3	-811.0	4,600.6	4,651.0	0.00	0.00	0.00	
15,400.0	90.00	90.00	10,706.3	-811.0	4,700.6	4,750.7	0.00	0.00	0.00	
15,500.0	90.00	90.00	10,706.3	-811.0	4,800.6	4,850.4	0.00	0.00	0.00	
15,600.0	90.00	90.00	10,706.3	-811.0	4,900.6	4,950.0	0.00	0.00	0.00	
15,700.0	90.00	90.00	10,706.3	-811.0	5,000.6	5,049.7	0.00	0.00	0.00	
15,800.0	90.00	90.00	10,706.3	-811.0	5,100.6	5,149.4	0.00	0.00	0.00	
15,900.0	90.00	90.00	10,706.3	-811.0	5,200.6	5,249.1	0.00	0.00	0.00	
16,000.0	90.00	90.00	10,706.3	-811.0	5,300.6	5,348.7	0.00	0.00	0.00	
16,100.0	90.00	90.00	10,706.3	-811.0	5,400.6	5,448.4	0.00	0.00	0.00	
16,200.0	90.00	90.00	10,706.3	-811.0	5,500.6	5,548.1	0.00	0.00	0.00	
16,300.0	90.00	90.00	10,706.3	-811.0	5,600.6	5,647.8	0.00	0.00	0.00	
16,400.0	90.00	90.00	10,706.3	-811.0	5,700.6	5,747.4	0.00	0.00	0.00	
16,500.0	90.00	90.00	10,706.3	-811.0	5,800.6	5,847.1	0.00	0.00	0.00	
16,600.0	90.00	90.00	10,706.3	-811.0	5,900.6	5,946.8	0.00	0.00	0.00	
16,700.0	90.00	90.00	10,706.3	-811.0	6,000.6	6,046.5	0.00	0.00	0.00	
16,800.0	90.00	90.00	10,706.3	-811.0	6,100.6	6,146.1	0.00	0.00	0.00	
16,900.0	90.00	90.00	10,706.3	-811.0	6,200.6	6,245.8	0.00	0.00	0.00	
17,000.0	90.00	90.00	10,706.3	-811.0	6,300.6	6,345.5	0.00	0.00	0.00	
17,100.0	90.00	90.00	10,706.3	-811.0	6,400.6	6,445.2	0.00	0.00	0.00	
17,200.0	90.00	90.00	10,706.3	-811.0	6,500.6	6,544.8	0.00	0.00	0.00	
17,300.0	90.00	90.00	10,706.3	-811.0	6,600.6	6,644.5	0.00	0.00	0.00	
17,400.0	90.00	90.00	10,706.3	-811.0	6,700.6	6,744.2	0.00	0.00	0.00	
17,500.0	90.00	90.00	10,706.3	-811.0	6,800.6	6,843.9	0.00	0.00	0.00	
17,600.0	90.00	90.00	10,706.3	-811.0	6,900.6	6,943.5	0.00	0.00	0.00	
17,700.0	90.00	90.00	10,706.3	-811.0	7,000.6	7,043.2	0.00	0.00	0.00	
17,800.0	90.00	90.00	10,706.3	-811.0	7,100.6	7,142.9	0.00	0.00	0.00	
17,900.0	90.00	90.00	10,706.3	-811.0	7,200.6	7,242.6	0.00	0.00	0.00	
18,000.0	90.00	90.00	10,706.3	-811.0	7,300.6	7,342.2	0.00	0.00	0.00	
18,100.0	90.00	90.00	10,706.3	-811.0	7,400.6	7,441.9	0.00	0.00	0.00	
18,200.0	90.00	90.00	10,706.3	-811.0	7,500.6	7,541.6	0.00	0.00	0.00	
18,300.0	90.00	90.00	10,706.3	-811.0	7,600.6	7,641.3	0.00	0.00	0.00	
18,400.0	90.00	90.00	10,706.3	-811.0	7,700.6	7,740.9	0.00	0.00	0.00	
18,500.0	90.00	90.00	10,706.3	-811.0	7,800.6	7,840.6	0.00	0.00	0.00	
18,600.0	90.00	90.00	10,706.3	-811.0	7,900.6	7,940.3	0.00	0.00	0.00	
18,700.0	90.00	90.00	10,706.3	-811.0	8,000.6	8,039.9	0.00	0.00	0.00	
18,800.0	90.00	90.00	10,706.3	-811.0	8,100.6	8,139.6	0.00	0.00	0.00	
18,900.0	90.00	90.00	10,706.3	-811.0	8,200.6	8,239.3	0.00	0.00	0.00	
19,000.0	90.00	90.00	10,706.3	-811.0	8,300.6	8,339.0	0.00	0.00	0.00	

Oasis Petroleum

Planning Report

Database:	OpenWellsCompass - EDM Prod	Local Co-ordinate Reference:	Well Wade Federal 5300 41-30 9B
Company:	Oasis	TVD Reference:	WELL @ 2070.0usft (Original Well Elev)
Project:	Indian Hills	MD Reference:	WELL @ 2070.0usft (Original Well Elev)
Site:	153N-100W-29/30	North Reference:	True
Well:	Wade Federal 5300 41-30 9B	Survey Calculation Method:	Minimum Curvature
Wellbore:	Wade Federal 5300 41-30 9B		
Design:	Plan #1		

Planned Survey										
Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)	
19,100.0	90.00	90.00	10,706.3	-811.0	8,400.6	8,438.6	0.00	0.00	0.00	
19,200.0	90.00	90.00	10,706.3	-811.0	8,500.6	8,538.3	0.00	0.00	0.00	
19,300.0	90.00	90.00	10,706.3	-811.0	8,600.6	8,638.0	0.00	0.00	0.00	
19,400.0	90.00	90.00	10,706.3	-811.0	8,700.6	8,737.7	0.00	0.00	0.00	
19,500.0	90.00	90.00	10,706.3	-811.0	8,800.6	8,837.3	0.00	0.00	0.00	
19,600.0	90.00	90.00	10,706.3	-811.0	8,900.6	8,937.0	0.00	0.00	0.00	
19,700.0	90.00	90.00	10,706.3	-811.0	9,000.6	9,036.7	0.00	0.00	0.00	
19,800.0	90.00	90.00	10,706.3	-811.0	9,100.6	9,136.4	0.00	0.00	0.00	
19,900.0	90.00	90.00	10,706.3	-811.0	9,200.6	9,236.0	0.00	0.00	0.00	
20,000.0	90.00	90.00	10,706.3	-811.0	9,300.6	9,335.7	0.00	0.00	0.00	
20,100.0	90.00	90.00	10,706.3	-811.0	9,400.6	9,435.4	0.00	0.00	0.00	
20,200.0	90.00	90.00	10,706.3	-811.0	9,500.6	9,535.1	0.00	0.00	0.00	
20,300.0	90.00	90.00	10,706.3	-811.0	9,600.6	9,634.7	0.00	0.00	0.00	
20,400.0	90.00	90.00	10,706.3	-811.0	9,700.6	9,734.4	0.00	0.00	0.00	
20,500.0	90.00	90.00	10,706.3	-811.0	9,800.6	9,834.1	0.00	0.00	0.00	
20,600.0	90.00	90.00	10,706.3	-811.0	9,900.6	9,933.8	0.00	0.00	0.00	
20,700.0	90.00	90.00	10,706.3	-811.0	10,000.6	10,033.4	0.00	0.00	0.00	
20,729.4	90.00	90.00	10,706.3	-811.0	10,030.0	10,062.7	0.00	0.00	0.00	
TD at 20729.4										

Design Targets									
Target Name	Dip Angle (°)	Dip Dir. (°)	TVD (usft)	+N/-S (usft)	+E/-W (usft)	Northing (usft)	Easting (usft)	Latitude	Longitude
- hit/miss target									
- Shape									
WADE 9B PBHL	0.00	0.00	10,706.3	-811.0	10,030.0	393,806.78	1,219,626.30	48° 2' 19.640 N	103° 33' 43.220 W
- plan hits target center									
- Point									

Casing Points					
Measured Depth (usft)	Vertical Depth (usft)	Name	Casing Diameter (")	Hole Diameter (")	
2,050.0	2,050.0	9 5/8"	9-5/8	13-1/2	
11,529.0	10,706.3	7"	7	8-3/4	

Oasis Petroleum

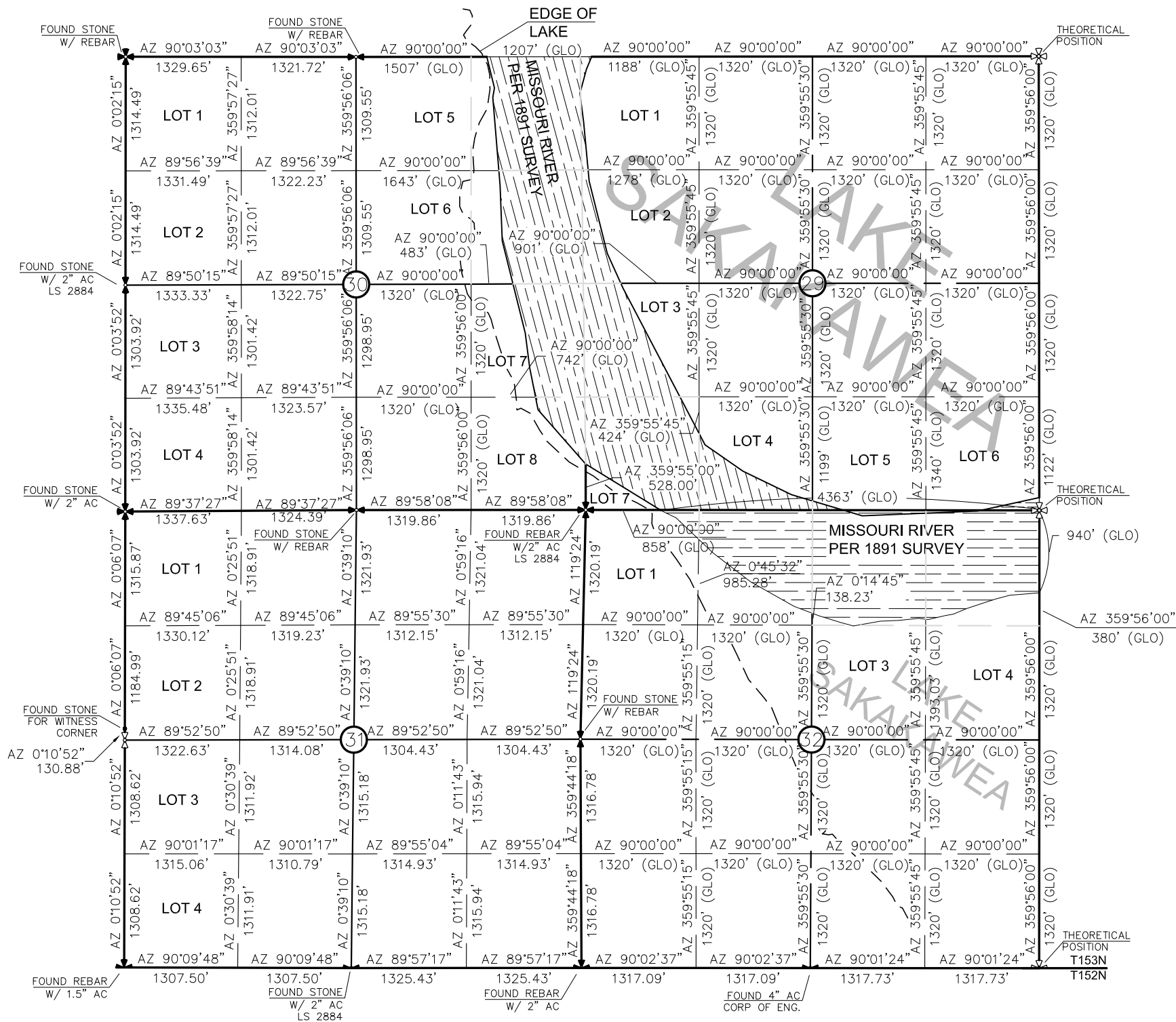
Planning Report

Database:	OpenWellsCompass - EDM Prod	Local Co-ordinate Reference:	Well Wade Federal 5300 41-30 9B
Company:	Oasis	TVD Reference:	WELL @ 2070.0usft (Original Well Elev)
Project:	Indian Hills	MD Reference:	WELL @ 2070.0usft (Original Well Elev)
Site:	153N-100W-29/30	North Reference:	True
Well:	Wade Federal 5300 41-30 9B	Survey Calculation Method:	Minimum Curvature
Wellbore:	Wade Federal 5300 41-30 9B		
Design:	Plan #1		

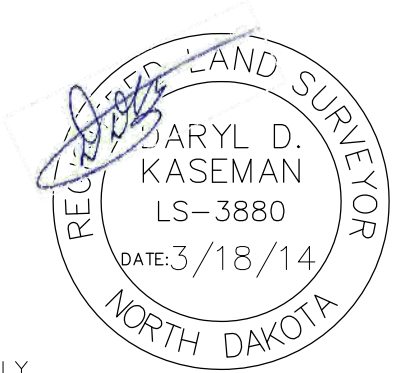
Formations						
Measured Depth (usft)	Vertical Depth (usft)	Name	Lithology	Dip (°)	Dip Direction (°)	
1,920.0	1,920.0	Pierre				
4,566.0	4,566.0	Greenhorn				
4,969.0	4,969.0	Mowry				
5,391.0	5,391.0	Dakota				
6,407.0	6,407.0	Rierdon				
6,896.0	6,896.0	Dunham Salt				
6,942.0	6,942.0	Dunham Salt Base				
7,205.0	7,205.0	Pine Salt				
7,229.0	7,229.0	Pine Salt Base				
7,291.0	7,291.0	Opeche Salt				
7,371.0	7,371.0	Opeche Salt Base				
7,615.0	7,615.0	Amsden				
7,771.0	7,771.0	Tyler				
7,994.0	7,994.0	Otter/Base Minnelusa				
8,336.1	8,336.0	Kibbey Lime				
8,484.1	8,484.0	Charles Salt				
9,163.1	9,163.0	Base Last Salt				
9,377.1	9,377.0	Mission Canyon				
9,926.1	9,926.0	Lodgepole				
10,757.8	10,656.0	False Bakken				
10,786.3	10,668.0	Upper Bakken Shale				
10,874.7	10,695.0	Middle Bakken (Top of Target)				
10,961.1	10,706.0	Middle Bakken (Base of target)				

Plan Annotations					
Measured Depth (usft)	Vertical Depth (usft)	Local Coordinates			
		+N/-S (usft)	+E/-W (usft)	Comment	
10,225.0	10,224.8	-44.6	0.0	Start Drop -12.50	
10,229.0	10,228.8	-44.6	0.0	Start Build 12.00	
10,979.0	10,706.3	-297.6	404.9	Start 550.0 hold at 10979.0 MD	
11,529.0	10,706.3	-589.1	871.3	Start DLS 3.92 TFO 270.00	
12,344.7	10,706.3	-811.0	1,645.3	Start 8384.7 hold at 12344.7 MD	
20,729.4	10,706.3	-811.0	10,030.0	TD at 20729.4	

SECTION BREAKDOWN
OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"WADE FEDERAL 5300 41-30 9B"
811 FEET FROM SOUTH LINE AND 280 FEET FROM WEST LINE
SECTIONS 29 & 30, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA

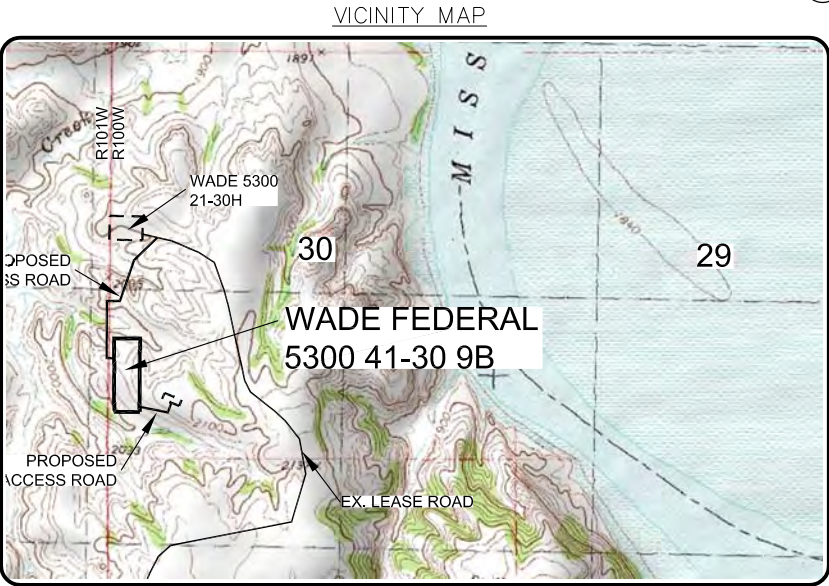
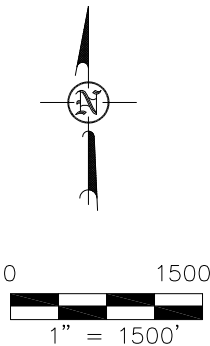


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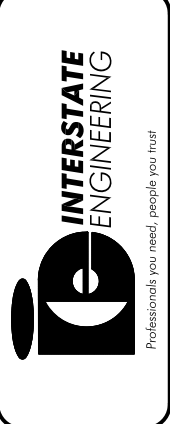


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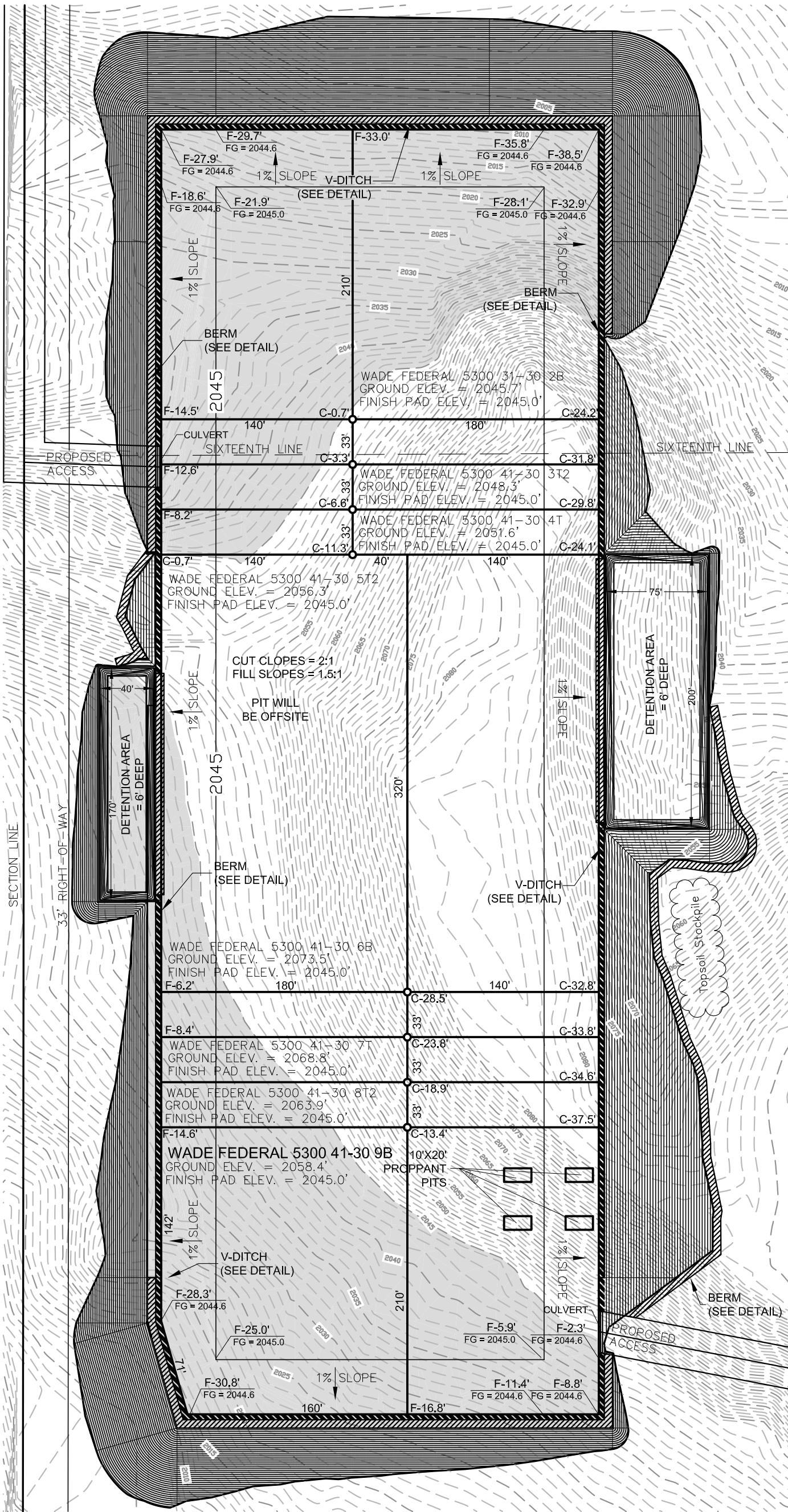
Revision No.	Date	By	Description
REV 1	3/17/14	JJS	MOVED WELLS ON PAD & ACCESS ROAD

OASIS PETROLEUM NORTH AMERICA, LLC SECTION BREAKDOWN SECTIONS 29 & 30, T153N, R100W MCKENZIE COUNTY, NORTH DAKOTA	Project No.: S13-09-381.08 Date: JAN. 2014
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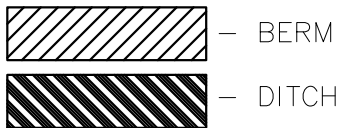
PAD LAYOUT
OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"WADE FEDERAL 5300 41-30 9B"
811 FEET FROM SOUTH LINE AND 280 FEET FROM WEST LINE
SECTION 30, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



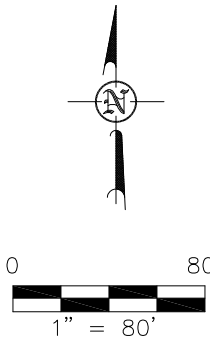
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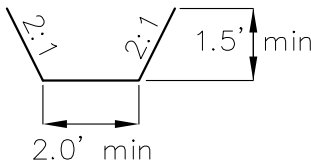
NOTE: Pad dimensions shown are to usable area, the v-ditch and berm areas shall be built to the outside of the pad dimensions.



Proposed Contours
Original Contours



V-DITCH DETAIL



NOTE: All utilities shown are preliminary only, a complete utilities location is recommended before construction.

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OASIS PETROLEUM NORTH AMERICA, LLC
PAD LAYOUT
SECTION 30, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: B.H.H. Project No.: S13-09-381.08
Checked By: D.D.K. Date: JAN. 2014

Revision No.	Date	By	Description
REV 1	3/17/14	JJS	MOVED WELLS ON PAD & ACCESS ROAD

© 2013/013-08-1081.08 Oasis Petroleum - Wade 5300 41-30 9B Well Location & Well Pad (CAD) REVISED: WADE 9B.dwg - 3/18/2014 7:11 AM joan schriener

WELL LOCATION SITE QUANTITIES
 OASIS PETROLEUM NORTH AMERICA, LLC
 1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
 "WADE FEDERAL 5300 41-30 9B"
 811 FEET FROM SOUTH LINE AND 280 FEET FROM WEST LINE
 SECTION 30, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA

WELL SITE ELEVATION	2058.4
WELL PAD ELEVATION	2045.0
EXCAVATION	149,966
PLUS PIT	<u>0</u>
	149,966
EMBANKMENT	113,402
PLUS SHRINKAGE (25%)	<u>28,351</u>
	141,753
STOCKPILE PIT	0
STOCKPILE TOP SOIL (6")	7,955
BERMS	2,533 LF = 821 CY
DITCHES	1,655 LF = 253 CY
DETENTION AREA	4,219 CY
STOCKPILE MATERIAL	3,909
DISTURBED AREA FROM PAD	10.18 ACRES

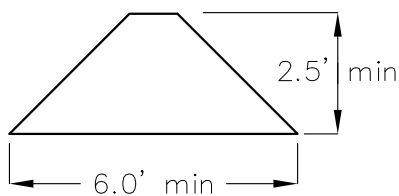
NOTE: ALL QUANTITIES ARE IN CUBIC YARDS (UNLESS NOTED)
 CUT END SLOPES AT 2:1
 FILL END SLOPES AT 1.5:1

WELL SITE LOCATION

811' FSL

280' FWL

BERM DETAIL



DITCH DETAIL



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OASIS PETROLEUM NORTH AMERICA, LLC
 QUANTITIES
 SECTION 30, T153N, R100W
 MCKENZIE COUNTY, NORTH DAKOTA

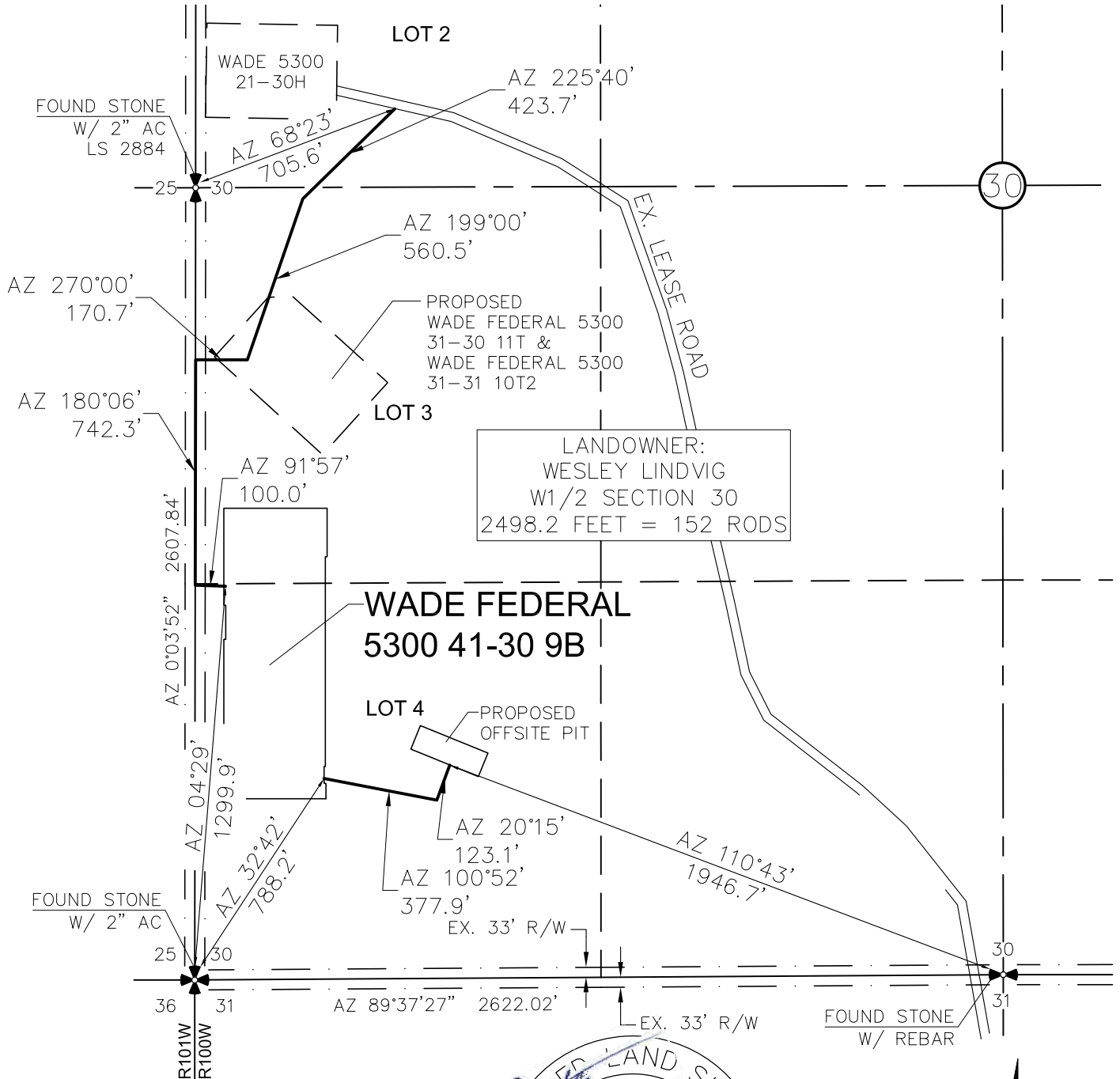
Drawn By: B.J.H. Project No.: S13-09-381,08
 Checked By: D.D.K. Date: JAN, 2014

Revision No.	Date	By	Description
REV 1	3/17/14	JJS	MOVED WELLS ON PAD & ACCESS ROAD

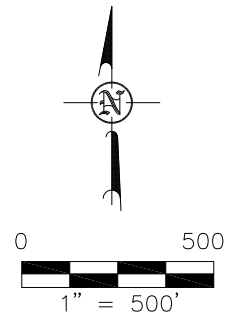
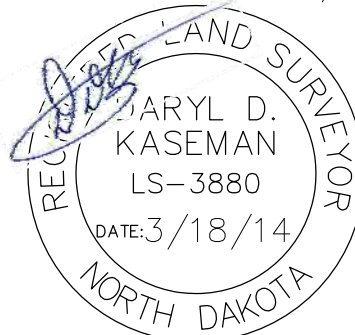
ACCESS APPROACH

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"WADE FEDERAL 5300 41-30 9B"

811 FEET FROM SOUTH LINE AND 280 FEET FROM WEST LINE
SECTION 30, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



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NOTE: All utilities shown are preliminary only, a complete
utilities location is recommended before construction.

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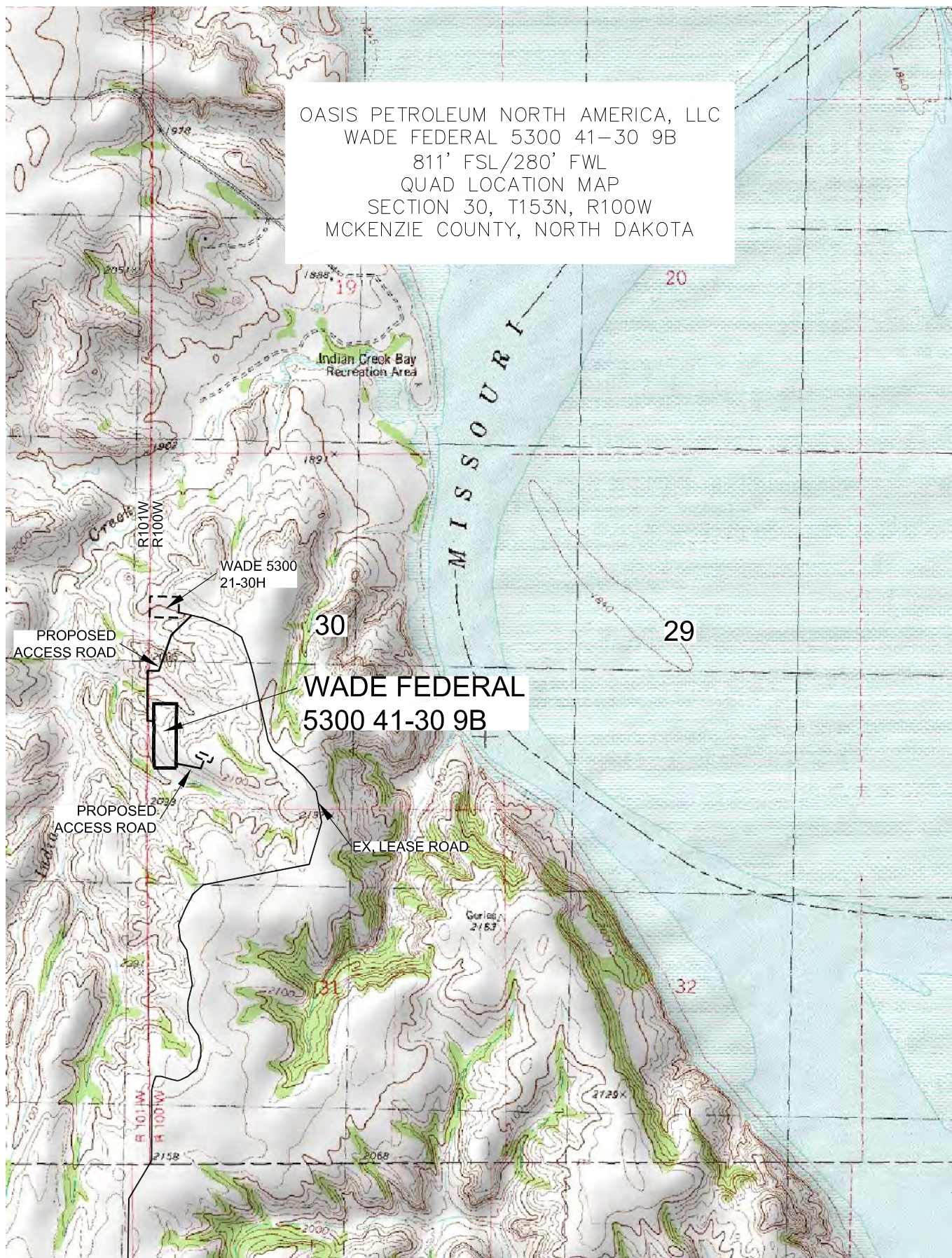
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OASIS PETROLEUM NORTH AMERICA, LLC
ACCESS APPROACH
SECTION 30, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: B.H.H. Project No.: S13-09-381_08
Checked By: D.D.K. Date: JAN, 2014

Revision No.	Date	By	Description
REV 1	3/17/14	JJS	MOVED WELLS ON PAD & ACCESS ROAD

OASIS PETROLEUM NORTH AMERICA, LLC
 WADE FEDERAL 5300 41-30 9B
 811' FSL/280' FWL
 QUAD LOCATION MAP
 SECTION 30, T153N, R100W
 MCKENZIE COUNTY, NORTH DAKOTA



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OASIS PETROLEUM NORTH AMERICA, LLC
 QUAD LOCATION MAP
 SECTION 30, T153N, R100W
 MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: B.J.H. Project No.: S13-09-381.08
 Checked By: D.D.K. Date: JAN, 2014

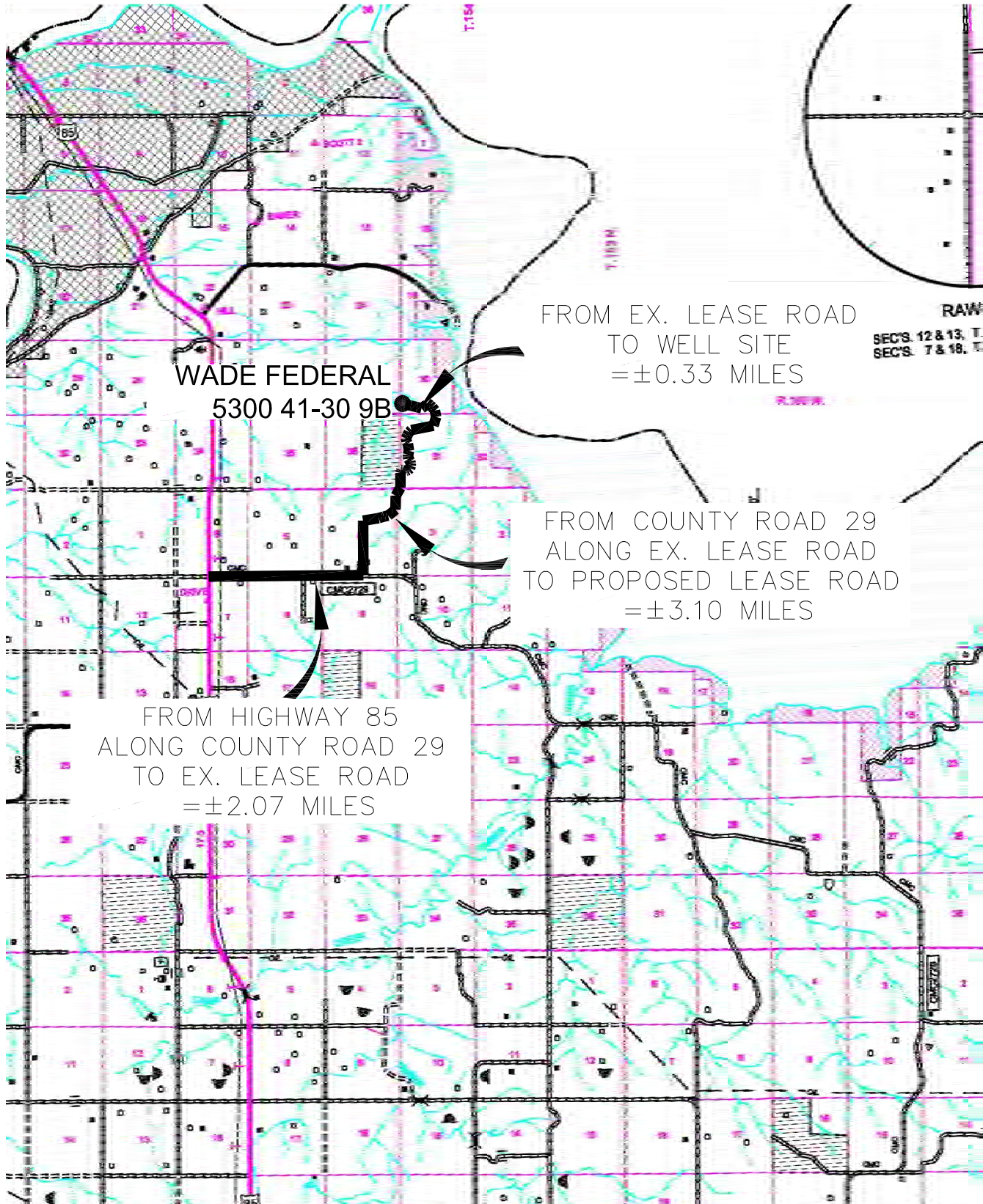
Revision No.	Date	By	Description
REV 1	3/17/14	JJS	MOVED WELLS ON PAD & ACCESS ROAD

COUNTY ROAD MAP

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002

"WADE FEDERAL 5300 41-30 9B"

811 FEET FROM SOUTH LINE AND 280 FEET FROM WEST LINE
SECTION 30, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



**WADE FEDERAL
5300 41-30 9B**

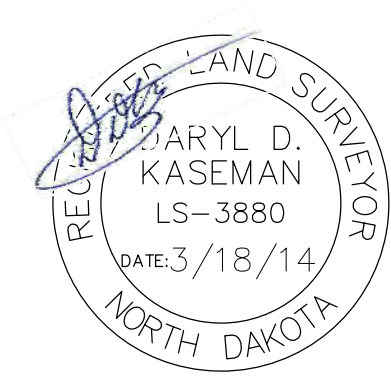
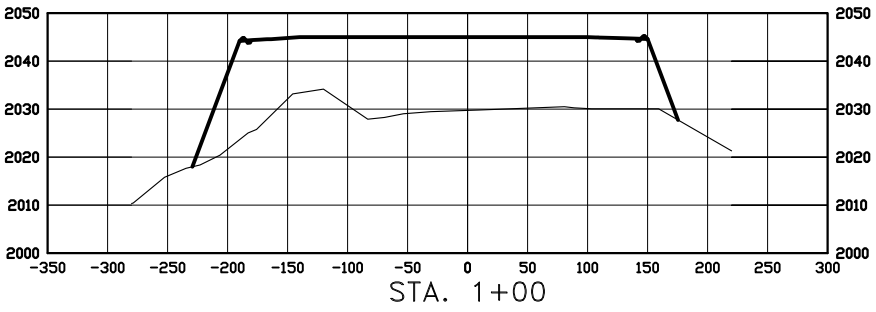
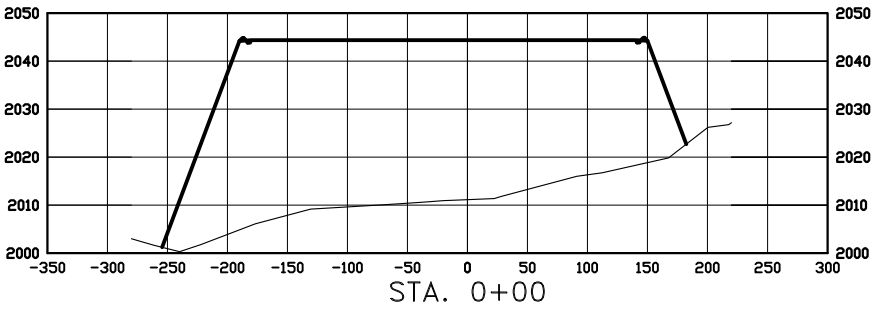
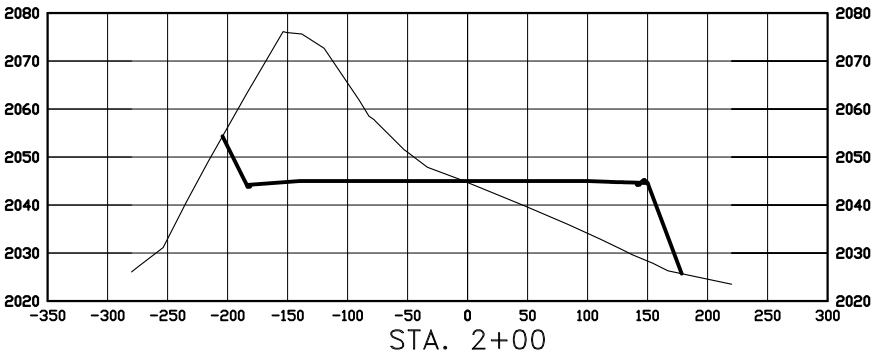
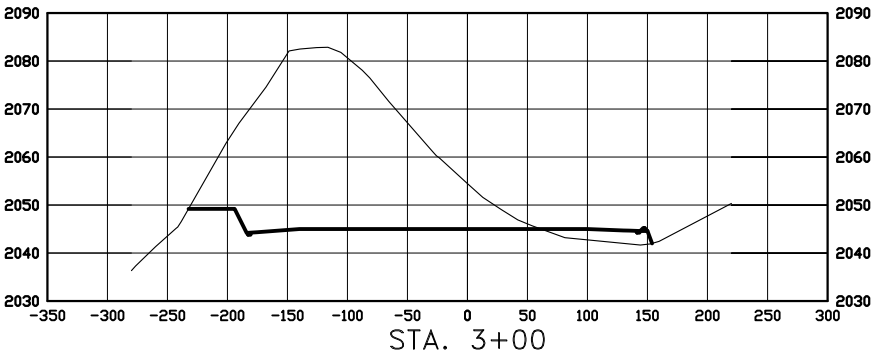
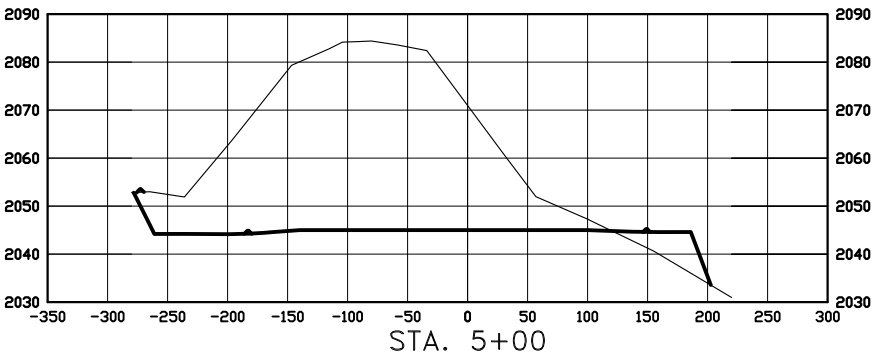
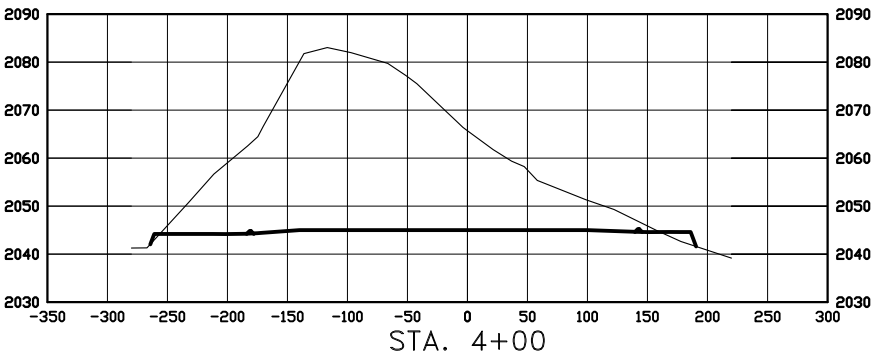
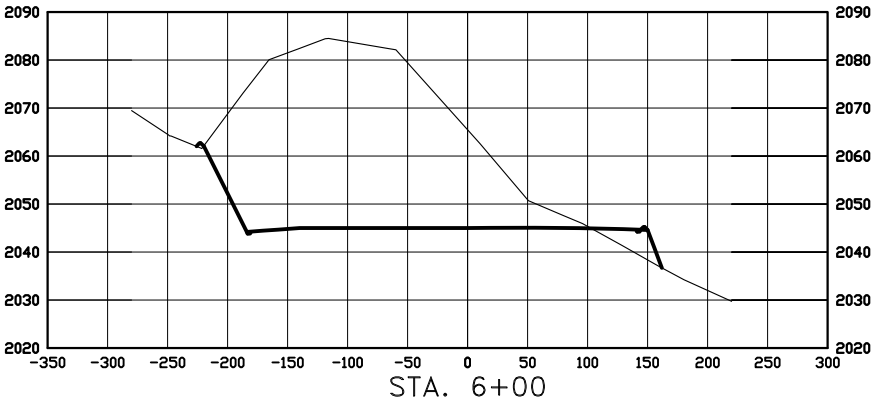
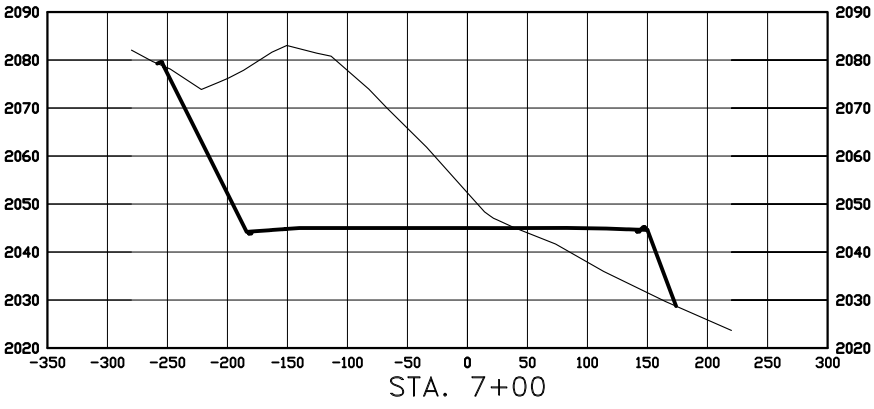
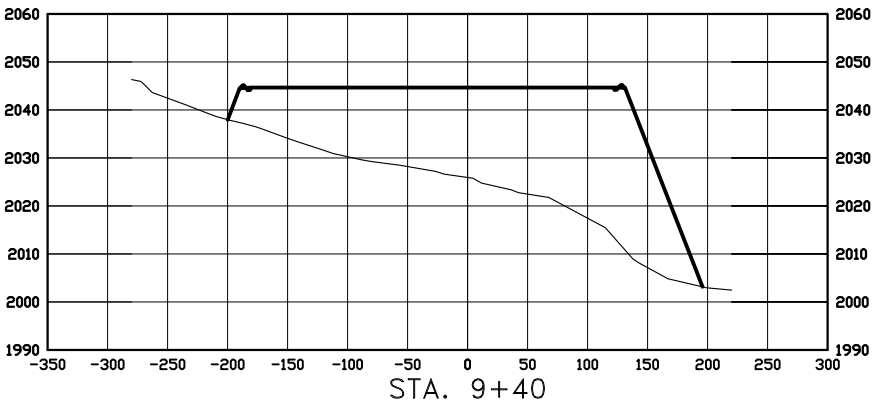
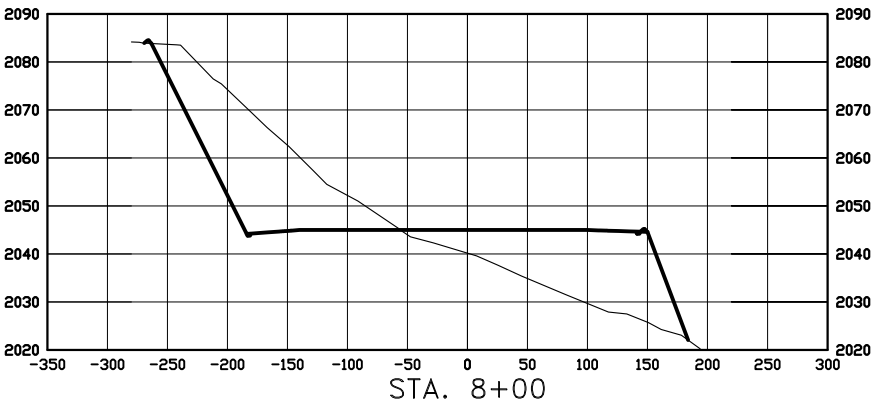
FROM EX. LEASE ROAD
TO WELL SITE
= ±0.33 MILES

RAW
SEC'S. 12 & 13, T.
SEC'S. 7 & 18, T.

FROM COUNTY ROAD 29
ALONG EX. LEASE ROAD
TO PROPOSED LEASE ROAD
= ±3.10 MILES

FROM HIGHWAY 85
ALONG COUNTY ROAD 29
TO EX. LEASE ROAD
= ±2.07 MILES

CROSS SECTIONS
OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"WADE FEDERAL 5300 41-30 9B"
811 FEET FROM SOUTH LINE AND 280 FEET FROM WEST LINE
SECTION 30, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



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SCALE
HORIZ 1"=160'
VERT 1"=40'

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CROSS SECTIONS
SECTION 30, T153N, R100W

MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: B.H.H.

Project No.: S13-09-381.08

Checked By: D.D.K.

Date: JAN, 2014

Revision No.	Date	By	Description
REV 1	3/17/14	JJS	MOVED WELLS ON PAD & ACCESS ROAD
C:\2013\ST13-09-381.08 Oasis Petroleum - Wade 5300 41-30 9B Well Location Well Pad\CAD\REVISED WADE 9B.dwg - 3/19/2014 7:12 AM jsh schimeler			

STATEMENT

This statement is being sent in order to comply with NDAC 43-02-03-16 (Application for permit to drill and recomplete) which states (in part that) "confirmation that a legal street address has been requested for the well site, and well facility if separate from the well site, and the proposed road access to the nearest existing public road". On the date noted below a legal street address was requested from the appropriate county office.

McKenzie County

Aaron Chisolm – McKenzie County Dept.

Wade Federal 5300 31-30 2B

Wade Federal 5300 31-30 3T2

Wade Federal 5300 41-30 4T

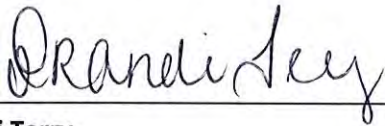
Wade Federal 5300 41-30 5T2

Wade Federal 5300 41-30 6B

Wade Federal 5300 41-30 7T

Wade Federal 5300 41-30 8T2

Wade Federal 5300 41-30 9B

A handwritten signature in black ink, reading "Brandi Terry". The signature is written in a cursive, flowing style. Below the signature is a horizontal line.

Brandi Terry

Regulatory Specialist

Oasis Petroleum North America, LLC

From: [Heather McCowan](#)
To: [Webber, Alice D.](#); [Chelsea Covington](#); [Brandi Terry](#)
Subject: RE: Wade Federal 8 Well Pad
Date: Thursday, June 05, 2014 9:25:20 AM
Attachments: [image002.png](#)

Hello Alice,

The cuttings will be buried offsite for this well. I believe a sundry was sent in.

Thanks,

Heather McCowan

Regulatory Assistant | 1001 Fannin, Suite 1500, Houston, Texas 77002 | 281-404-9563 Direct |
hmccowan@oasispetroleum.com



From: Webber, Alice D. [mailto:adwebber@nd.gov]
Sent: Thursday, June 05, 2014 9:12 AM
To: Chelsea Covington; Brandi Terry; Heather McCowan
Subject: RE: Wade Federal 8 Well Pad

Good morning Ladies,

Just checking in to see if we know where the cuttings are being hauled to for this pad. That is the only thing we are waiting for and I would like to get these permitted for Oasis today.

Thanks,
Alice
